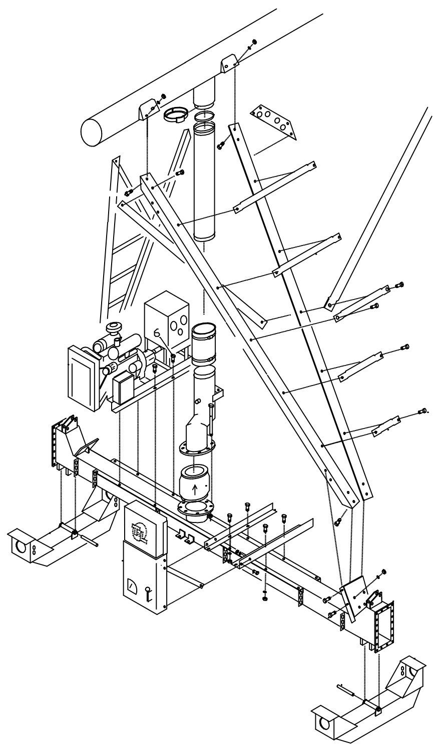
Linear Irrigation Systems
The choice is simple.

HOSE DRAG LINEAR INSTALLATION MANUAL

With Above Cable & Buried Guidance for Manual Control Linear



Use in conjunction with T-L Pivot Installation Manual.

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CD90451
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CONTENTS

OTHER LINEAR MANUALS:

CD90452 PLC Linear Installation Manual
CD90472 PLC Above & Buried Operators Manual
CD90479 Linear GPS Navigation for Manual Control
CD90480 Linear GPS Navigation for PLC Control

CONTENTS

CONTENTS

**** IMPORTANT ****

Read the Installation Manual thoroughly before assembly. The linear system has many special items that must be installed correctly. Reassembly is frustrating and takes valuable time. Note the location of special items such as: Oil lines and End Tower By-Pass.

- The Linear Installation Manual is to be used in conjunction with the standard T-L PIVOT INSTALLATION MANUAL.

- Throughout this Manual, several terms are used: Forward, Reverse, Left, Right, Tractor unit and End (or Lead) Tower. The tractor unit supports the engine, fuel tank, hydraulic control panel and the guidance system. On an End Feed Linear, the Tractor unit is the Left End and the Right End is the Lead Tower. On a Center Feed Linear, there are 2 lead towers located on the ends of the system. (See Figures 1 and 2)

- The roadway the tractor runs on, must be kept dry and free from any side slope. Part circle sprinklers or drops may have to be used.

- The guidance cable (below or above ground) must be installed straight and at the correct depth or height. See later sections on cable installation

FLUSHING See Pivot Installation manual for Starting and flushing procedure. Make sure no span lines are crossed and that no alignment valves are in bypass.

Center Feed **-** Flush each wing of the system separately. The linear system pressure and return and the lead tower lines will also need to be flushed separately. It works best to flush the lead tower lines first. Jumper the lead tower 1"lines at the top of last tower for flushing. Jumper the pressure and return at the next to end tower for flushing.

End Feed **-** The linear system pressure and return and the lead tower lines need to be flushed separately. It works best to flush the lead tower lines first. Jumper the lead tower 1" lines at the top of last tower for flushing. Jumper the pressure and return at the next to end tower for flushing.

UNLOADING

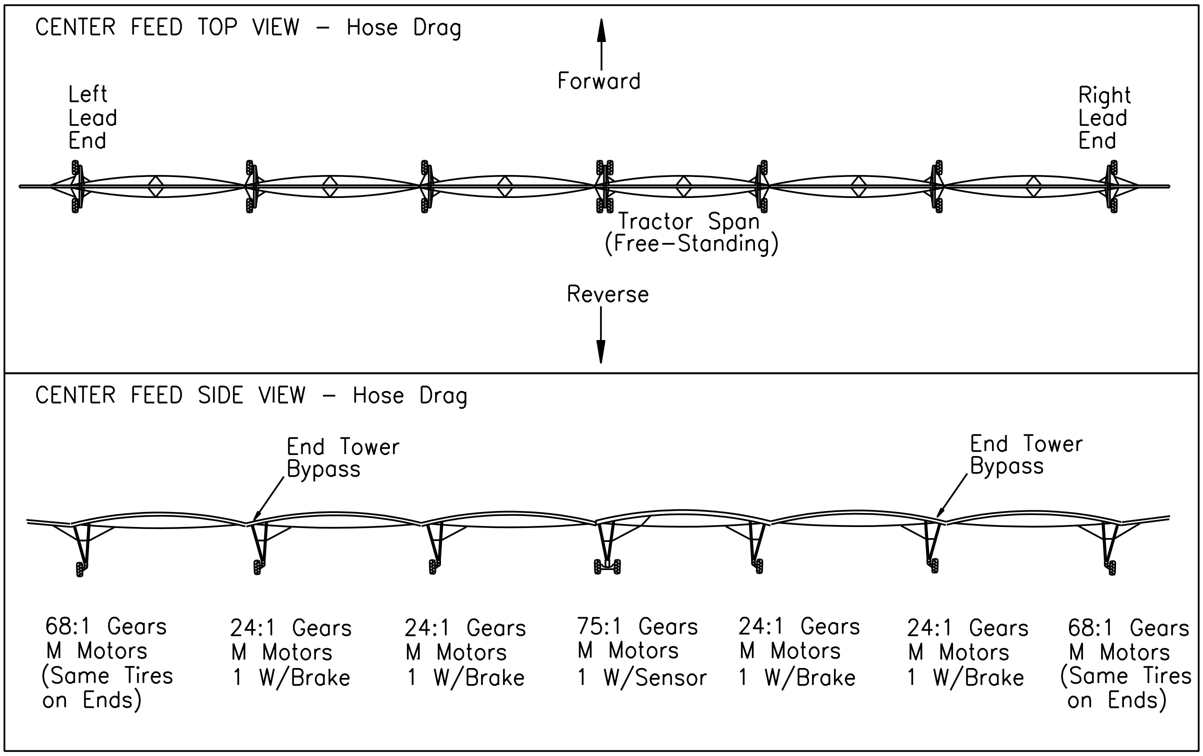
The unloading procedure for a linear is somewhat like unloading two pivots. The free standing span acts as a pivot point. The spans for each side of the system are unloaded from the free standing span out the same as if you were unloading a pivot. Note the Tractor placement on the Center Feed vs. the End Feed. (See Figures 1 and 2)

NOTE: For systems with Pivoting Option and 5 spans or less, locate the Free Standing Span as the first span.

Caution: See Figures 1 and 2 for the gear/motor combinations to be used at each tower and the tractor. “M” motors are painted blue. The 75:1 gears are used on the Tractor. The tires on the Tractor should be either 14.9 or 16.9 x 24. The end towers must have the same size tires and motors.

NOTE: If gears and tires are not located properly, guidance tracking problems will occur.

Center Feed Hose Drag Linear Layout



02/07/22 M Motors on End Towers, were G Motors

Figure 1. Center Feed Hose Drag Linear Layout

End Feed Hose Drag Linear Layout

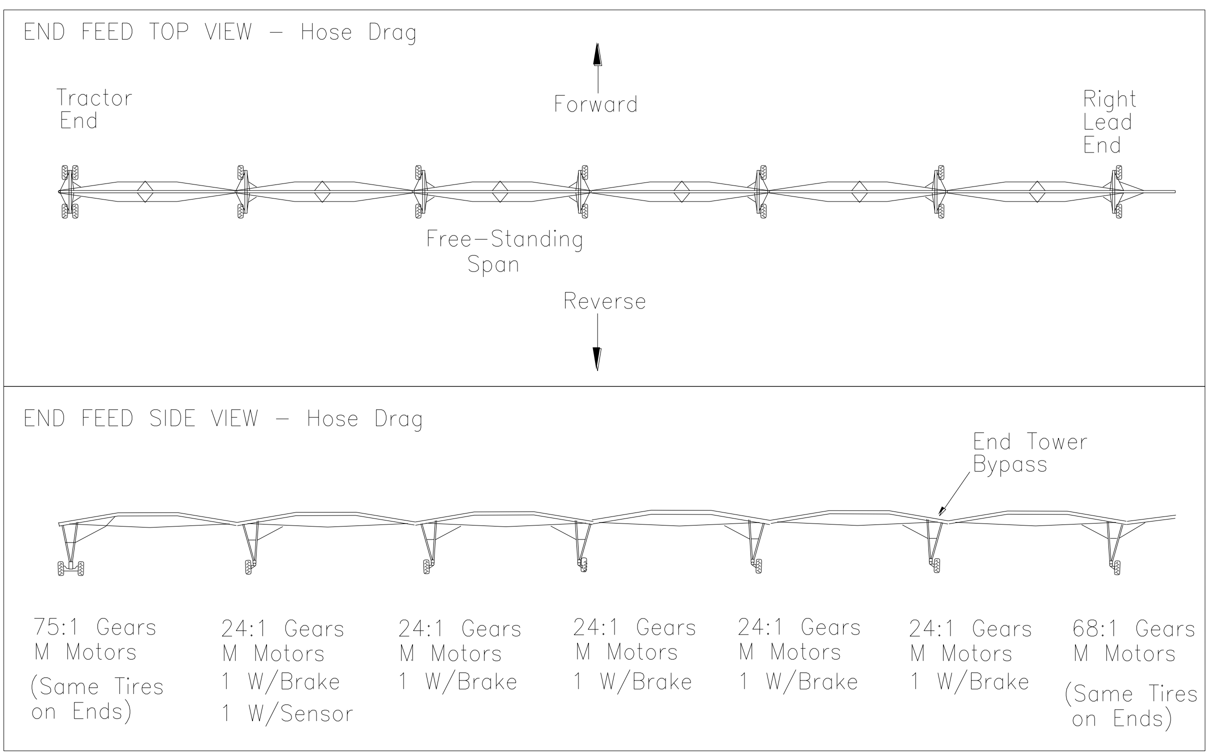
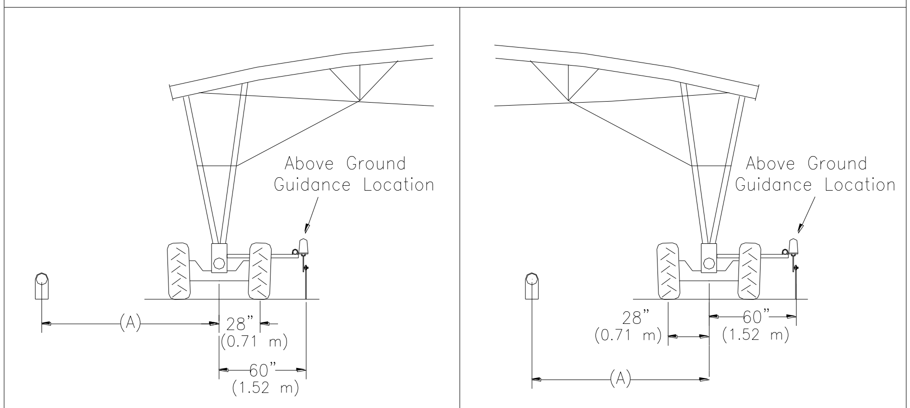


Figure 2. End Feed Hose Drag Linear Layout

TRACTOR INLET TO RISER DIMENSIONS (Figure 2a)

4 WHEEL LINEAR TRACTOR: INLET TO RISER DIMENSIONS
PIVOTING: SYSTEM PIVOTS ABOUT THE CENTER OF THE TRACTOR



Hose	(A) Hose Loop
4" Soft	5 ft (1.52 m)
4 1/2" Soft	7 ft (2.13 m)
5" Soft	8 ft (2.44 m)
5" Hard	18 ft (5.49 m)
6" Hard	22 ft (6.71 m)

Figure 2a. Tractor Inlet to Riser and Guidance Dimensions

TYPICAL RISER INSTALLATION WITH T-L PART NUMBERS (Figure 2b)

IV75931 VOE 6 X 4 BLF
IV75932 VOE 6 X 5 BLF
IV7593 VOE 6 X 6 BLF
IV75934 VOE 6 X 4 BLF
w/chains (soft hose)
IV75935 VOE 6 X 5 BLF
w/chains (soft hose)

IV76045 Valve Stub, 6 X 6 for
PVC

IC6877 Draw Band 6"

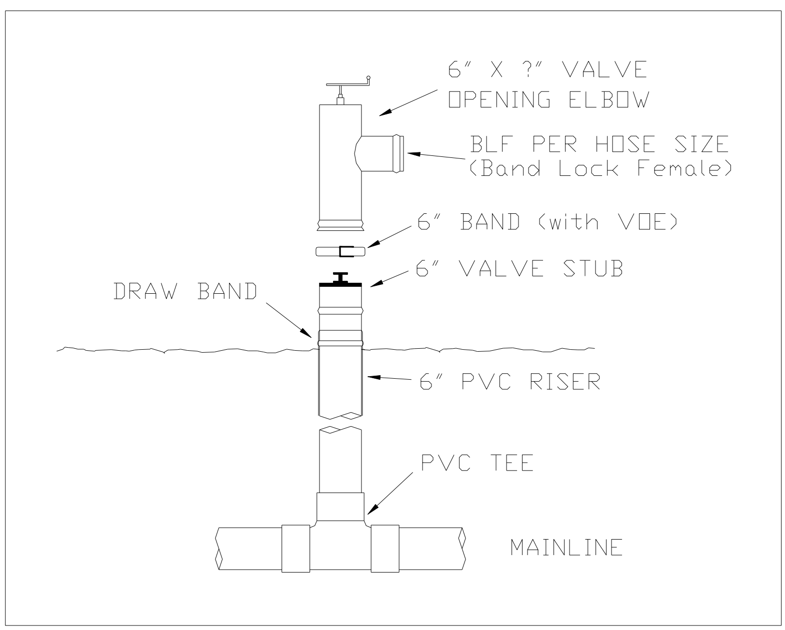


Figure 2b. Typical Riser Installation for Hose Drag Linears.

TYPICAL HOSE LAYOUT (Figure 2c)

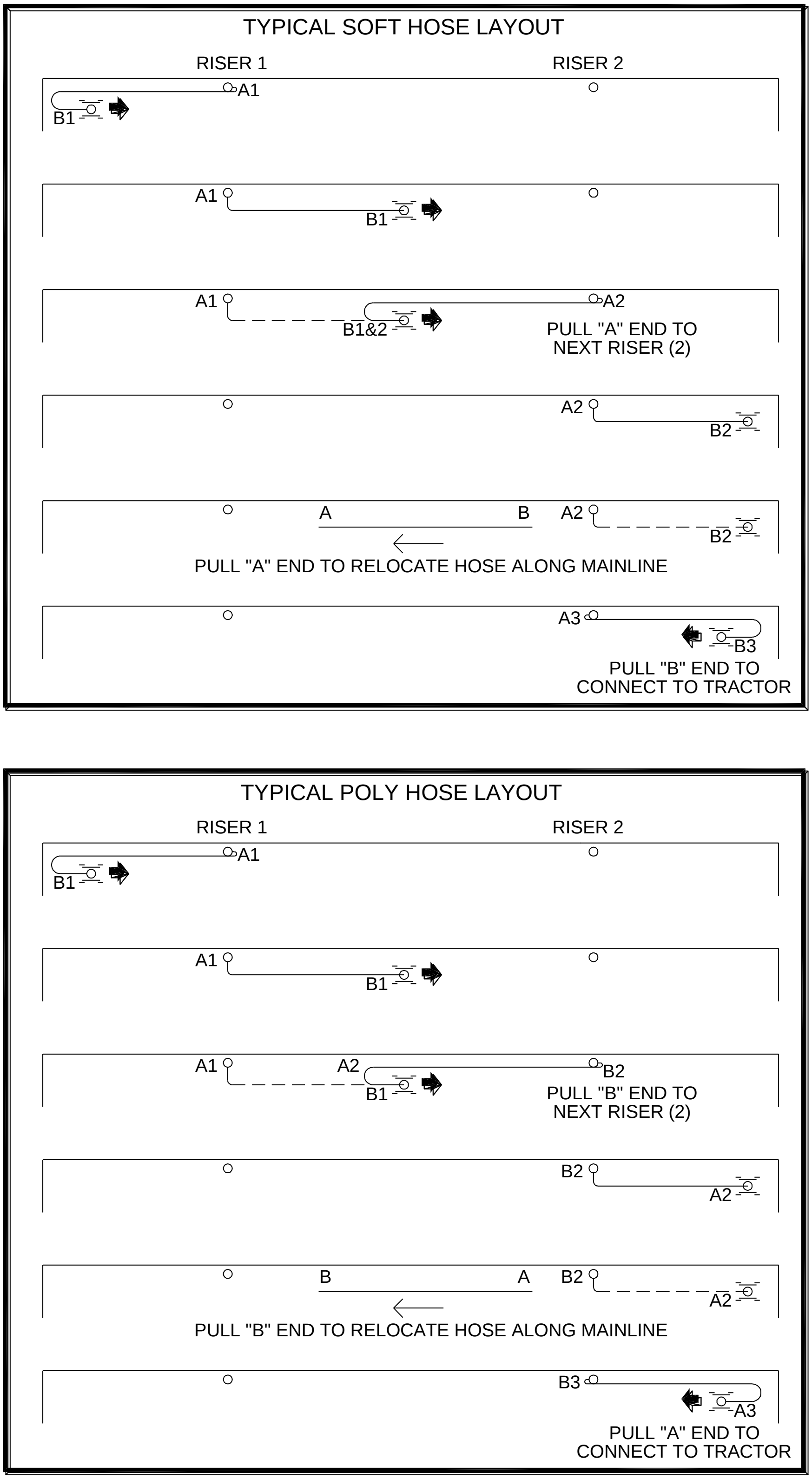
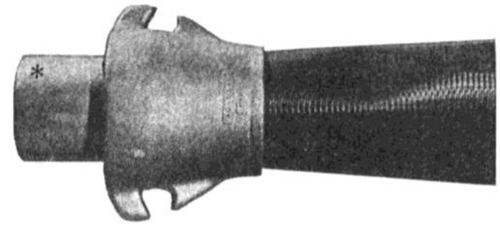


Figure 2c. Typical Hose Layouts for Soft and Poly (Hard) Hose

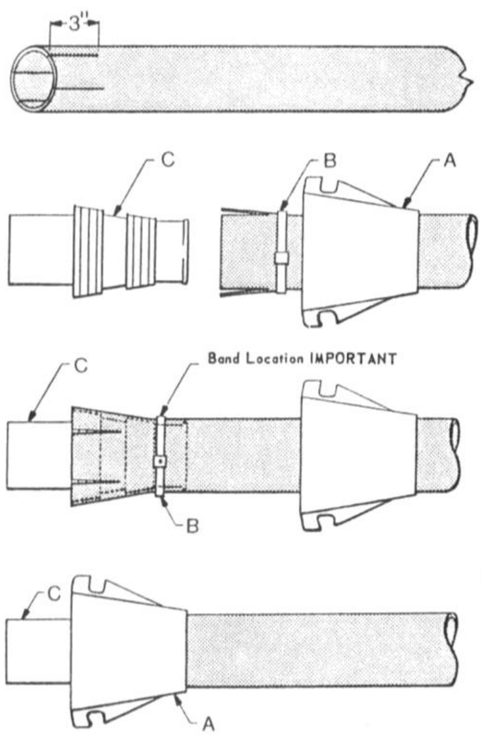
Soft Hose End Installation (Figure 2d)

Hose Pull Couplers designed and patented specifically for use with soft ose travelers, hose fed linears, slurry injection hoses or any high pressure application.

Hose Dia	Discharge* O.D.
5.0 ins	5 ins
4.5	5
4.5	4
4/4⅛	4
3.5	4
3.0	4
3.0	3
2.5	3



INSTALLATION INSTRUCTIONS



Step 1:

Cut the end of the hose off square, then measure 3-inches from the hose end and mark all the way around. Cut four evenly spaced slits on even quarters exactly 3-inches deep to the mark you made.

CAUTION: Be sure to make no less than four slits and make them evenly spaced. Tabs wider than 1/4 of the hose circumference will wrinkle on the coupler causing uneven tension on the hose strands resulting in premature failure. Also make sure the slits are of equal 3-inch depth. Uneven depth of slits will cause the coupler to "cock" to one side, concentrating pull on one side of the hose, resulting in premature failure. Slits less than 3-inches deep permit insufficient contact between the hose and coupler resulting in decreased pull capacity.

Step 2:

Place the outer coupler member (A) and Punch-Lok Band (B) over the prepared hose end as shown. Do not tighten the Punch-Lok Band yet.

Step 3:

Insert the inner coupler member (C) into the hose end as far as possible. Now position the band around the untapered recess of inner coupler member (C) as shown next to the tapered shoulder making sure the slits are well forward of the band. Slits under the band will cause leaking.
CAUTION: Never install the band between the tapered shoulders on the inner coupler member, as immediate failure will result. When pull is applied to a properly installed coupler some slippage will occur as the coupler seats to the hose. For this reason, make sure the band is properly placed as instructed. Otherwise, the band may cut the hose.

Step 4:

Tighten the Punch-Lok Band (B) securely with the Punch-Lok Tool.
CAUTION: Don't strike the locking punch so hard as to dent the inner coupler. Don't bend the end of the band back over the buckle. Doing so can create leaks.

Pull the outer coupler member (A) up tight over the assembled band and hose on the inner coupler member (C). The finished installation should appear as shown.

OPERATING HINT: Never pull the hose coupler by only one ear. Doing so can break off the ear or loosen the coupler and band resulting in the hose being cut by the band.

Figure 2d. Soft Hose End Installation

Hard Hose Installation (Figure 2e & 2f)

Figure 2e. Hard Hose Welding

Figure 2f. Hard Hose Ends

FIELD DIMENSIONS (Figure 3)

Dimension information for the Guidance Cable, either Above Ground Cable or Buried Wire Cable, is required to properly locate the guidance with respect to the Linear Spans. See Figure 3 for dimension information. More information can be found in the Linear Design Section of the T-L Dealer Book. The actual installation of the Guidance Cable is covered in later sections of this manual.

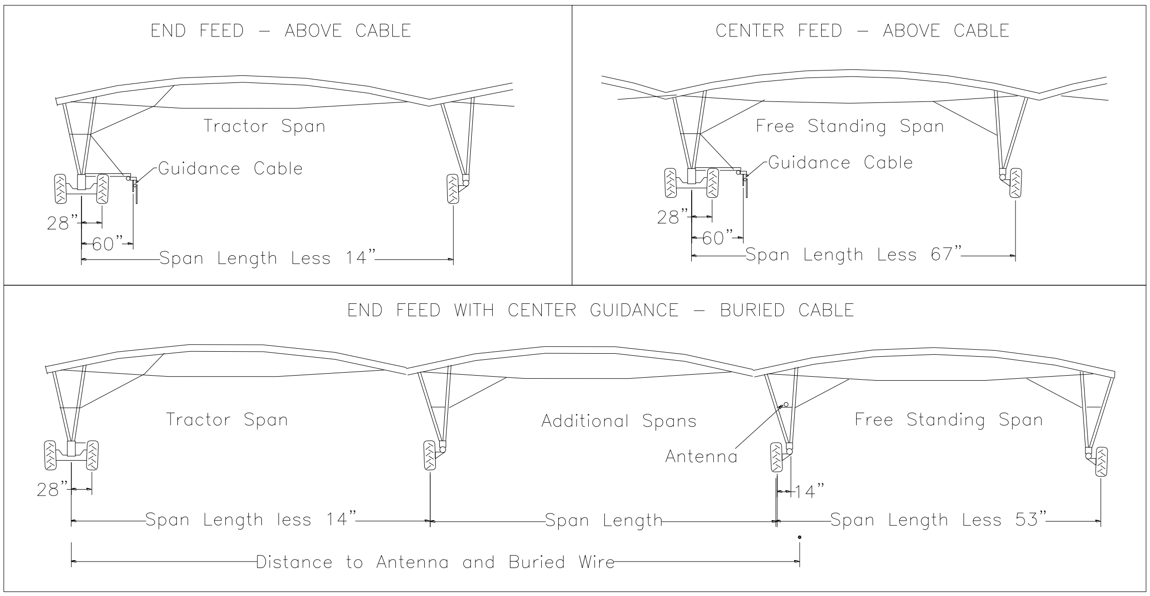


Figure 3. Field Dimensions

SENSOR MOTOR LOCATION – Overwater Timer (Figure 3a)
Overwater Timer used on Manual Control Linears, not PLC Linears.

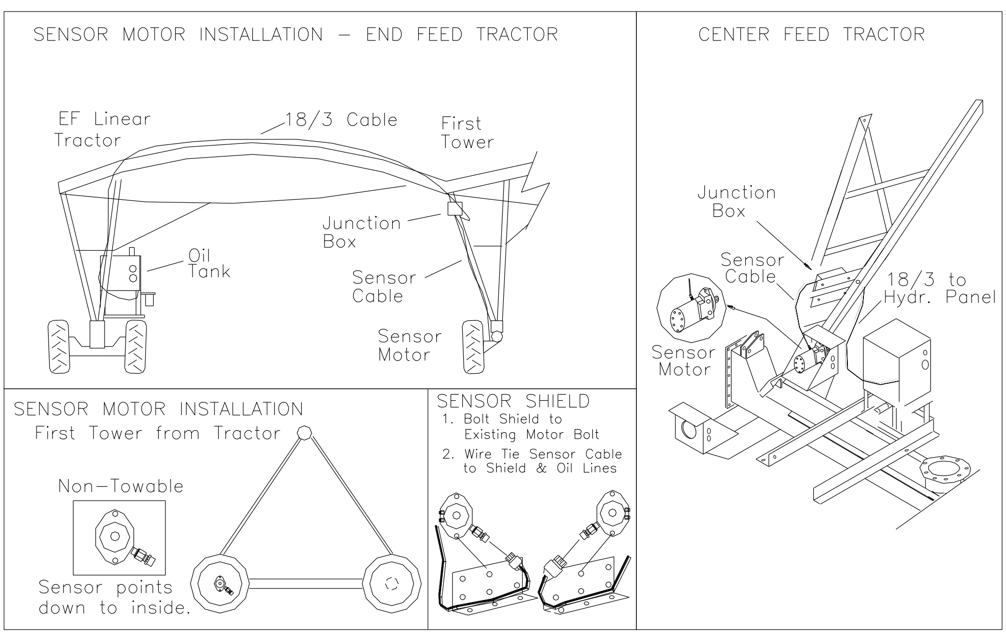


Figure 3a. Speed Sensor Motor Location – Center Feed & End Feed Tractors

TRACTOR SPAN – END FEED SYSTEM: (Figure 4)

On an End Feed Linear, the tractor span will have a couple of special parts: the Tractor A-Pipe has a six inch inlet and 2 Truss Support Braces are installed. The Truss Support Brace (3, with 15’ 8-3/8” hole centers) is placed between the #1truss and #2 truss to strengthen the span. During the Tractor A-pipe assembly, bolt two Small Truss Braces (3) with offset toward center, along with the #1 AV Truss Brace (10), Tower Brace (1) and Truss Rods (11,12) using two ¾” x 2 ¾” bolts w/LW, nuts (13). Add a second ¾”nut (15) to the bolt holding the Tower Brace (1) to lock the bolt from working loose. Bolt the opposite end of the Small Truss Brace (3) onto the Tractor A-pipe at the #2 Truss Brace Bracket along with the #2 Truss Brace, not shown.

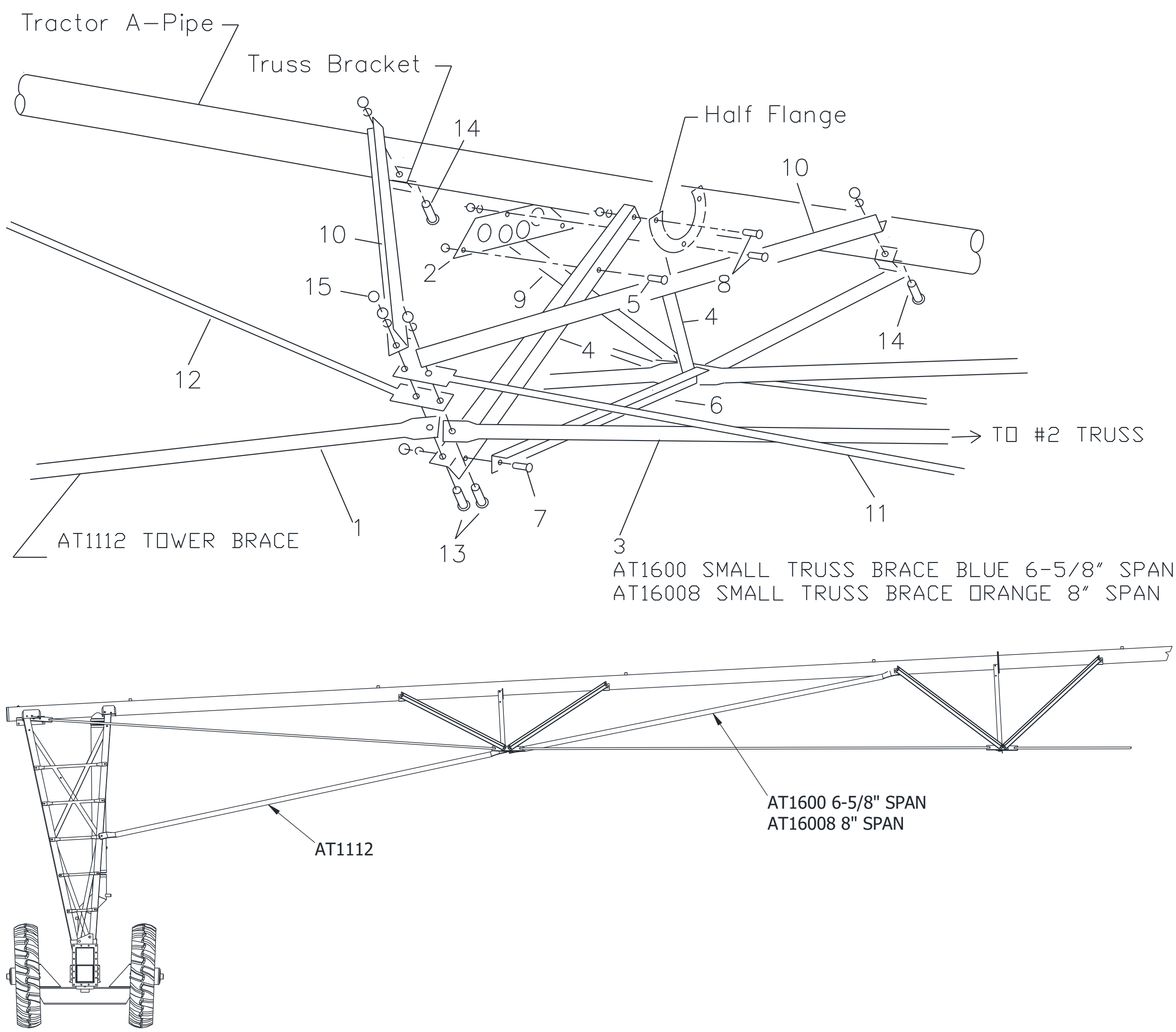


Figure 4. Truss Support Brace Installation

PANEL ASSEMBLY: (Figure 5)

1. Mount Electric control panel (1) to mount angles (2, one each left and right) using four rubber mounts (3) with eight 5/16" nut and lock washer (4,5).

2. Insert spacer washer (7) and mounting strap (8) between Hydraulic panel (6) and mount angles (2). Bolt together using four 3/8" x 2" bolts (9)with locks and nuts (10,11).

3. Bolt reversing switch (12) to back side of Hydraulic panel face (6) using two 10-32 x ½ screws (13). Wire switch into electric panel as shown in the wiring diagram..

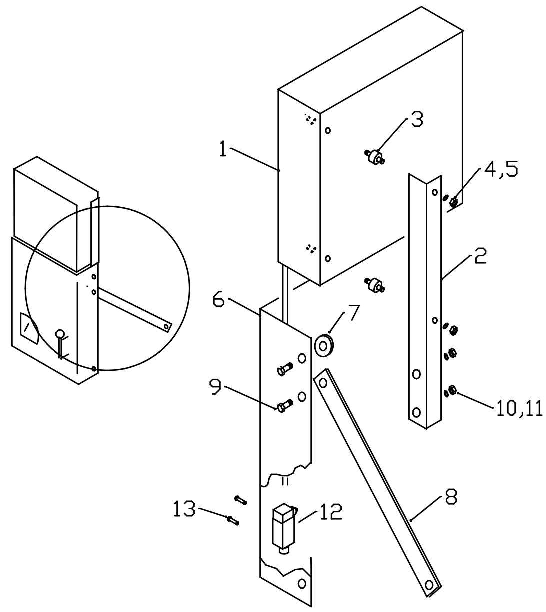


Figure 5. Control Panel Assembly

COMPONENT LOCATION ON TRACTOR BASE (Figure 5a)

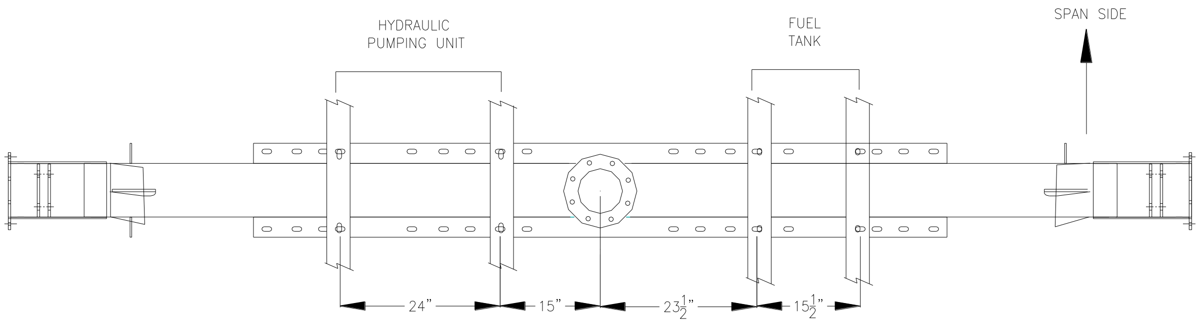


Figure 5a. Location of Components on Tractor Base.

FREE STANDING SPAN CONSTRUCTION (Figure 5b)

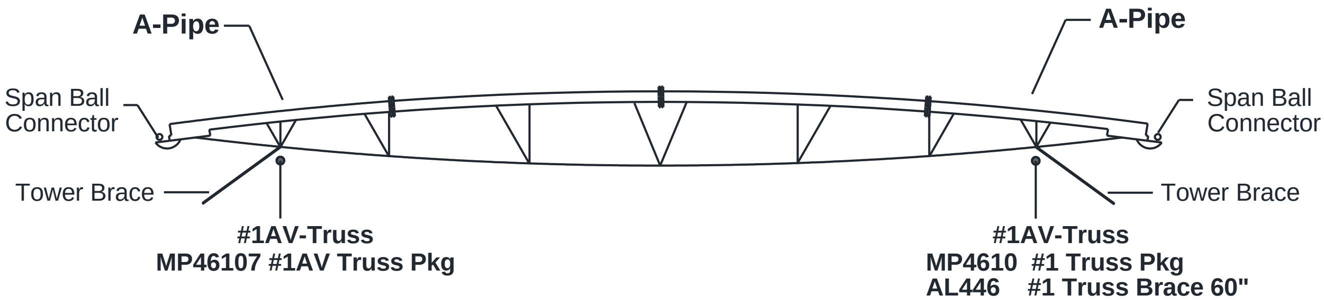


Figure 5b. Free Standing Span Construction.

TRACTOR STRUCTURE: (Figure 6)

1. Slide the Leg Gusset (1) over the 1" Oil Lines. Install rubber grommets (2) in the Leg Gusset holes.
2. Lift the end of the span approximately 7 feet high. Bolt on 4 tower legs (3) to the outer holes of the two and three hole bracket on the tractor A-pipe using four - 5/8 x 1 1/2 bolts w/nut and lock (4). Cross the legs until following step is complete.
3. Bolt top end of Diagonal Braces, ¼" thick material, (5) and Diagonal Cross Straps (12) to top end of Tower Legs (3) with four - 5/8 x 1 1/4 bolts with nut and lock washer (6). Bolt four Tower steps (7,8,9,10) to the right side Tower leg (3) using four - 3/8 x 1 bolts with nut (11).
4. Lift the span up high enough to uncross the legs. Bolt the bottom ends of the Diagonal Braces (5), Diagonal Straps (12), Lower Diagonal Braces and Straps (2,22), the Tower Braces (13) and the Middle Steps (14) to the Tower Legs (3) with two 5/8" x 2" Bolts with locks and nuts (15). Bolt the Middle Steps (14) to the other Tower Leg (3) with two 5/8" x 2" Bolts with locks and nuts (15). Bolt four Tower steps (7,8,9,10) to the other Tower leg (3) using four - 3/8 x 1 bolts with nut (11).
5. **NOTE: The Deck must be turned so that the tall side of the auxiliary brackets at the top of the ends of the deck are toward the free standing span. See note on Figure 2.** Lift span and tower legs high enough to bolt on deck (16). Bolt deck (16) to Tower legs (3), both on top and on the sides using eight - 5/8 x 1 1/2 bolts with nut and lock washer (4) on the sides and eight - 5/8" x 2" Bolts with locks and nuts (15).
6. (See Detail B) Attach Non-Pivoting Wheel Mounts (17) to deck (16) with two 1" x 12 1/2" Axle Shaft (18), two 1" flat washers (19) and two 3/16" x 2" cotter pins (20). *NOTE: Grease Axle shaft before assembly*. Make sure motor port holes on Wheel Mount face toward the center of the deck. *For Pivoting type mounts see "Pivoting Option" installation section*.
7. Bolt Planetary Drive Assemblies (40) to the Wheel mounts using 32 - 5/8" x 2" bolts with nuts and lock washers (6). Line up motor ports with holes in wheel mount gusset. Bolt the Wheels to the Gears with 36 - Taper Lug Nuts. (Tires should be mounted so that the lugs point to the center of the Deck with valve stems out.)
8. Bolt 6" Water shut off valve (21), and Lower drop pipe (23) to the water outlet located at the center of the deck (16) with eight 5/8" x 12" Bolts with nuts (24). Electric solenoid on 6" valve should be turned toward engine. Rotate Lower drop pipe (23) so off set is toward free standing span. Slide 6 5/8" x 16" Water Boot (25) over Lower Drop Pipe (23). Fit rubber gasket (26) into Upper Drop Pipe (27) and slide upper drop pipe down into lower water drop. Attach Upper Drop Pipe (27) to Tractor A-Pipe inlet (28) using a 6" band lock band (29). Center Water Boot (25) on the Lower Drop Pipe (23) and Upper Drop Pipe (27). Clamp in place with two 6 5/8" Tee Bolt Bands (30).
9. Bolt Hydraulic Pumping Unit (31) and Fuel Tank Rails (32) to the Deck (16) using eight-5/8" x 1 1/4" bolts with lock washers and nuts (6).
10. Mount Electric control panel and hydraulic control panel assembly to the end of fuel tank rails using four 3/8" x 1 ¼" bolts with lock washers and nuts (11).

TRACTOR ASSEMBLY

Ref	PART #	DESCRIPTION	Qnt	Ref	PART #	DESCRIPTION	Qnt
1	AG729	LEG GUSSET LINEAR 4H (Man cntrl)	1	20	FP5741	COTTER PIN 3/16" X 2"	4
	AG10833	LEG GUS 6-5/8" OL CLAMP (PLC ONLY) (11/24 -)	1	21	IA98902	6" NELSON VALVE 12VDC	4
				22	AP16532	LOWER DIAG CROSS STRAP	2
2	AL27855	LOWER DIAG TWR BRACE	2	23	AW27871	WATER DISCHARGE PIPE	1
3	AL1109	TOWER LEG	4	24	FB54970	HHCS 5/8" X 11"	8
4	FB5455	HHCS 5/8" X 1 1/2" W/L&N	32	25	IH6142	FLEX HOSE 6-5/8" X 16"	1
5	AL13924	UPPER TWR BRACE-1/4" THICK	2	26	IG5805	6" MIDWEST GASKET	1
6	FB5454	HHCS 5/8" X 1 1/4" W/L&N (non-eff-06/16)	18	27	AW14151	UPPER DROP PIPE	1
7	TZ1379	#4 TOWER STEP	2	28	AW1046	TRACTOR A-PIPE (QT & OSL)	1
8	TZ1378	#3 TOWER STEP	2	29	IG4937	6" BANDLOCK BAND	1
9	TZ1376	#2 TOWER STEP	2	30	FC4889	T BOLT CLAMP	4
10	AB16530	#1 TOWER STEP 4WT (Eff. 6/17/24)	2	31	HA40636	HYDRAULIC PUMPING UNIT	1
11	FB5439	HHCS 3/8" X 1" W/NUT	16	32	AL29730	FUEL TANK RAIL 4WT	2
12	AP16535	UPPER DIAG CROSS STRAP	2	34	ASSEMBLY	HYD & ELEC. PANEL ASSY.	1
13	AT1112	TOWER BRACE	2	35	AW14154	SPACER 4W LINEAR W/O VLV	1
14	TZ1377	MIDDLE TOWER STEP	2	36	FW5622	5/8" LOCKWASHER	30
15	FB5456	HHCS 5/8" X 2" W/L&N	44	37	FW5663	5/8" FLATWASHER	24
16	AW27860	TRACTOR SQUARE BASE	1	38	FN5406	5/8" NUT	30
17	AW27880	NON PIVOTING AXLE	2	39	FB5442	HHCS 3/8" X 1-1/4" W/ LW	8
18	AR27838	AXLE SHAFT	2	40	FB5725	U-BOLT 5/16" X 1-3/8" X 2-3/16" (PLC ONLY)	2
19	FW5673	FLAT WASHER PLATED 1"	4	41	FN5400	NUT HEX 5/16-18 (PLC ONLY)	4

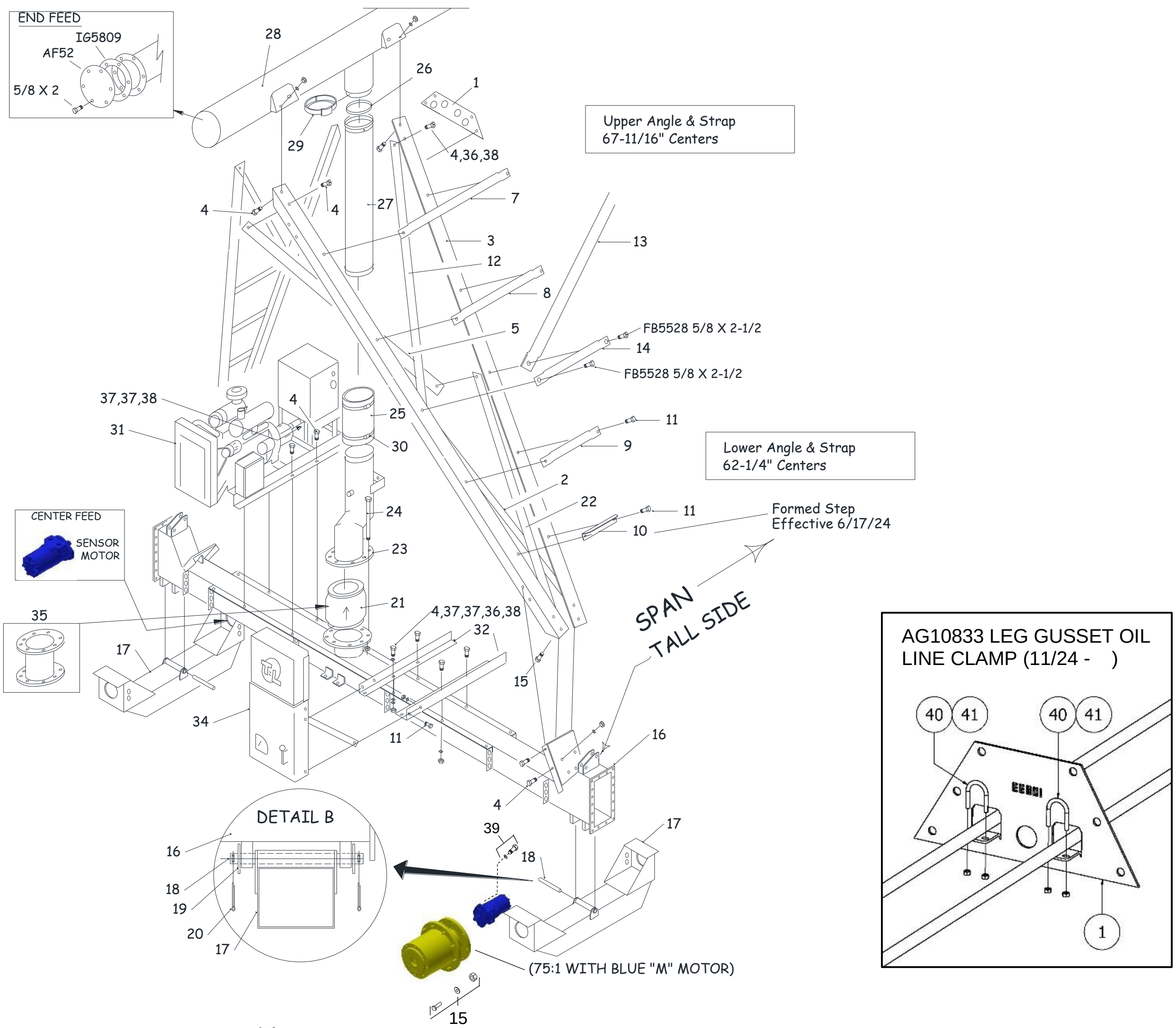


Figure 6. Tractor Structure Assembly

WATER INLET: (Figure 7)

Attach a Water Inlet Plate (40) and Rectangular Gasket (41) to each end of the Tractor Deck (16) using eighteen- 1/2" x 1 1/4" bolts with nuts (42,43).

1. Attach two Pull Chains (44) to a Single Inlet Plate (45) or four to a Double Inlet Plate (40) using one each-1/2" x 2 1/4" grade 5 bolts with lock and nut (46,47,48).

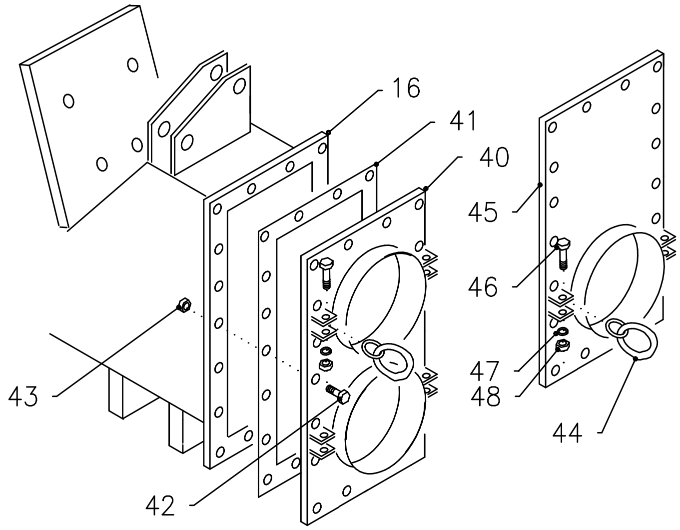
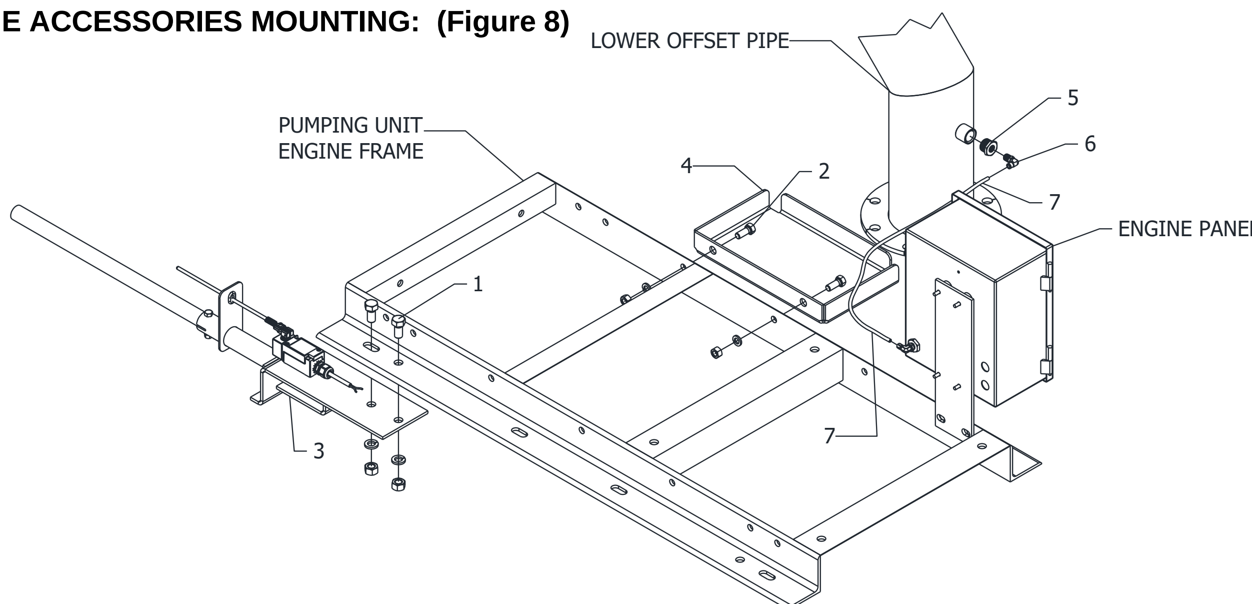


Figure 7. Water Inlet Installation

ENGINE ACCESSORIES MOUNTING: (Figure 8)



REF.	PART NO.	DESCRIPTION	QTY
1	FB5454	HHCS 5/8” X 1-1/4” GR5 W/ LOCK WASHER & NUT	2
2	FB5446	HHCS 1/2” X 1-1/4” GR5 W/ LOCK WASHER & NUT	2
3	EA52690	POSITION STOP SWITCH ASSY LINEAR TRACTOR	1
4	AW2561	BATTERY BOX FRAME	1
5	PB80412	BUSHING 3/4” X 1/4” GALV	1
6	TM96372	1/4" MALE ELBOW, PUSH IN (8/22 -)	1
7	TW6119	NYLON TUBE .25 OD .04 WALL	AR

Figure 8. Engine Accessory Installation

FUEL TANK (Figure 9)

1. Position the fuel tank (100) onto the Fuel Tank Rails (32). NOTE: Face the tank with the lower outlet on the engine side of the deck . Secure the Fuel Tank to the rails using two Fuel Tank Straps (116) Bolt the straps to the rails using two 3/8” x 1” bolts w/nuts (9). At the top bolt the two straps together using one 3/8” x 3” bolt w/nut (117).
2. Assemble the following plumbing onto bottom outlet of tank (100). Two 1” close nipple (101), 1” elbow (102), 1” Gate Valve (103), 1" x 1/8" Bushing (105) and 1/8" x 5/16" barbed adapter (106).
3. Assemble fuel filter assembly plumbing including: 1” x 6” nipple (109), 1” elbow (102), 1" x 1/8" Bushing (105) and 1/8" x 5/16" barbed adapter (106) to the fuel filter (104) inlet. Attach 1" x 1/8" Bushing (105) and 1/8" x 5/16" barbed adapter (106) to fuel filter (104) outlet. Bolt plumbing assembly to the Fuel Tank Rail (32) using one 5/16” x 2 ½” U-bolt w/nuts (118).
4. Connect plumbing assemblies using a short section of fuel line (108). Connect to adapters (106) using Hose clamp (107). Route 5/16" fuel line (110) to engine Clamp fuel line to adapter (106) using Hose clamp (107).
5. Install a 2" x close nipple (111) between the fuel tank (53) and tank cap (112). Attach a 1/8" nipple (113), 1/8" elbow (114) and 1/8" x 5/16" barbed adapter (106) to small coupler on tank top. Clamp fuel line to adapter (106) using Hose clamp (107). Attach the opposite end of 5/16”fuel line to the fuel return on the engine.
6. Install fuel gauge (115).

NOTE: ***Use thread sealant HS61024 on all threaded fuel connections*.**
(Sept. 2016 on)

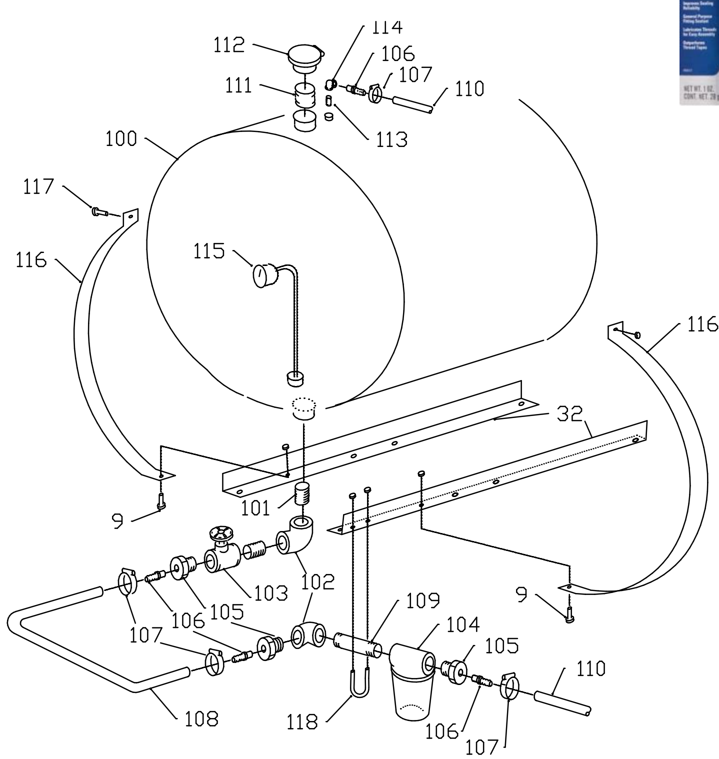


Figure 9. Fuel Tank Installation

AUTO DRAIN DIVERSION (Figure 10)

1. Locate the 1 1/2" Coupler that is welded to the bottom of the Tractor A-pipe (1). Thread the Low Pressure Drain (2) into this coupler. Thread one 90 Deg. Hose Barb x Female Threaded Elbow (3) onto the Drain Valve.
2. Complete the Assembly by attaching the 1" Flex Hose (4) to the Elbow using one 1" Hose Clamp (5).
3. Route the Flex Hose over the Tower Leg and up along the Truss Rod. Secure the Hose to the Truss Rod by using six Plastic Wire Ties (6).

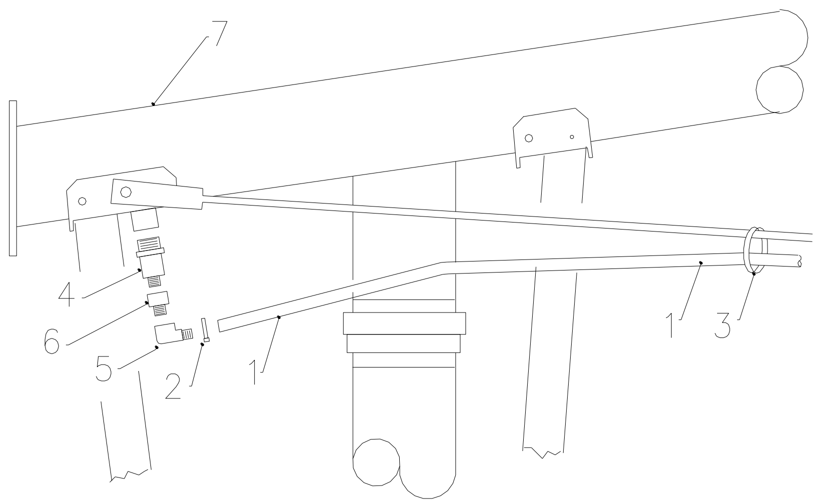


Figure 10. Auto Drain Diversion Installation

MP47903 Auto Drain Diversion Kit

Ref	P/N	Item	Qty	
1	IH61485	1" Flex 80 Utility Tb /Ft"	20.0	Ft
	MB47904	Drain Diversion Fittings		
2	FC4883	Hose Clamp B12	1.0	Ea
3	FT5702	Wire Tie 0.27" X 13" Bk	6.0	Ea
4	IV6732	1-1/2" Low Pressure Drain	1.0	Ea
5	IY76032	1” Hose Barb X ¾” FPT Elbow	1.0	Ea
6	PA82075	Adptr Garden X 3/4 Mpt	1.0	Ea
7	AW1046-XX	6-5/8” Inlet A-Pipe, XX outlet spacing	1.0	Ea

CENTER FEED HOSE DRAG TRACTOR PLUMBING (Figure 11)

1. Insert Grommets into all holes in Deck Gussets where a line or hose will be routed.
2. Insert 144" Tractor Deck Line (10) through bottom hole in gussets on the Tractor Base opposite the Hydraulic Control Panel. Insert a Towable lower tower line (20) lines through the top gusset hole on each end of the Base on the side of the Control Panel. Attach a 1/2" Tubing Union (13) and ½” Oil Line Elbow (1) to each of the 4 ends of the 1/2" Base Lines.
3. Connect ½” Oil Line Elbow (1) to Hydraulic Motor Ports using four 3/8" x 22" Hoses- ½ ERM x ½ MPX (14).
4. Connect remaining Motor Ports under the Base using two 1/2" x 66" Hoses- ½ MPX x ½ MPX (15). NOTE top to bottom plumbing. Tie hose up to Base Gusset using wire ties.
5. Connect 1” Pressure and Return span lines together at top of Tractor. Connect two assemblies each consisting of one Hose ¾” x 23” - ¾ MP x 1 ERM (21), ¾” tee (5) and ¾” x 1”adapter (24), Hose ¾” x 23” - ¾ MP x 1 ERM (21), 1” x 6” oil line (22), Hose ¾” x 26” - 1 ERM x 1 ERM tee (29). Cross hoses (21) on the span side of the tractor after they go through the tower gusset.
6. Cut span lines to required length to match up to hose (21) Reuse nut from scrap tubing and place new sleeve (3) on shortened span line. Hold adapter tight to end of 1" tube while tightening to insure correct setting of sleeve.
7. Connect plumbing above to 1” Tractor lines (23) at the ¾” x 1”adapter (24). Also connect to alignment valve (not shown) at top of tower the same as other intermediate towers.
8. Connect 1” Right End Tower lines to ½” Tractor center lines (19) using 90 degree adapter (18). Tie ½” Tractor lines (19) and 1” Tractor lines (23) to Discharge Goose neck (28) with oil line holder (27) using one ½” x 2” bolt (not shown).
9. Connect two ½” x 92” hoses 8FP x 8MPX (16) to ½” Tractor lines (19) using ½” tubing adapters (2). Connect opposite end of hoses (16) to Ports #1 and #2 on Hydraulic Control Panel.
10. Connect 1” Left End Tower lines to ½” Tractor leg lines (19) using two Hose ¾” x 23” - ¾ MP x 1 ERM (21), ¾” x ½” elbow (30) and ½”adapter (2). Tie ½” tractor line (19) to leg with wire ties.
11. Connect two ½” x 72” hoses 8FP x 8MPX (17) to ½” Tractor tower lines (19) using ½” tubing adapters (2). Connect opposite end of hoses to Ports #4 and #5 on Hydraulic Control Panel.
12. From the Hydraulic Pumping Assembly Return filter connect ¾” Street Elbow (34), 3/4” nipple (4), 3/4” tee (5), 3/4" x 1/2” bushing (6), and 1/2” Street Elbow (7). Add a 3/4” nipple (4), and 3/4” Female Quick Coupler End (8) to the 3/4" tee (5).
13. Bolt Tank plumbing mounting plate (11) to side of tank and tank leg using existing ½” bolt. On the Hydraulic Pumping Assembly Pressure side connect one 3/4” x 16” Hose MP x MP (26), ¾” x ¾” x ½” tee (31), a 3/4” nipple (4), and 3/4” Male Quick Coupler End (8). Add a ½” Street Elbow (7) to the ¾” x ¾” x ½” tee (31). Secure Pressure plumbing with 5/16 x 1 ½” U-bolt (12).
14. Connect the Pump Pressure and Return to Ports #3 and #6 on Hydraulic Control Panel. Route hoses under Base, through Gussets. Connect the Pressure using a 1/2" x 116" Hose- ½ MPX x ½ MPX (9) to #6. Connect the Return using a 1/2" x 132" Hose- ½ MPX x ½ MPX (33) to #3.
15. Connect the Pumping Unit to Span Pressure and Return through 1” Tractor lines (23). Connect Pressure with Female Quick Coupler (8) and one ¾" x 36" Hose- 3/4 MPT x 1 ERM (25). Connect Return with Male Quick Coupler (8) and one ¾" x 46" Hose- 3/4 MPT x 1 ERM (25). NOTE: With back to the tractor, left = Pressure, right = Return.
16. Connect ½” Oil Lines (20) to upper tower Valve Lines (not shown) using ½” tubing unions. Tie tower lines into tower leg using line clips as on intermediate towers, per Pivot Installation Manual.
17. NOTE: Flush each pair of 1" span Lines separately. Right wing end tower, Left wing end tower, Right wing Pressure-Return, Left wing Pressure-return.

Center Feed Tractor Plumbing, Figure 11

Ref	PART #	DESCRIPTION	Qty	Ref	PART #	DESCRIPTION	Qty
1	TX1047	ALIGN VLV RET LINE ½" OIL LINE EL	6	18	HA8335	1" TB x 1/2" TB RED EL A	2
2	TC9614	B7205 X 8 X 8 MALE CNECTR	4	19	TX1409	END TOWER UPPER TWR LINE	4
3	TS9609	7165 X 16 SLEEVE	4	20	TX1099	TOWABLE LOWER TOWER LINE	2
4	PN8206	NIPPLE ¾" X CLOSE BEH	3	21	HH6130	HOSE 12C2AT-12MP-16ERM-23"	6
5	PT8153	TEE ¾ BAAR	1	22	TY542	1" X 6" OIL LINE	2
6	PB7998	BUSH ¾x1/2 STEEL	1	23	TY14582	CF TRACTOR LINE	2
7	PE81255	ST ELBOW ½ X 90 DEG PLATED	2	24	MD9615	B7205 x 16 X 12 MALE CONTR	2
8	HV8040	QUICK COUPLER SET ¾"	2	25	HH61370	HOSE 12C2AT-12MPX-16ERM-36"	1
9	HH61261	HOSE 8CIT-8MPX-8MPX-116"	1	26	HH612951	HOSE 12C2AT-12MP-12MP-16"	1
10	TX10877	TRACTOR DECK LINE 144"	1	27	AP85	OIL LINE HOLDER	1
11	AB14352	TANK PLUMBING MOUNT PLATE	1	28	AW27871	DISCHARGE GOOSE NECK	1
12	FB57255	U BOLT 5/16 X 1 ½ X 2 ½	1	29	HH6182	HOSE ¾-16ERM-16ERM-TEE-26"	2
13	TU9622	B7305 X 8 UNION	6	30	PE8092	ELBOW ¾ X ½ 90 DEG BAAR	2
14	HH61730	HOSE 6C1T-8ERM-8MPX-22"	4	31	PT8152	TEE ¾ x ¾ x ½ BEH	1
15	HH61260	HOSE 8CIT-8MPX -8MPX-66"	2	32	HH61822	HOSE 12C2AT-12MPX-16ERM-46"	1
16	HH61263	HOSE 8CIT-8FPT-8MPX-92"	2	33	HH61235	HOSE 8CIT-8MPX-8MPX-132"	1
17	HH61325	HOSE 8CIT-8FPT-8MPX-72"	2	34	PE8113	Street Elbow ¾ x 90 BEH	1
				35	TX1072	Upper Tower Valve ½" Line	2

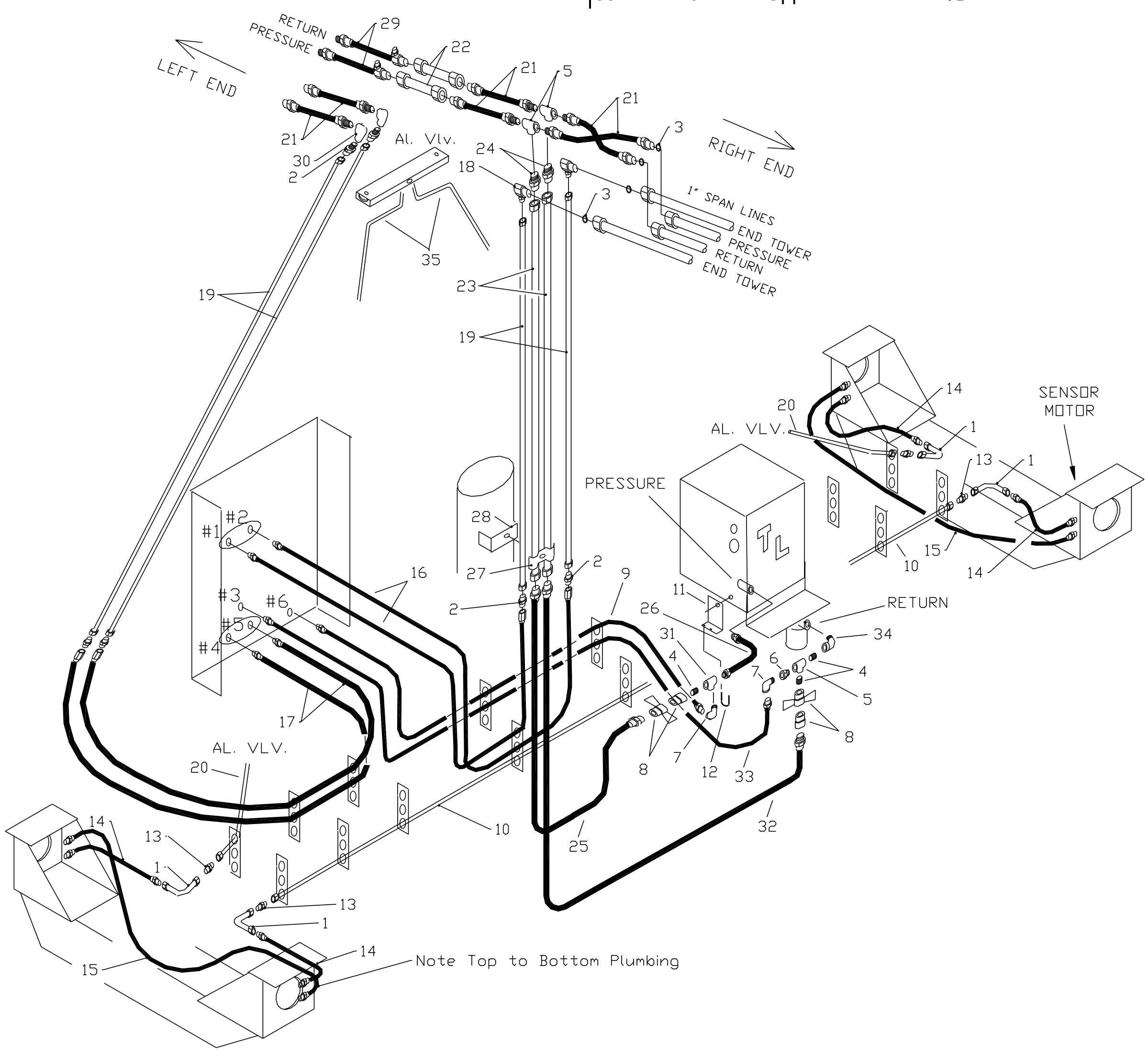


Figure 11. Center Feed Tractor Plumbing

END FEED HOSE DRAG TRACTOR PLUMBING (Figure 12)

1. Insert Grommets into all holes in Deck Gussets.
2. Bolt Tank plumbing mounting plate (11) to side of tank and tank leg using existing ½" bolt. On the Hydraulic Pumping Assembly Pressure side connect one 3/4" x 16" Hose MPX x MP (26), ¾" x ¾" x ½" tee (31), a 3/4" nipple (4), and 3/4" Male Quick Coupler End (8). Add a ½" Street Elbow (7) to the ¾" x ¾" x ½" tee (31). Secure Pressure plumbing with 5/16 x 1 ½" U-bolt (12).
3. From the Hydraulic Pumping Assembly Return filter connect ¾" Street Elbow (34), 3/4" nipple (4), 3/4" tee (5), 3/4" x 1/2" bushing (6), and 1/2" Street Elbow (7). Add a 3/4" nipple (4), and 3/4" Female Quick Coupler End (8) to the 3/4" tee (5).
4. Insert 144" Tractor Deck Line (10) through bottom hole in gussets on the Tractor Base opposite the Hydraulic Control Panel. Insert the 38" (11) and 92" (12) ½ Oil Lines through the bottom hole in gussets on the Base on the side of the Control Panel. Attach a 1/2" Tubing Union (13) and Alignment Valve return line (1) to each end of the 1/2" Base Lines.
5. Connect Base Lines (10,11,12) to Hydraulic Motor Ports using four 3/8" x 22" Hoses- ½ ERM x ½ MPX (14).
6. Connect remaining Motor Ports using two 1/2" x 66" Hoses- ½ MPX x ½ MPX (15). Route hoses under the Base. NOTE top to bottom plumbing. Tie hose up to Base Gusset using wire ties.
7. Connect two 1" formed Tractor Tower Lines (23) to the 1" span Pressure and Return Lines using two 1" Unions (24). Cut span lines to length. Reuse nut from scrap tubing and place new sleeve (3) on shortened span line. Hold Union tight to end of 1" tube while tightening to insure correct setting of sleeve.
8. Connect the Pumping Unit to Span Pressure and Return through 1" Tractor lines (23). Connect Pressure with Female Quick Coupler (8) and one ¾" x 36" Hose- 3/4 MPT x 1 ERM (25). Connect Return with Male Quick Coupler (8) and one ¾" x 46" Hose- 3/4 MPT x 1 ERM (25). NOTE: With back to the FREE STANDING SPAN, left = Pressure, right = Return.
9. Connect two ½" x 92" tower lines (19) to 1" End Tower span lines using 1" Erm x ½" Erm 90 degree adapters (18). Cut 1" lines to length as described above if needed. Route two ½" x 92" hoses ½ FPT x ½ MPX (16) around under the Base. Using ½" tubing adapters (2) connect the tower line (19) closest to the Panel to Port # 1 and the tower line (19) furthest form the Panel to Port # 2.
10. Connect the Pump Pressure and Return to Ports #3 and #6 on Hydraulic Control Panel. Route hoses under Base, through Gussets. Connect the Pressure using a 1/2" x 116" Hose- ½ MPX x ½ MPX (9) to #6. Connect the Return using a 1/2" x 132" Hose- ½ MPX x ½ MPX (33) to #3.
11. Connect from Hydraulic control panel Port #5 to 92" Deck Line (12) and Port #4 to 38" Deck Line (11) using two 1/2" x 28" Hoses- ½ FPT x ½ MPT (17) two 1/2" adapters (2) are required to adapt to the tubing.
12. Secure 1"Tractor Tower Lines (23) and ½" tower lines (19) to the lower drop pipe mount (28) with Oil Line Holder (27) and a 1/2" x 2" bolt with lock and nut (not shown).
13. NOTE: Flush each pair of 1" Span Lines separately: 2 End Tower Lines, System Pressure & Return.

NOTE: If Pivoting Option is used, see Figure 25a for Hydraulic Tank Plumbing.

End Feed Tractor Plumbing, Figure 12

Ref	PART #	DESCRIPTION	Qty	Ref	PART #	DESCRIPTION	Qty
1	TX1047	ALIGN VALVE RETURN LINE	6	16	HH61263	HOSE 8CIT-8FPT-8MPX-92"	2
2	TC9614	B7205 X 8 X 8 MALE CNECTR	4	17	HH61256	HOSE 8CIT-8MP-8FP-28"	2
3	TS9609	7165 X 16 SLEEVE	4	18	HA8335	1" TB x 1/2" TB RED EL A	2
4	PN8206	NIPPLE 3/4" X CLOSE BEH	3	19	TX1409	END TWR UPPER TOWER LINE	2
5	PT8153	TEE 3/4 BAAR	1	20	TX10879	TRACTOR DECK LINE 38"	1
6	PB7998	BUSH 3/4x1/2 STEEL	1	21	TX10878	TRACTOR DECK LINE 92"	1
7	PE81255	ST.ELBOW 1/2 X 90 DEG PLATED	2	23	TY14581	TRACTOR PRES/RET LINE	2
8	HV8040	QUICK COUPLER ¾"	2	24	TU9623	B7305 x 16 UNION	2
9	HH61261	HOSE 8CIT-8MPX-8MPX-116"	1	25	HH61370	HOSE 12C2AT-12MPX-16ERM-36"	1
10	TX10877	TRACTOR DECK LINE 144"	1	26	HH612951	HOSE 12C2AT-12MP-12MP-16"	1
11	AB14352	TANK PLUMBING MOUNT PLATE	1	27	AP85	OIL LINE HOLDER	1
12	FB57255	U BOLT 5/16 X 1 1/2 X 2 1/2	1	31	PT8152	TEE ¾ x ¾ x ½ BEH	1
13	TU9622	B7305 X 8 UNION	6	32	HH61822	HOSE 12C2AT-12MPX-16ERM-46"	1
14	HH61730	HOSE 6C1T-8ERM-8MPX-22"	4	33	HH61235	HOSE 8CIT-8MPX-8MPX-132"	1
15	HH61260	HOSE 8CIT-8MPX -8MPX-66"	2	34	PE8113	Street Elbow ¾ x 90 BEH	1

NOTE: If Pivoting Option is used, see Figure 25a for Hydraulic Tank Plumbing.

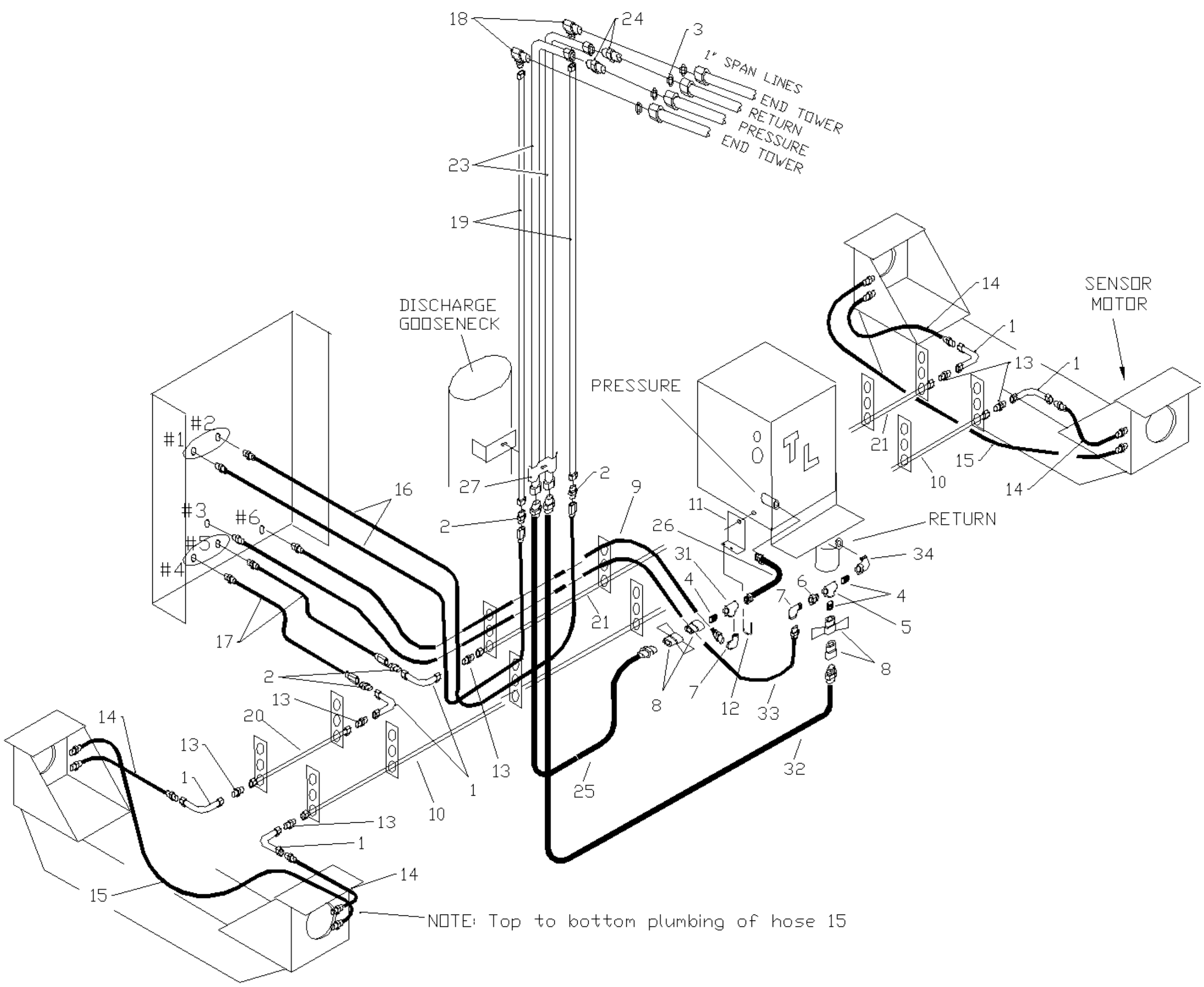
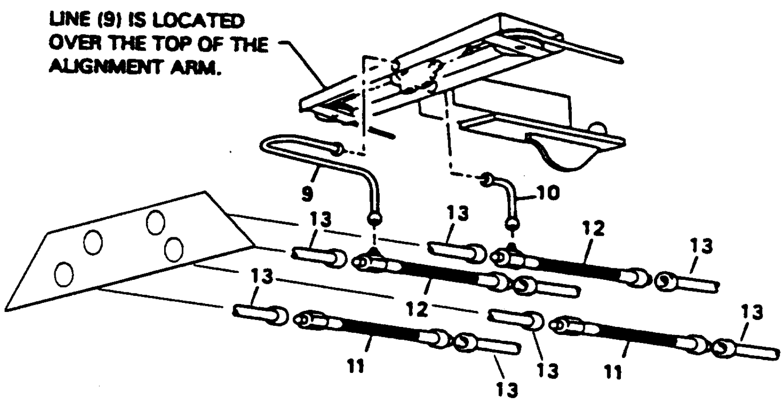


Figure 12. End Feed Tractor Plumbing

SPAN OIL LINE INSTALLATION (Figure 13)

1. All spans will have four 1' oil lines strung the length of the span except the last span on each end. This span will have two 1" oil lines. The spans and towers will require four hole gussets to accommodate the four oil lines.
2. **System Pressure and Return – End Feed**: The two center lines are the system pressure and return line. These lines will run the length of the system and terminate at the End Tower By-pass on the next to last tower. These two lines will **cross in the middle** of the Free Standing span. The pressure line will be on the left; determined while standing with your back to the center of the Free Standing Span. The return line will be on the right.
3. **System Pressure and Return – Center Feed**: The two center lines are the system pressure and return line. These lines will run the length of the system and terminate at the End Tower By-pass on the next to last tower on each end. The pressure line will be on the left; determined while standing with your back to the tractor unit. The return line will be on the right.
4. Lead Tower Motor Lines - The two outer lines supply oil to the lead tower(s) only. Do Not Cross these.
5. Join all oil lines on span with tubing unions. **Use Caution Do Not Cross Any Lines, except as noted above.** At each tower connect the two center lines (13); using a 26" hose with a 1/2" tee adapter (12) connect the pressure side to the alignment valve using line (9) and return side to the alignment valve using line (10), as on a standard pivot. Connect the two outside lines (13); lead tower lines, with a 26" straight hose (11).
6. Use linear oil line hangers (14) in pairs offset opposite each other at the same locations along the span where standard oil line are normally placed.



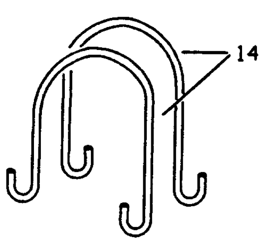
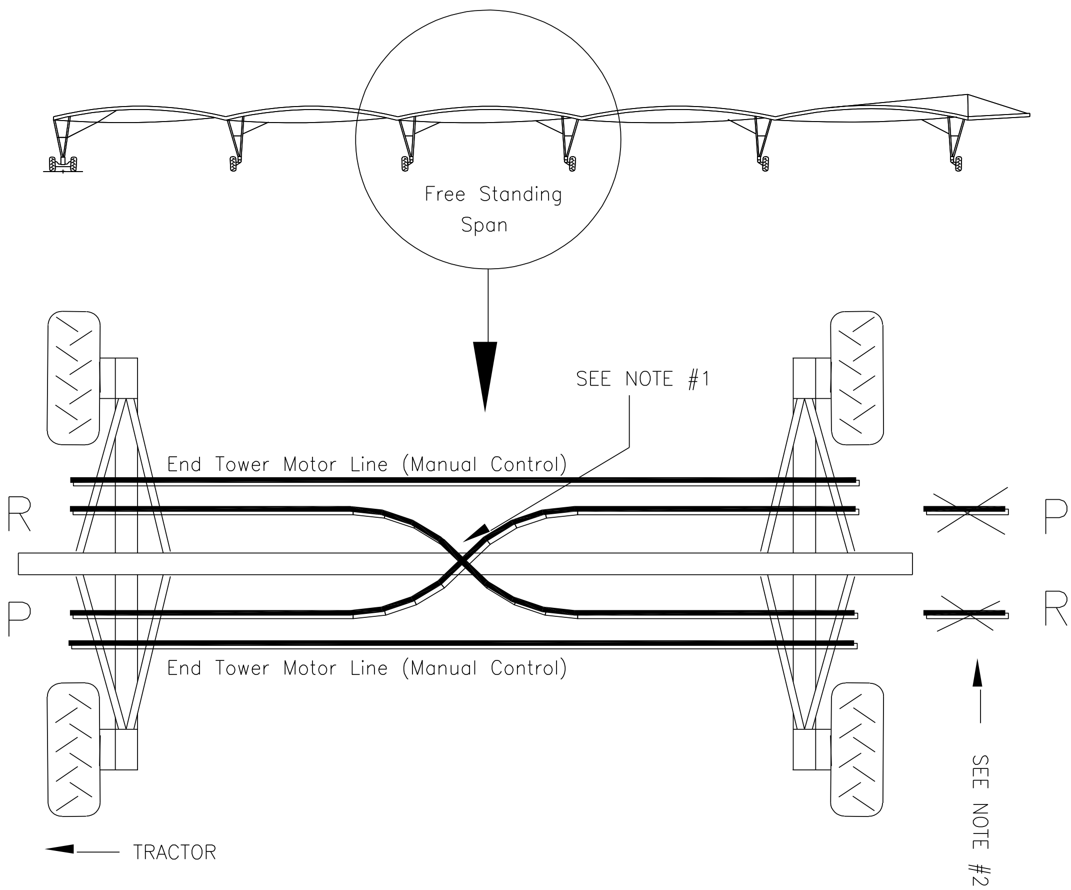


Figure 13. Span Oil Line Installation

Free Standing Span Oil Line Installation

– will need to cut back one end of Pressure/Return lines



NOTES:

1. Pressure and Return Oil lines must cross in the middle of the Free Standing Span.
2. The Pressure and Return Oil lines will have to be cut shorter at the end of the Free Standing Span furthest from the tractor. If the span is equipped with 1–1/4” Oil lines, the last section of Oil Line will have to be exchanged with 1” Oil Line from End Span.

Figure 13.1. Free Standing Span Oil Line Installation

END (LEAD) TOWER BYPASS PLUMBING (Figure 14)

1. Attach the End Tower Bypass Valve Plate (ETBP valve plate has two valves attached) in place of a Standard Valve Plate at the next to last tower with two 5/8" x 1 1/4" bolts with nuts and locks.
2. Remove the hex head screws from the outside ends of the Alignment Valve Spools, save the 5/16" Spool Washers (1), and place them on the 5/16" threaded end of the ETBP Alignment Cable Assemblies (2), attach the cables to Alignment Valves. Check spool movement of valves to insure that valves move smoothly and reach the end of spool travel at the same point. Adjust linkage between valves if adjustment is necessary.
3. Attach 1/2" Return Oil Line (7) and 1/2" Pressure Oil Line (3) to alignment Valve (5) and connect to 1" Oil Lines with two 1" x 1/2" Reducing Elbow Assemblies (9).
4. Attach 1/2" oil lines (6,4) to the ETBP Valve (10; valve with plugs in Motor ports) and to 1" Lead Tower Control lines using two Span Hoses (8).
5. Connect Alignment Cables (2) to the Alignment arm and tighten so springs are fully compressed and valves are centered.
6. Connect upper Tower Lines (14) to Alignment Valve (5).

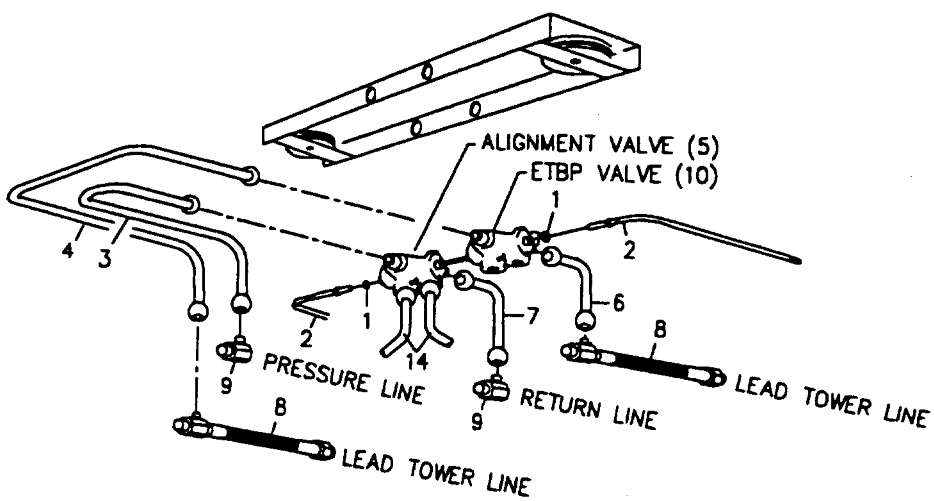


Figure 14. End Tower Bypass Plumbing

END TOWER PLUMBING (Figure 15)

1. Attach one Reducing Elbow Assembly and one Reducing Tee Assembly to 1" Oil Line. Attach 1/2" planetary lines to these assemblies.
2. Install Motor Adapter Fittings (1/2" MPT X ½" Erm) in Motor Ports if not already installed.
3. Install base line to top motor ports.

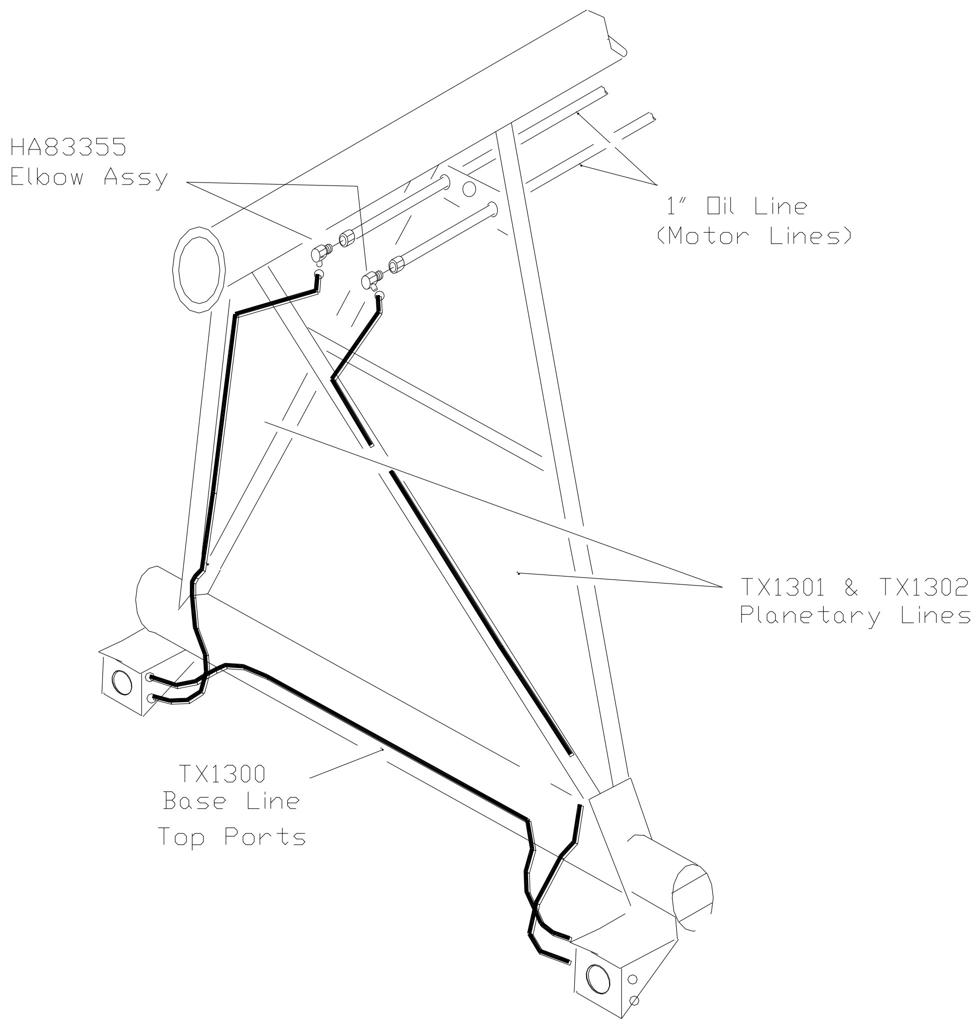
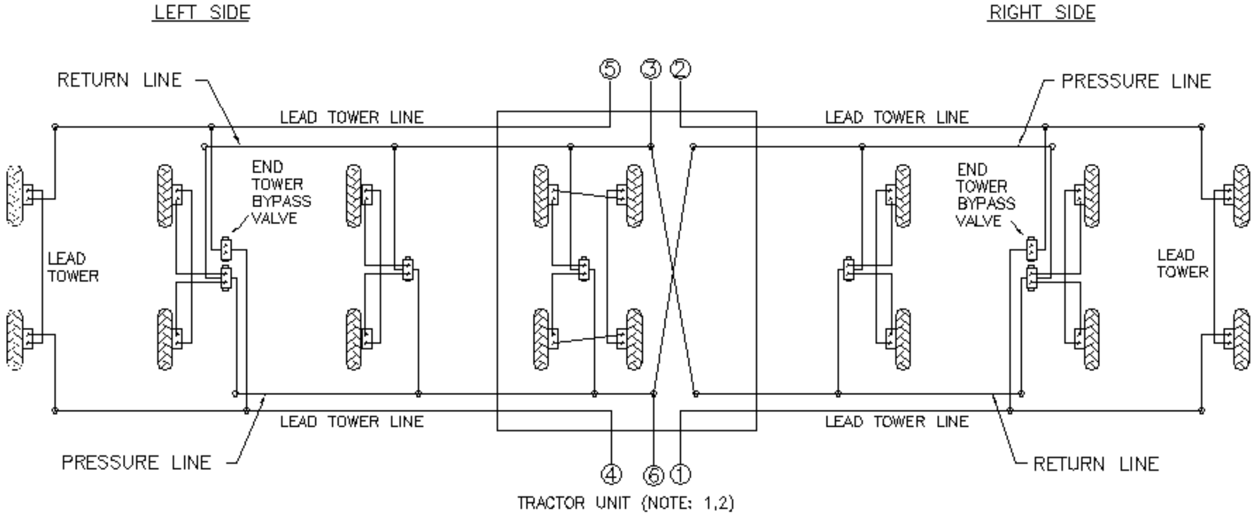
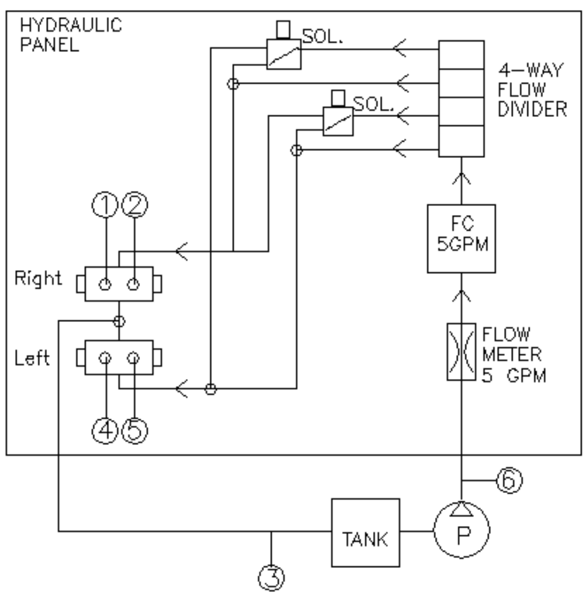


Figure 15. End Tower Plumbing

HYDRAULIC CIRCUIT, CENTER FEED TRACTOR (Figure 16)



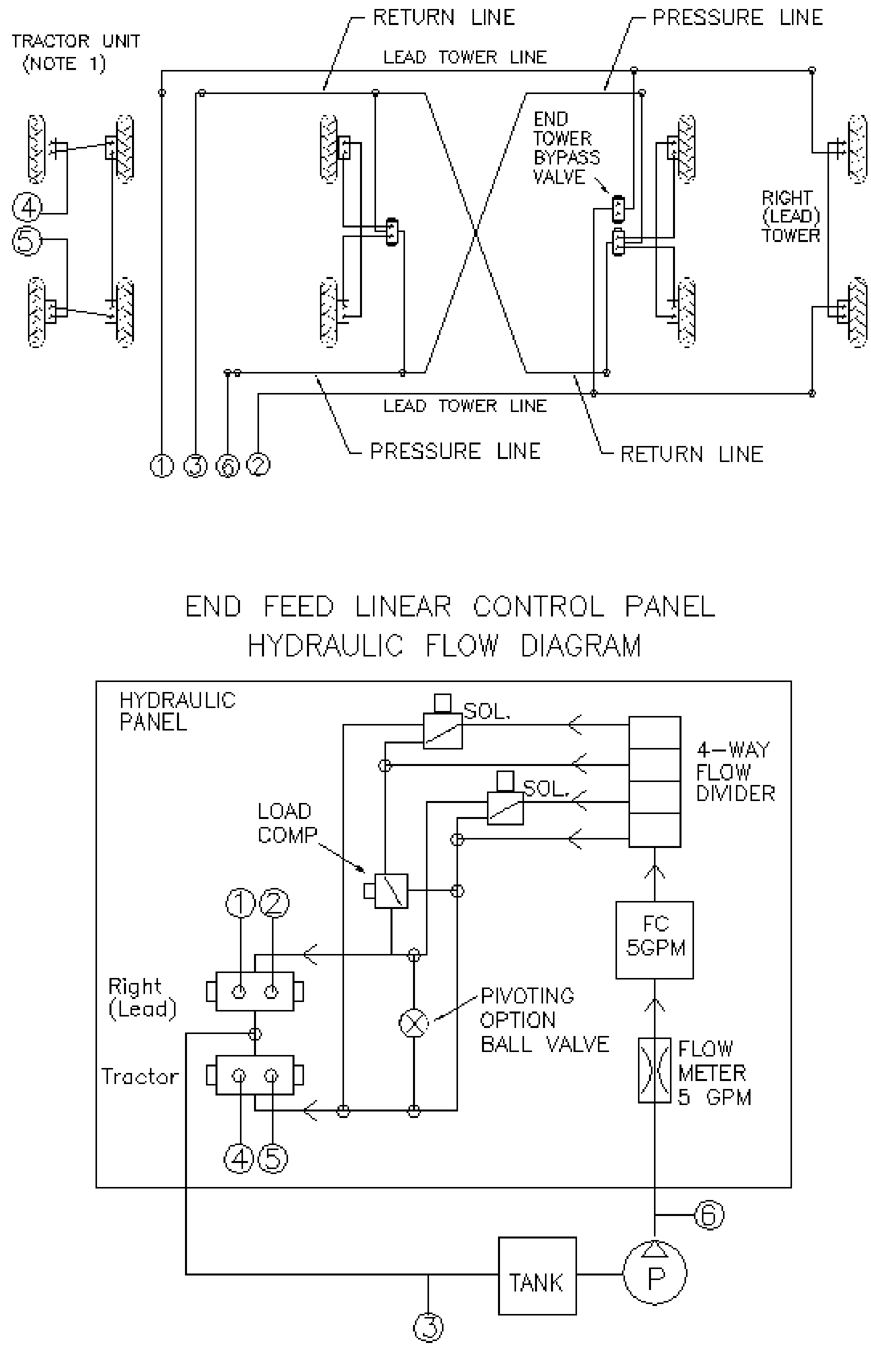
CENTER FEED LINEAR CONTROL PANEL
HYDRAULIC FLOW DIAGRAM



NOTES:
1. PANEL, PUMP, TANK AND ENGINE ASSEMBLY ARE LOCATED ON TRACTOR UNIT
2. SYSTEM PRESSURE AND RETURN LINES CROSS ABOVE TRACTOR
3. BOTTOM MOTOR PORTS ARE SHOWN ON OUTBOARD SIDE – TYPICAL FOR ALL TOWERS
4. ① ② ④ ⑤ ARE CONNECTED TO RESPECTIVE TERMINALS ON TRACTOR UNIT.
5. ③ ⑥ ARE CONNECTED TO PUMP PRESSURE AND TANK RETURN .

Figure 16. Center Feed Hydraulic Flow Diagram

HYDRAULIC CIRCUIT, END FEED TRACTOR (Figure 17)



NOTES:

1. PANEL, PUMP, TANK AND ENGINE ASSEMBLY ARE LOCATED ON TRACTOR UNIT
2. PRESSURE AND RETURN LINES CROSS IN MIDDLE OF SYSTEM
3. BOTTOM MOTOR PORTS ARE SHOWN TOWARD THE OUTSIDE ON DIAGRAM
4. 1 2 4 5 ARE CONNECTED TO RESPECTIVE PORTS ON TRACTOR UNIT.
5. 3 6 ARE CONNECTED TO PUMP PRESSURE AND TANK RETURN.

Figure 17. End Feed Hydraulic Flow Diagram

LEAD TOWER MOVED IN ONE

On longer systems, it may be advantageous to move the lead tower(s) in one for better guidance. Consult
T-L about this option. See Figures 17a & 17b for installation.

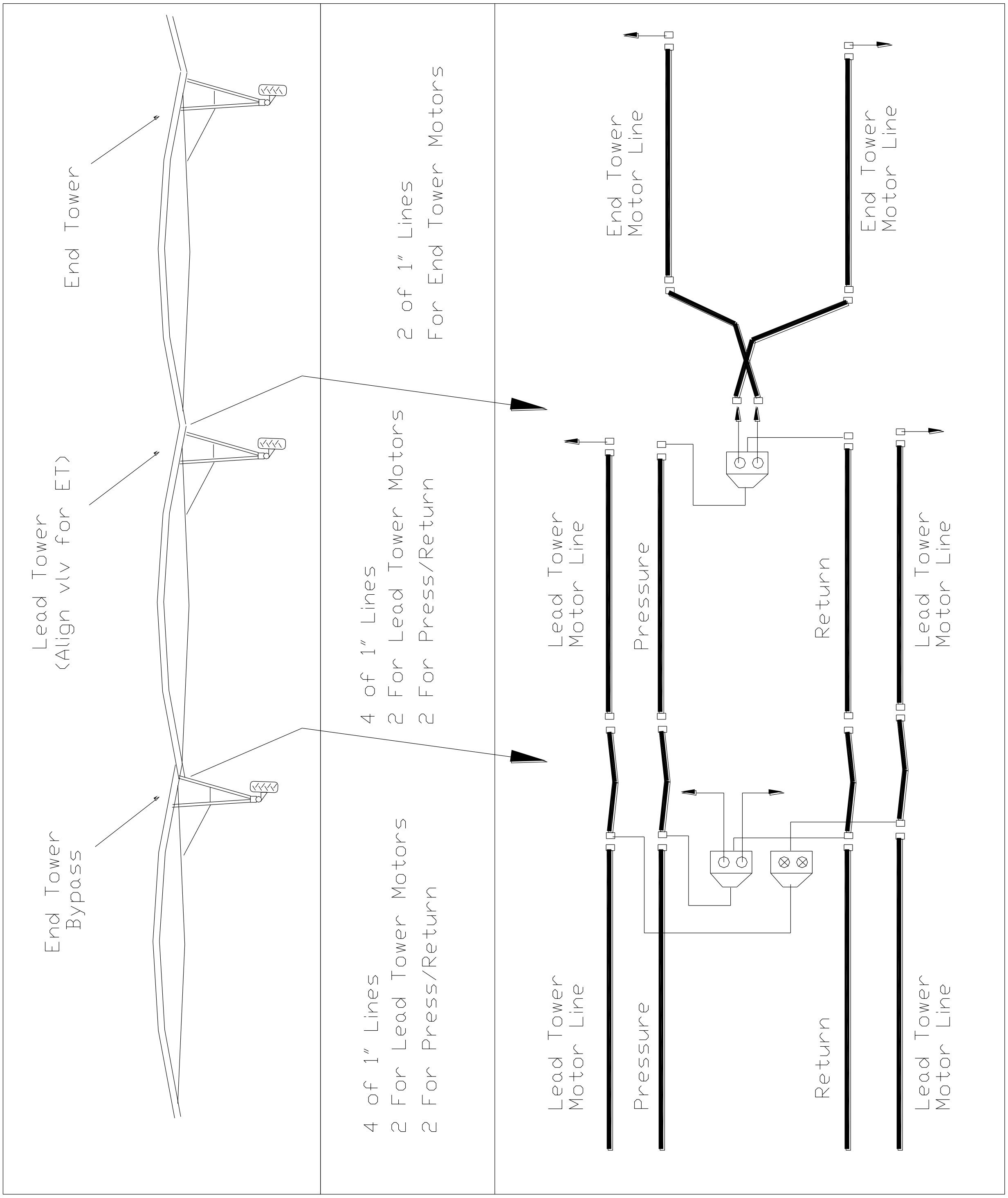


Figure 17a. Lead Tower Moved in One - Layout

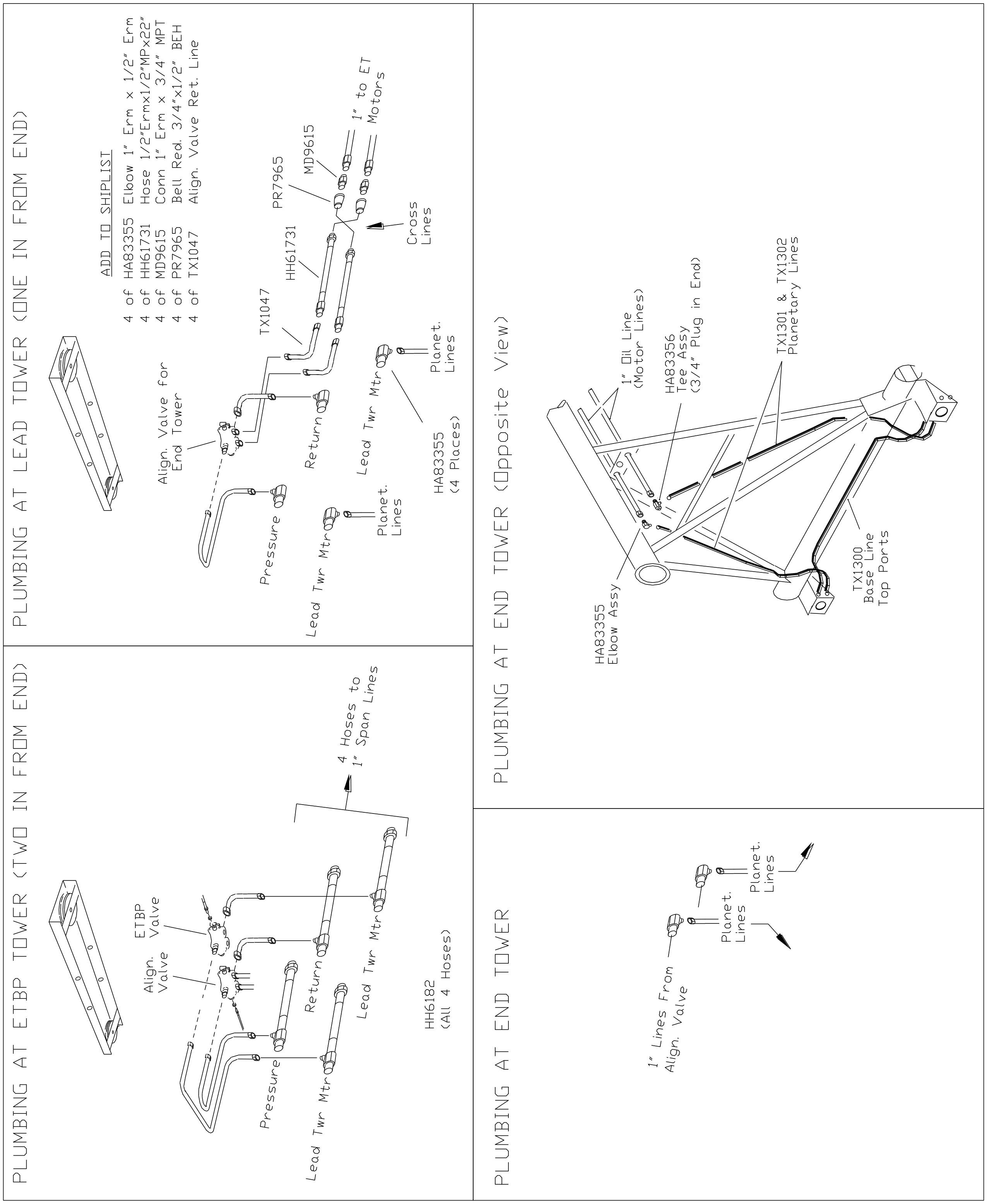


Figure 17b. Lead Tower Moved in One - Plumbing

ABOVE GROUND GUIDANCE CONTROLLERS - Non-Pivoting (Figure 18)

Refer to the Pivoting option section for pivoting guidance mounting

1. The Guidance will normally be located on the field side of the tractor if possible, although it can be mounted on either side (see Figure 18).
2. Bolt two Control Tube Extension Angles (60) to Tractor Deck (16) on the mounting brackets nearest the end of the deck using four 5/8" x 11/2" bolts with locks and nuts (6). Bolt Support Straps (61) to the side of the Control Tube Extension Angles (60) using two 5/8" x 11/2" bolts with locks and nuts (6). Bolt the upper end of the Support Straps (61) to the middle tower step (14) using two 5/8" x 2" bolts with locks and nuts (15).
3. Attach Controller Tube (62) to Extension Angles (60) using two 3/8" U-Bolts (63).
4. Attach Secondary Safety Assembly (64) to center of Controller Tube using 3/8" U-Bolt (63).
5. Attach a Wand (65) to each Controller using two ¼" x ¾" bolts with double nuts (66).
6. Attach the Proximity Controllers (67,68) to each end of the Controller Tube using two 3/8" x 1" bolts with nuts and locks (9) as shown. Note: The Controllers will be marked for shipment as "Left" and "Right." Stand with your back to the Tractor to determine the correct side to mount them on. The Controller Wand should rotate so the torsion spring compresses and the wand is spring loaded away from the Tractor and toward the cable (see Figure 18).
7. Route the electrical cable from the Controller, through the controller tube, out the port hole and to the control panel. There will be one 16/8 (16 gauge, 7 wire) cable that extends from each controller to the Linear Electric Panel. Cut to length. NOTE: *Wiring of the Controllers is explained later. Excess cable can be removed when wiring controller cables. Tie cables into place with wire ties.*

Refer to the "Wiring Diagram" for all electrical connections.

GUIDANCE CABLE INSTALLATION (Figure 18)

1. The Guidance cable (76) should be installed approximately 21" above the ground and should be supported by a stake every 75 to 100 feet depending on field conditions. It is recommended that the cable ride 1/2 way up on the controller wand The cable should be tight enough so it does not sway in the wind nor sag excessively between supports. The support stakes (77) should be aligned with a transit to insure straightness. The cable bracket (78) is attached at the top of the stake (77) by sliding it over the stake and clamping in place with the retaining nut.
2. Locate Anchor Posts (79) at the end of the run. Mount a 2500lb winch (80) to the top of one anchor post using three - 3/8" x 1" bolts with nut and lock (9). Set the anchor posts into concrete if soil is light or cannot be well packed. Locate top of anchor post or center of winch 21" above ground level.
3. Connect cable (76) to anchor post using two 1/8" cable clamps (81) on one end and to the winch on the other. If splices are required, use 2 - 1/8" cable crimp splices to fasten the cables together. NOTE: *Proper cable tension can be maintained with the winch. Cable may require tightening in warm weather but should be left slightly slacked over winter or non-use periods. Keep area within three feet of cable free of crop and weeds*.
4. Attach Position Stop Loop (82) at each end of run. Place loop over cable and extend toward tractor at a 45 degree angle. Press into ground until loop touches cable. Additional position stops can be place along cable if desired. Check operation of the position stop with system running at maximum speed.

   NOTE: *Stops are one directional and must be placed to push wand off of cable.*

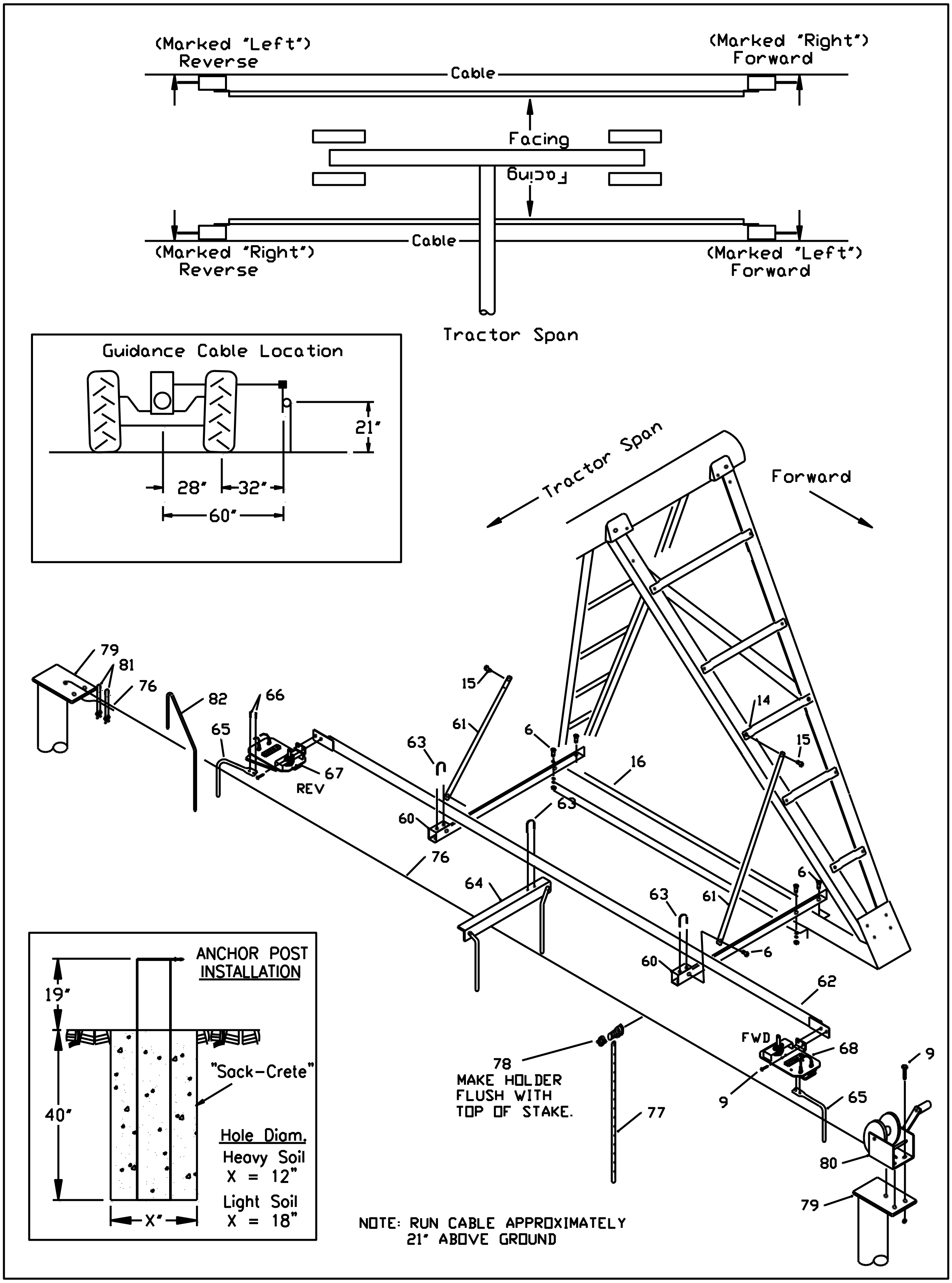


Figure 18. Above Ground Cable Installation (Non-Pivoting)

SYSTEM ALIGNMENT – ABOVE GROUND CABLE (Figures 18a, 18b)

This procedure is best performed after system installation is complete and linear is ready to run.

STEP 1: Pick a common measuring point on both the left and right sides of the Guidance Control Tube and square the Control Tube to the Tractor. Dimension "A" should be equal.

STEP 2: Square the Tractor and Guidance Control Tube to the Guidance Cable. Dimension "B" should be equal.

STEP 3: Both Forward and Reverse Guidance Controllers should be in the "Neutral Steer" condition.

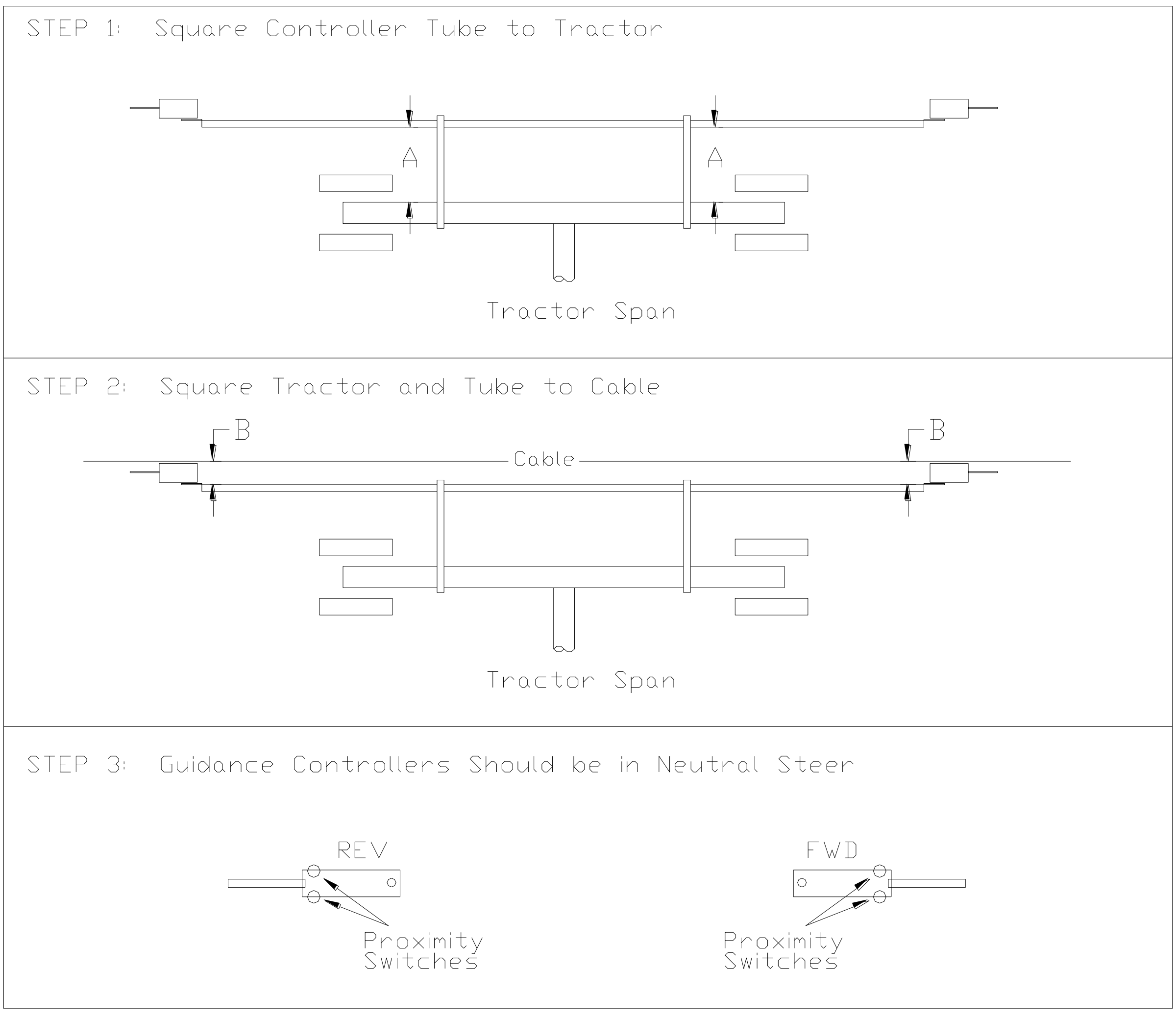


Figure 18a. Squaring Control Tube and Controllers to the Tractor and Cable

STEP 4: Align the entire system per alignment procedure. (Use caution if severe misalignment.) With the linear stopped, the towers should be in a straight line as sighted from end to end.

STEP 5: Repeat Step 2 if necessary, the system should now be square to the cable and in straight alignment.

END FEED LINEARS:

STEP 6: Operate the system in both forward and reverse directions for approximately 100 ft. The End Tower should have the same amount of lead in both directions. If the End Tower leads more in one direction than the other, make an adjustment to the OUTER alignment valve at the <u>Free Standing Span</u>.

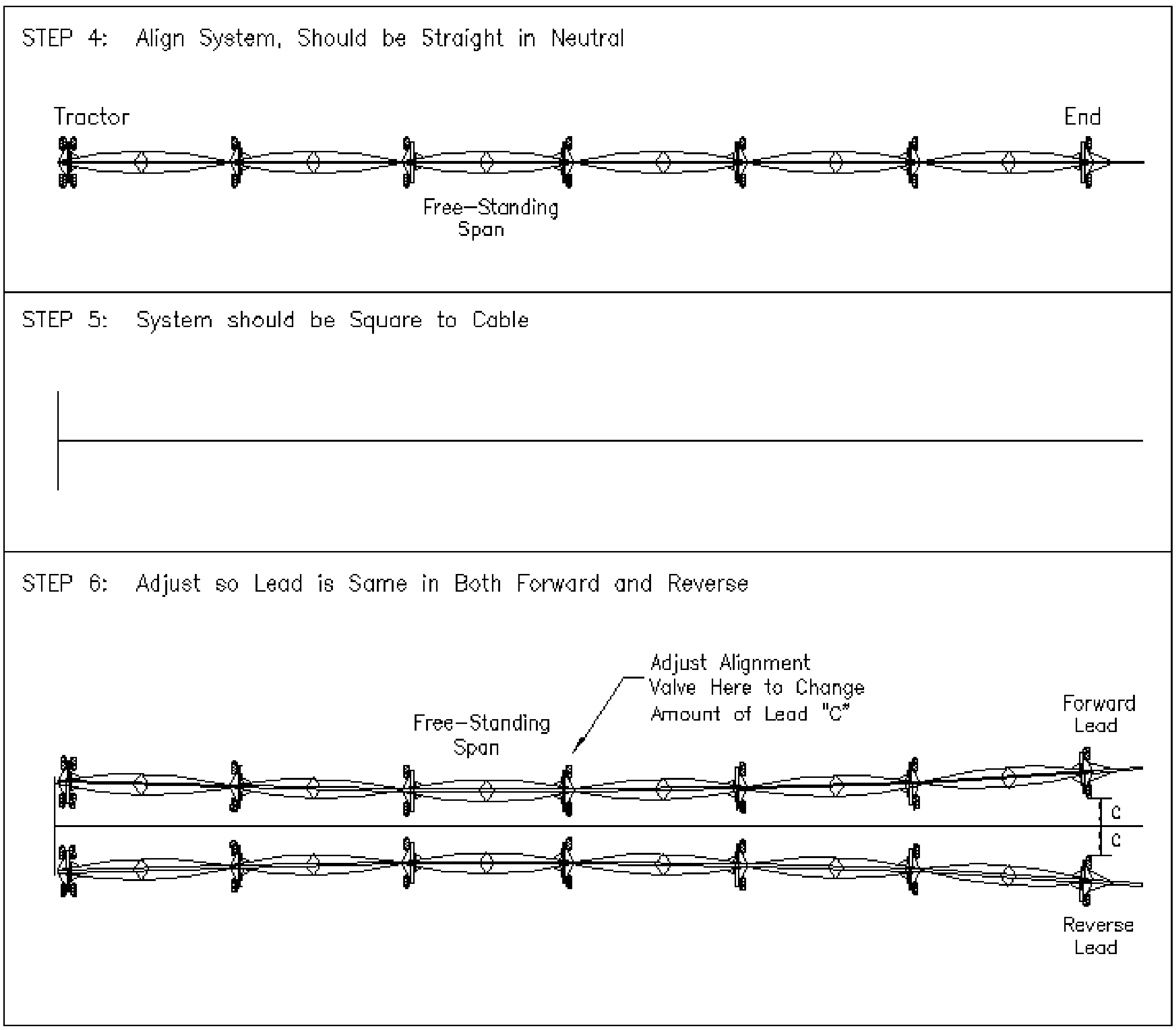
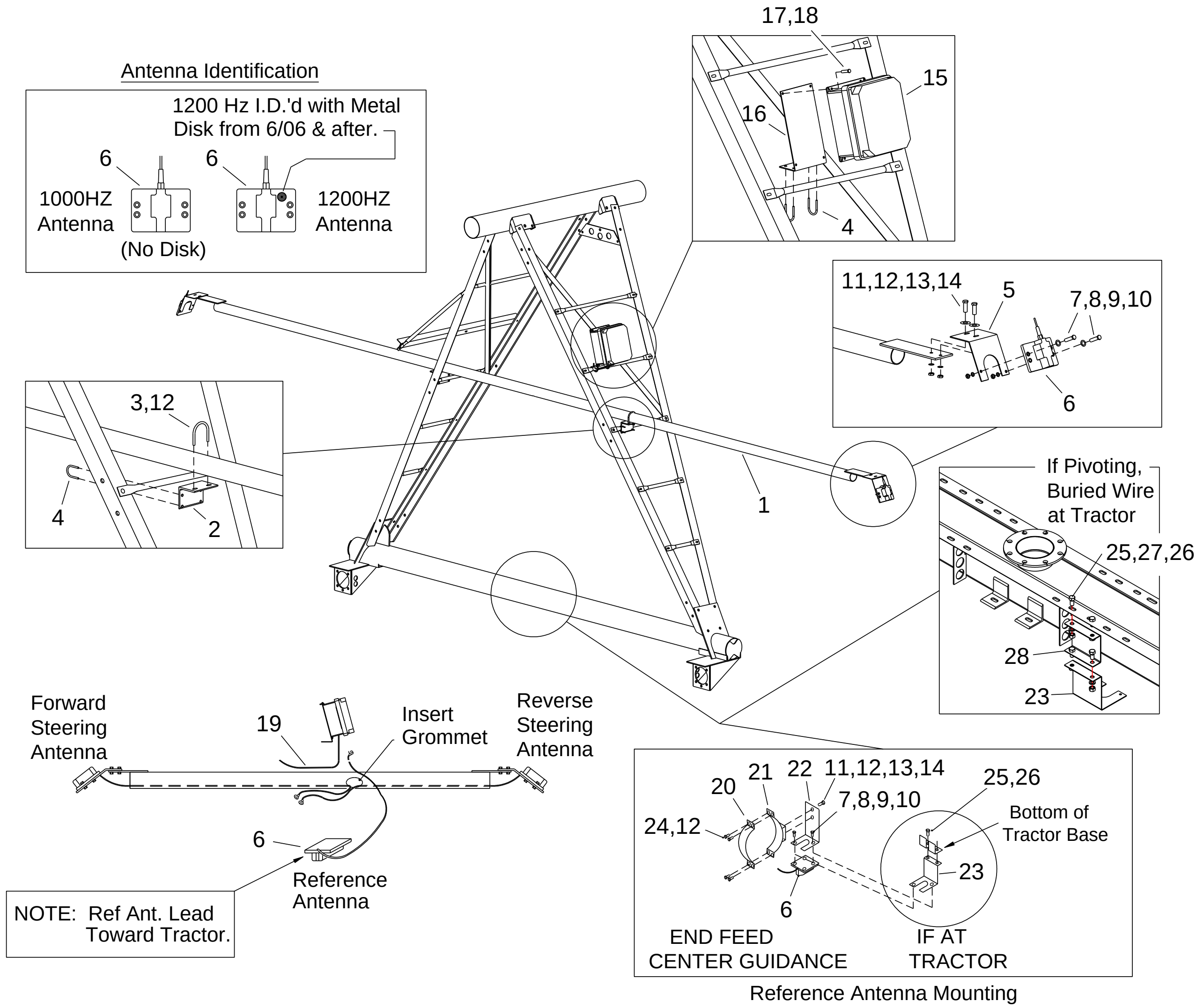


Figure 18b. System Alignment

Buried Wire Guidance II Component Installation (July 2008 -)



REF.	PART NO.	DESCRIPTION	#REQ'D.
1	AW25863	Control Tube	1
2	AL2481	Control Tube Mount Angle	1
3	FB57102	U-Bolt 3" x 3/8"	2
4	FB5725	U-Bolt 1 1/2" x 5/16" w/Nuts	6
5	AB2596	Antennal Mount Bracket	2
6	ED50704	1000HZ Antenna w/25ft. lead	3
	ED5079	1200HZ Antenna w/10ft. lead	3
7	FB5465	HHCS 5/16" x 1 1/4"	6
8	FN5400	Nut Hex 5/16"-18	6
9	FW5649	Flatwasher 5/16"	6
10	FW5650	Lockwasher 5/16"	6
11	FB5439	HHCS 3/8" x 1" Grd. 5	6
12	FN5401	Nut Hex 3/8"-16	10
13	FW5651	Flatwasher 3/8"	6
14	FW5652	Lockwasher 3/8"	6

REF.	PART NO.	DESCRIPTION	#REQ'D.
15	EA52060	Guidance Cntrl. Box II Assy	1
16	AB17010	Guidance Box Mount Plate	1
17	FB5435	HHCS 1/4" x 1" Grd. 5	4
18	FN5399	Nut Hex 1/4"-20	4
19	MD5374	Wire 16-8	AR
20	AP641	Half Band 6 5/8"	1
21	AW2723	Ref. Antenna Mount Clamp	1
22	AB2705	Ref. Antenna Mount EF or CG	1
23	AB27050	Ref. Antenna Mount at Tractor	AR
24	FB5444	HHCS 3/8" x 2"	4
25	FB5454	HHCS 5/8" x 1 1/4"	AR
26	FN5406	Nut Hex 5/8"-11	AR
27	FW5658	Lockwasher 5/8"	AR
28	AB27051	Rer. Antenna Pvt Ext Bracket/EFEG	AR

Figure 19. Installation of Buried Wire Components.

Squaring Antennas (Figure 20)

Center the Controller tube on the tower. Slide the Controller tube either direction until you achieve an equal distance from the tower step to the end of the tube on both sides. Adjust antenna supports so they are level to the ground. Snug the u-bolts to prevent any further movement. Adjust the control tube mount angles on the step so that the control tube is centered over the deck and perpendicular to the span. Check Square, antennas equal distance from center top of first flange..

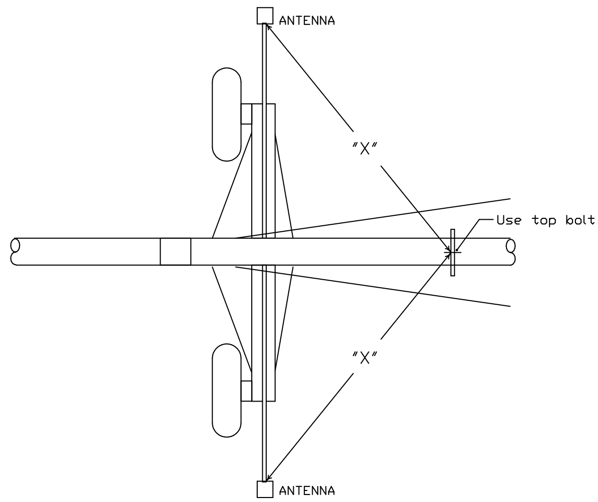


Figure 20. Squaring Antennas.

Wiring of Gd Box II for Linear (Figure 21)

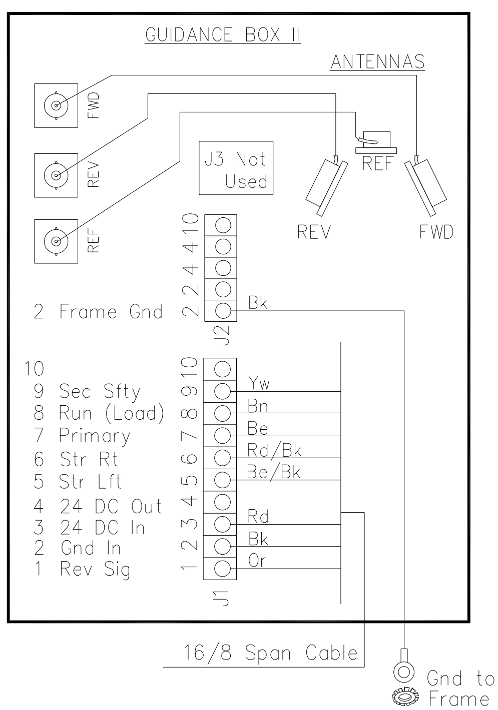


Figure 21. Wiring of Gd Box II.

Span Cable Installation (Buried Wire Guidance, End Feed with Center Guidance)

1. The 16/8 span cable, is attached to the top of the pivot spans, from the Guidance Box assembly to the Tractor (see Figure 12). The span cable is easiest to install after the system is spanned but the towers not yet stood.

2. Route Span Cable alternately between Sprinkler Outlets. At each Sprinkler Outlet (or every 9' 10" for multi-outlet pipe), secure span cable to Outlet with Span Cable Clip (4).

3. At each Span Pipe Flange, attach a Cable Holder (1). Use existing nut (or additional nut, not provided). Wire tie the Span Cable to the Cable Holder with the wire ties provided.

4. At each Tower, attach Span Cable with 32" Wire Tie (3).

5. At the tractor, allow enough cable to run down to the tractor deck and back up to the Linear Control Panel Assembly. Approximately 12" is needed inside the panel. Excess cable on both ends will be cut off during wiring of the system (see later section).

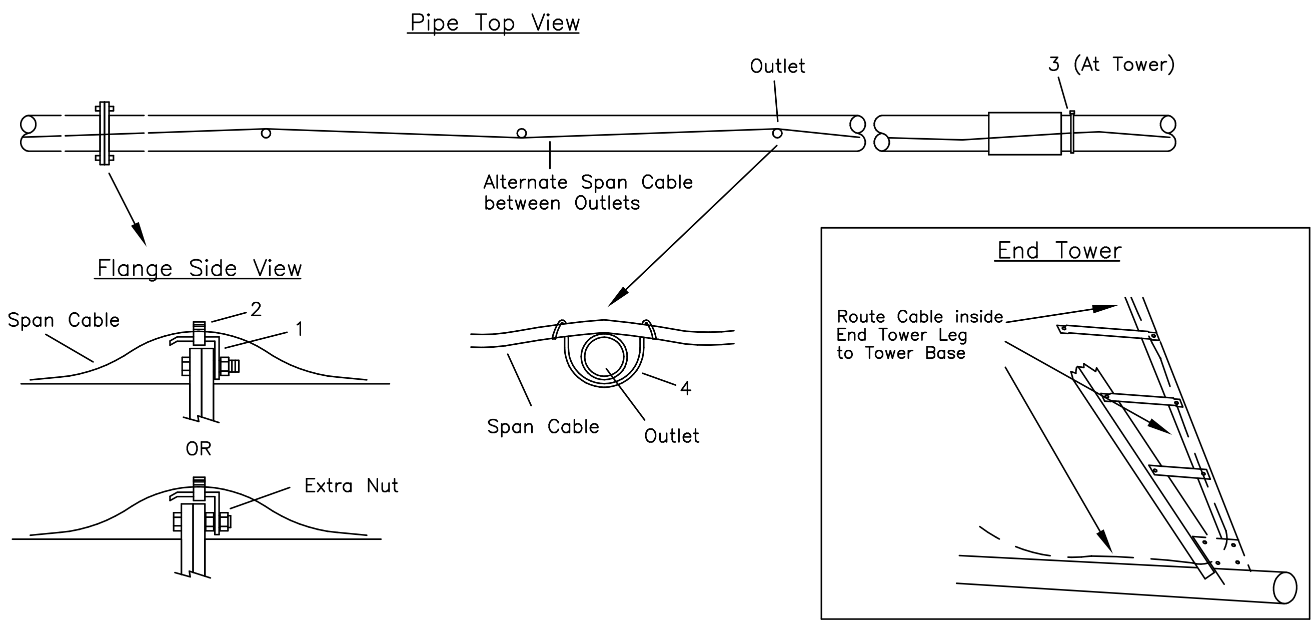


Figure 22. Span Cable Installation

Wheel Mount Installation (Figure 23)

1. Attach Pivoting Axle (120) in place of standard Wheel Mounts (shown earlier) to deck (16) with 1" x 12 1/2" Axle Shaft (18), two 1" flat washers (19) and two 3/16" x 2" cotter pins (20). Stationary Wheel mount should be placed to span side of tractor. *NOTE: Grease Axle shaft before assembly*. Make sure motor port holes on Wheel Mount gusset and tow arm are toward the center of the deck.
2. Attach Steering Arm Clevis to Short Steering Arm (122) using two 3/4" nuts (123). Adjust Clevis half way up threads. Attach to Pivoting Axle using a 1/2" x 3" bolt with nut (124). Tuck Arm against Axle when not in use.
3. Attach Pivoting Wheel Mount (125) to Pivoting Axle (120) using a 1" x 10" bolt (126) with Lock Nut (127). Tighten nut so Wheel Mount is snug but pivots freely by hand. Pin Wheel Mount arm into place with Steering Arm Pin (128) and Hitch Pin (137).
4. Attach Gears and Wheels as described in the “Tractor Assembly” Section.
5. Pin Jack Tie Strap (129) into Jack Top Tab (130) at hole (A) using one 1" Clevis Pin (131) with Hair Pin Clip. *Second Pin will be used only while operating Jack.* Pin Cylinder (132) onto Jack Tie Strap (129) at hole (B) and Jack Bottom outer Tab (135) using two 1" Cylinder Pins (133) with Cylinder ports toward center of deck.
6. Jack Safety Straps (134) will be stored until needed during actual pivoting operation. When used they will pin onto Jack Tie Strap (129) at hole (C) and Jack Bottom inner Tab (136) using two 1" Clevis Pin (131) with Hair Pin Clip. The second pin in Jack Top Tab (130) must also be in place.

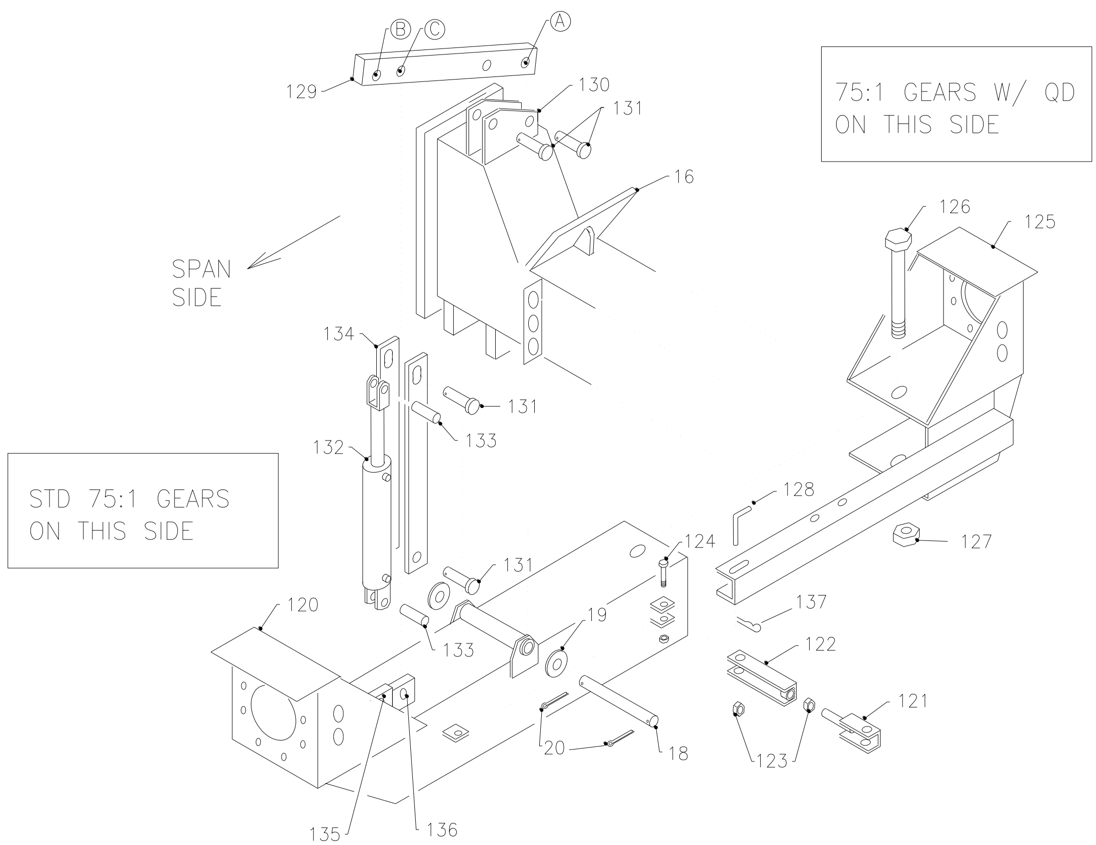


Figure 23. Pivoting Wheel Mount Installation

Pivoting 4 Wheel Tractor Pictures

Tower Brace Stiffener Package (Figure 24)

1. Install mid-way on Tower Braces that run between the Tractor and the #1AV Truss

Ref	PART #	DESCRIPTION	Qnt	Ref	PART #	DESCRIPTION	Qnt
13	AT1112	TOWER BRACE	2	38	FB5516	HHCS 3/8 X 3 1/2	9
35	AL15116	STIFFENER ANGLE	2	39	FN5401	NUT 3/8”	9
36	AP1839	STIFFENER ATTACH STRAP	4	40	FW5652	LOCK WASHER 3/8”	9
37	AT52698	SPACER TUBE	1				

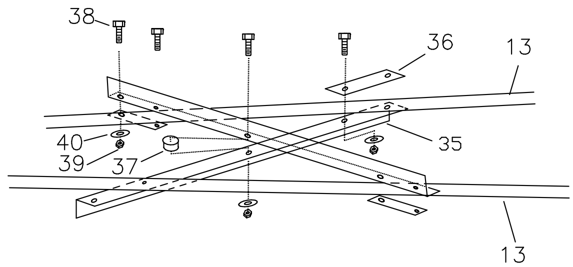


Figure 24. Tower Brace Stiffener Installation

Lift Option Plumbing (Figure 25a)

Use following in conjunction with the End Feed Tractor Plumbing shown in Figure 12.

1. Add the following into the Hydraulic Pumping Assembly Pressure line: one ¾ x ¾ x ½ Tee (31) and one ¾ X Close Nipple (4). To the Tee, add a Bushing ½ x ¼ (35).
2. Add the following into the Hydraulic Pumping Assembly Return line: one ½ X Close Nipple (36) and one ½ x ½ x ¼ Tee (37).
3. Attach Lift Valve Mount Plate (38) to the side of the Pumping Unit using existing 1/2" x 1 1/4" bolts. The Lift Valve Mount Plate is also used for the Tank Mounting Plate (#11, Figure 12).
4. Remove Adapter fitting from Directional Control Valve (39) if necessary. Plumb four 1/2" x 1/4" Bushings (35) into the ports on the Directional Control Valve (39). Plumb two 1/4" Street Elbows (40) into the Valve Pressure and return ports. Mount the Directional Control Valve (39) to the Lift Valve Mount (38) using three 1/4" x 1 3/4" Screws with nuts (41) and 5/16” Flat Washers (45).
5. Connect the Pressure Port from the Directional Control Valve (39) to the Hydraulic Pumping Unit Pressure Bushing (35) using one ¼” x 28” Hoses ¼ MP x ¼ MPX (42).
6. Connect the Return Port from the Directional Control Valve (39) to the Hydraulic Pumping Unit Return Tee (37) using one ¼” x 40” Hoses ¼ MP x ¼ MPX (43).
7. Attach Directional Control Valve (39) Motor ports to two ¼” x 28” Hoses ¼ MPT x ¼ MPX (42) and two 1/4" Tees (44). Connect two 1/4" x 112" Hoses ¼ MPT x ¼ MPX (46) and two 1/4" x 68" Hoses ¼ MPT x ¼ MPX (47) to the 1/4" Tees (44). Connect the four Hoses (46,47) to the Hydraulic Cylinders (48) on each Pivoting Wheel Mount with four 1/2" x 1/4" Bushings (35).

Pivoting Option Plumbing, Figure 25a

Ref	PART #	DESCRIPTION	Qty	Ref	PART #	DESCRIPTION	Qty
4	PN8206	NIPPLE 3/4" X CLOSE BEH	1	41	FS55376	SHCS ¼ x 1-3/4	3
12	FB57255	U BOLT 5/16 X 1 1/2 X 2 1/2	1	42	HH6154	HOSE 4CIT-4MP-4MPX-28"	3
31	PT8152	TEE ¾ x ¾ x ½ BEH	1	43	HH61764	HOSE 4CIT-4MP-4MPX-40"	1
35	PB7997	BUSH 1/2x1/4 STEEL PLATED	9	44	PT8164	TEE 1/4 PLATED	2
36	PN8204	NIPPLE 1/2" X CLOSE BEH	1	45	FW5649	FLAT WASHER 5/16”	2
37	PT8150	TEE ½ x ½ x ¼ BEH	1	46	HH61766	HOSE 4CIT-4MP-4MPX-112"	2
38	AB14351	LIFT VALVE MOUNTING PLATE	1	47	HH61767	HOSE 4CIT-4MP-4MPX-68"	2
39	HV9762	DIRECTIONAL CONTROL VALVE	1	48	HC98555	CYLINDER 3” X 8”	2
40	PE81111	ST.ELBOW 1/4 X 90 DEG PLATED	2				

Insert circled items into the Pressure and Return plumbing as shown below and in Figure 12.

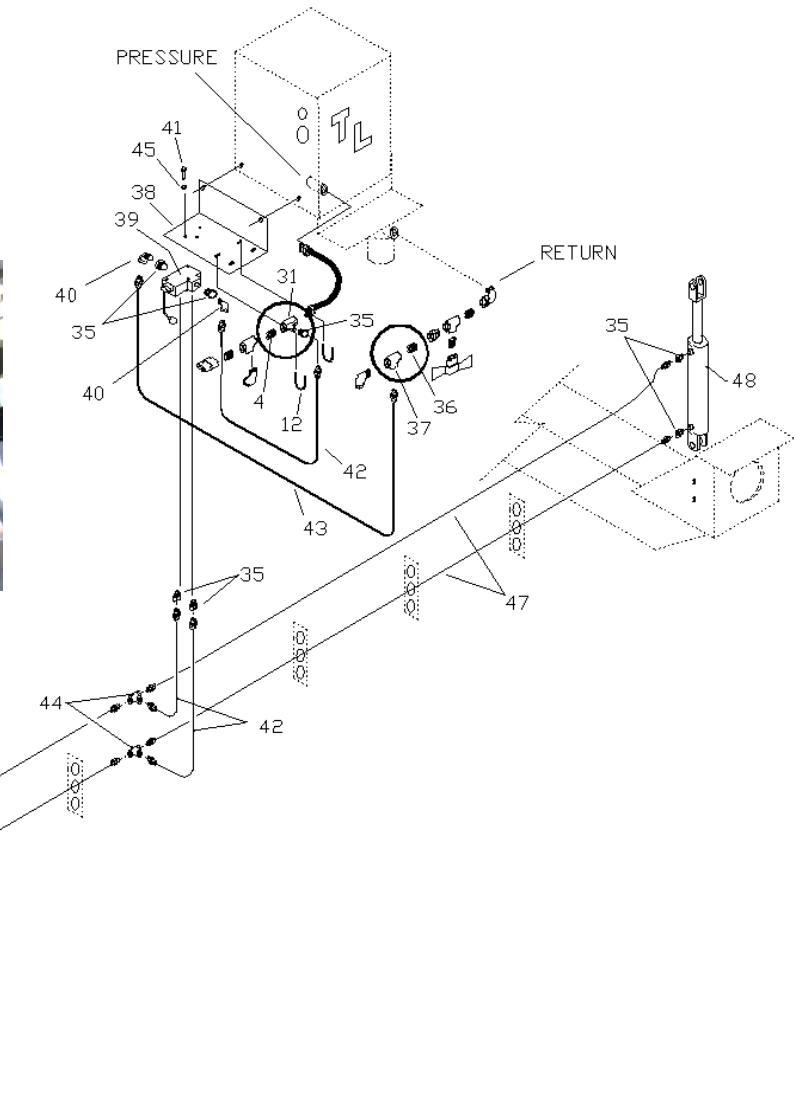


Figure 25a. Lift Option Plumbing

Pivot Ring Installation (Figure 25b)

Install the Pivot Ring (AA27840) on the Tractor Base using four 5/8 x 1 1/2" HHCS with FW,LW, nuts. Grease the Pivot Ring.

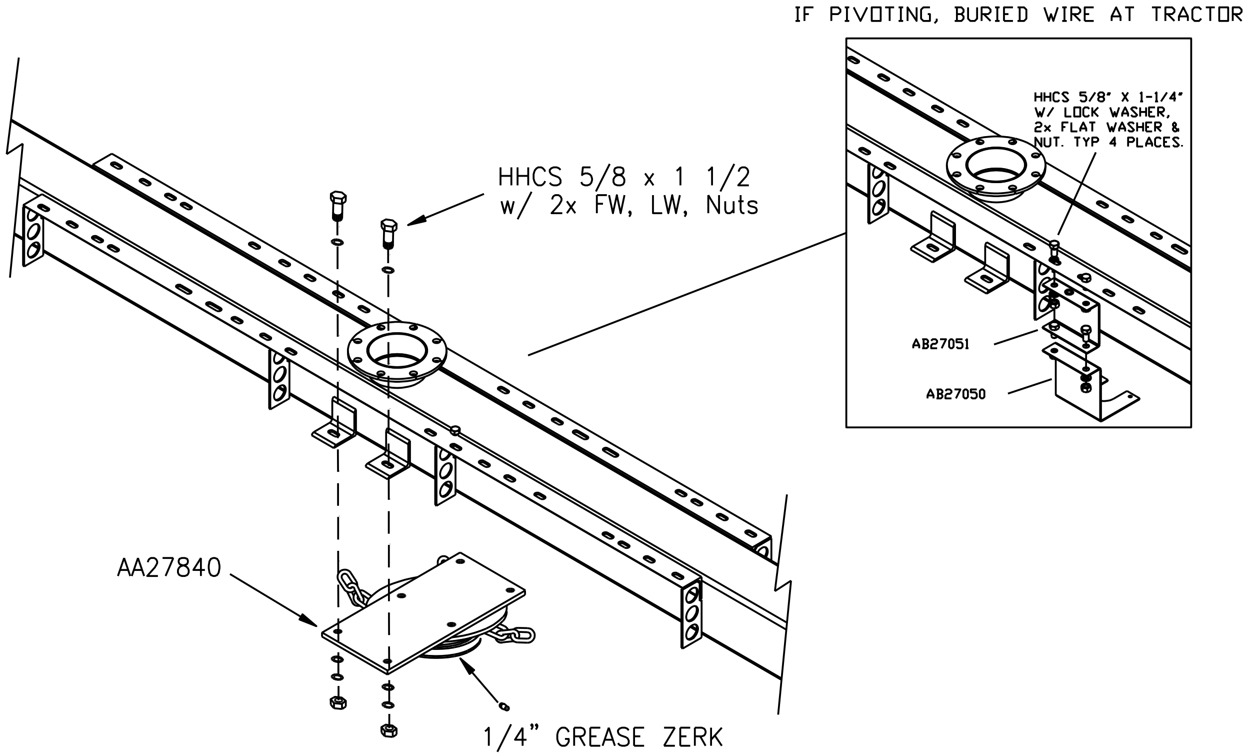


Figure 25b. Pivot Ring Installation

Pivot Chain Installation (Figure 25c)

The Pivot Chains shipped with the pivoting option (3/8" x 12 ft, field cut to 4 ft pieces) can be installed in concrete into the ground as shown below. Install in a 4 to 5 ft. radius at each pivoting location. Earth anchors can be used if conditions are suitable.

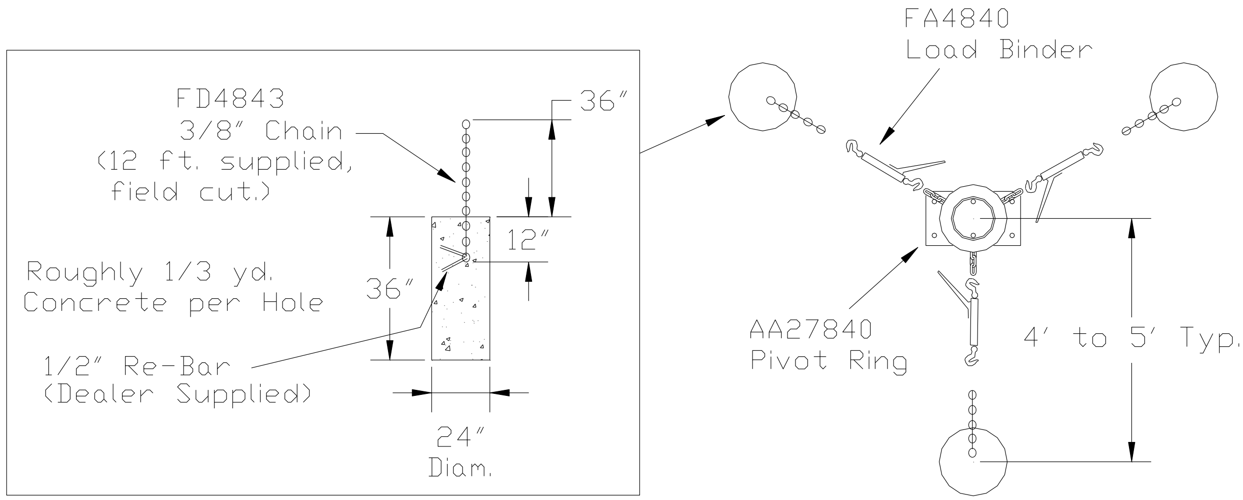


Figure 25c. Pivoting Chain Installation

Above Ground Cable Guidance Pivoting Option (Figure 26)

See earlier section on Guidance Installation for help.

1. Bolt two Double Controller Extension Angles (69) to Tractor Deck on the mounting brackets nearest the end of the deck using four 5/8" x 1 1/4" bolts with locks and nuts (6). Square Angles to deck making sure that the Angles are the same dimension apart on each side of the deck. Bolt Support Straps (71) to the sides of the Control Tube Extension Angles (69) on each side of tractor using 5/8" x 1 1/2" bolts with locks and nuts (6). Bolt the upper end of the Support Straps (71) to the middle tower step (not shown) using 5/8" x 2" bolts with locks and nuts (15).
2. Bolt four Socket Guide Brackets (73) to the ends of the Double Controller Extension Angles (69) using two each 3/8" x 1" bolts with flat washer, locks and nuts (9).
3. Attach Pin Guide Bracket (72) Controller Tube (70) using two 3/8" U-Bolts (63), do not tighten. Slide Pin Guide Bracket (72) into Socket Guide Bracket (73). Position end plates on Controller Tube (70) vertically and tighten U-bolts.
4. Attach secondary safety assembly (64) to center of Controller Tube (70) with one 3/8" U-Bolt (63).
5. Attach the Proximity Controllers to the Controller Tube as shown previously in the "Guidance Installation" Section. Note the Controllers will be marked "Left" and "Right" for shipment. Be sure to note how Forward and Reverse are determined.
6. Attach Pivoting Junction Box (74) to Mount Plate (75) using four Rubber Mounts (86). Attach Mount Plate (75) to Center of controller tube (70) using two 3/8" U-Bolts (63).
7. Route one Proximity cable (16 ga 7 wire) from each controller through the controller tube, out the side hole to the Junction Box (74).
8. Route 2 Junction Box Cables (85) along extension angle (69) to Linear Control panel.

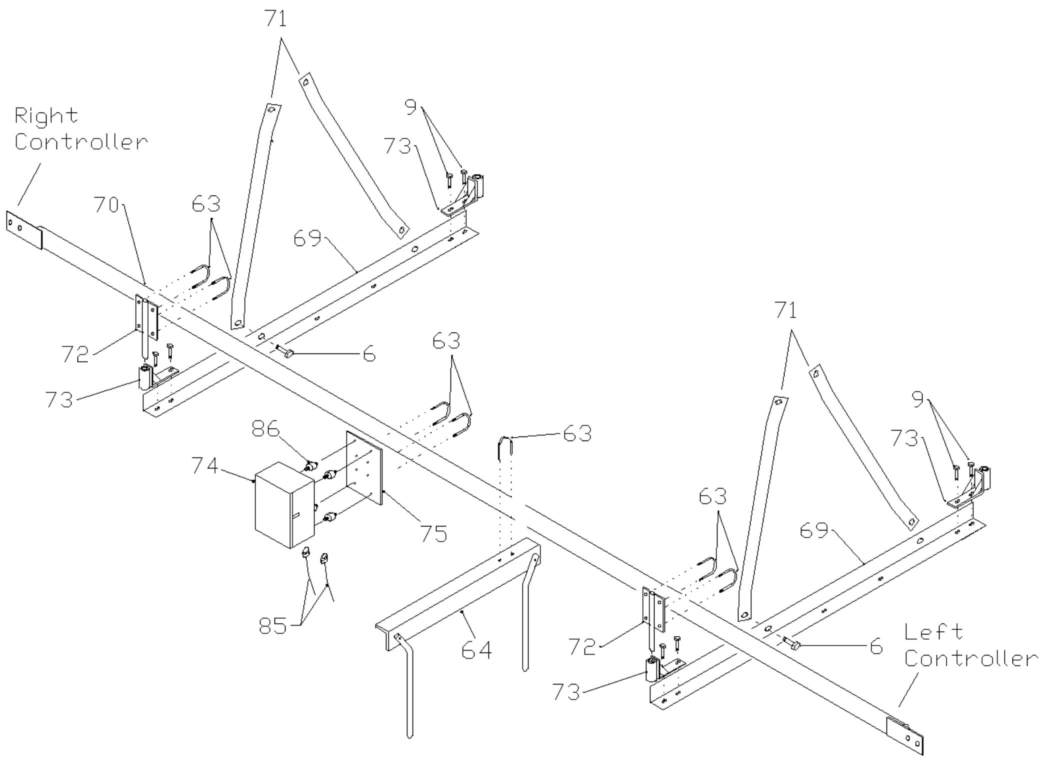


Figure 26. Above Ground Cable Pivoting Guidance

CD20330 Sheet 1 of 2Above Ground Cable Wiring Logic Board III-Non Pivoting
(Feb 2016 on = Linear Logic Board III)
(June 2022 on = Electroswitch Encoder)

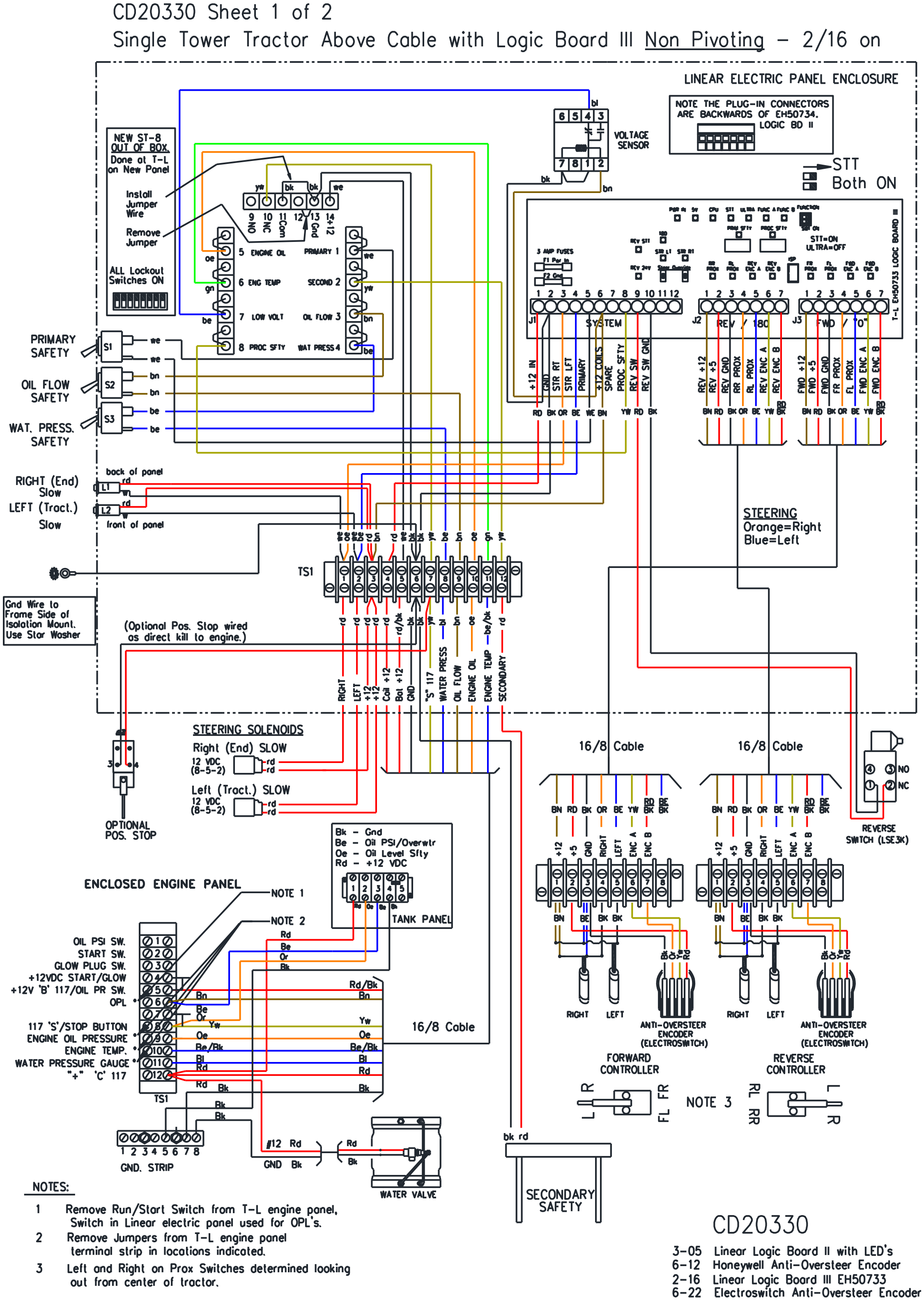


Figure 27. Wiring Diagram – Above Ground Cable Non-Pivot

CD20330 Sheet 2 of 2 – Field Wiring to Above Cable Panel with Linear Logic Board III

(Feb 2016 on = Linear Logic Board III)
(June 2022 on = Electroswitch Encoder)

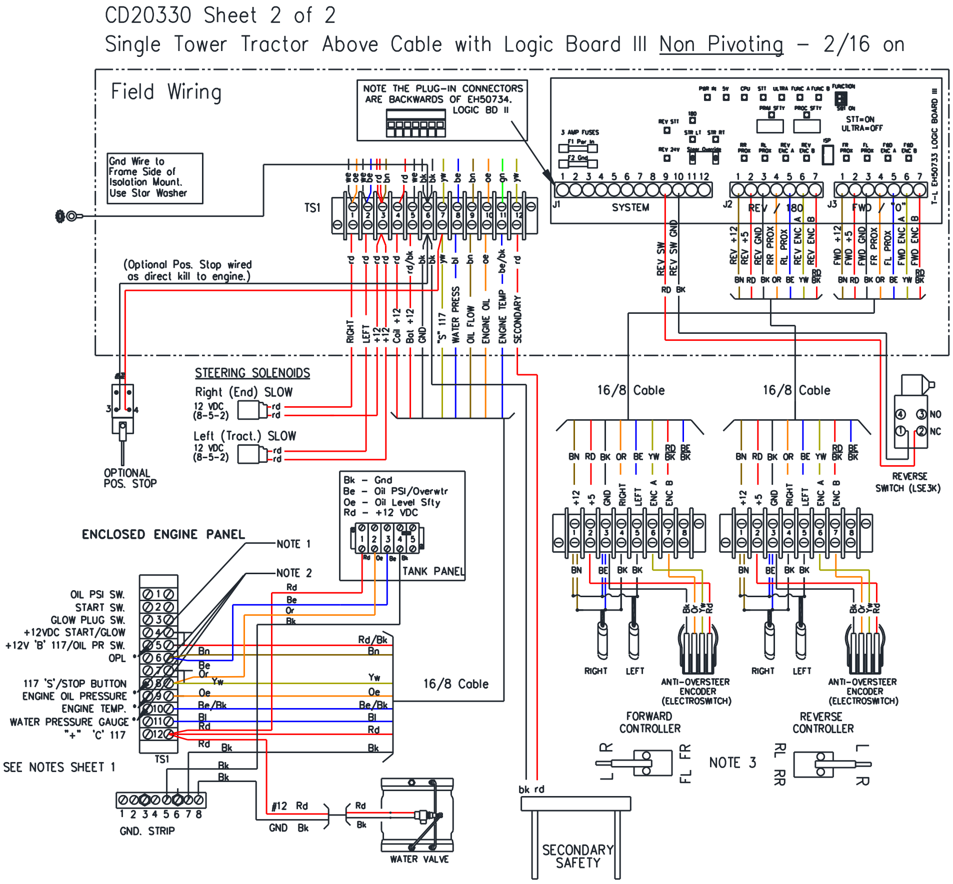


Figure 27.2 Field Wiring to Control Panel with Logic Board III

Guidance Controller Wiring

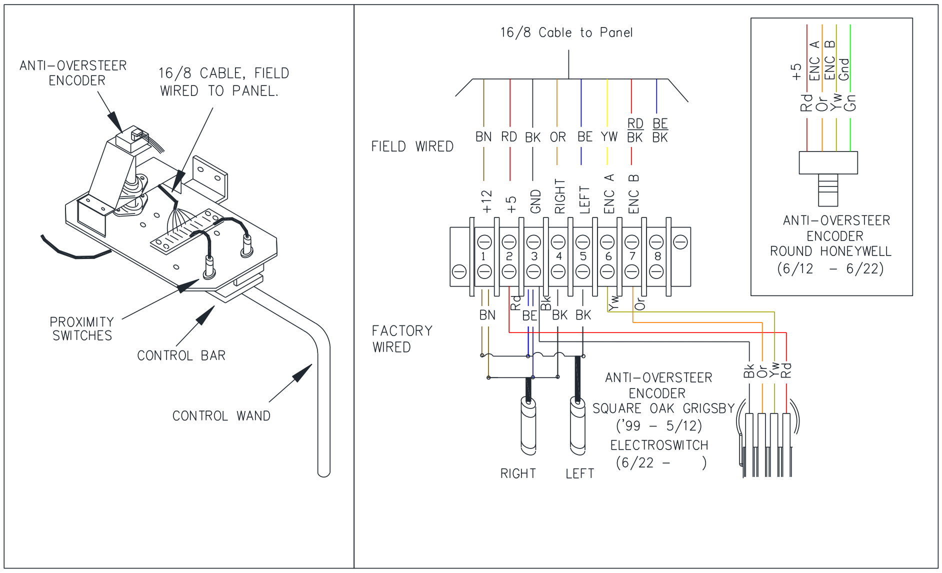


Figure 27.1. Above Ground Guidance Controller Wiring

Buried Wire - Non-Pivoting (Figure 28)

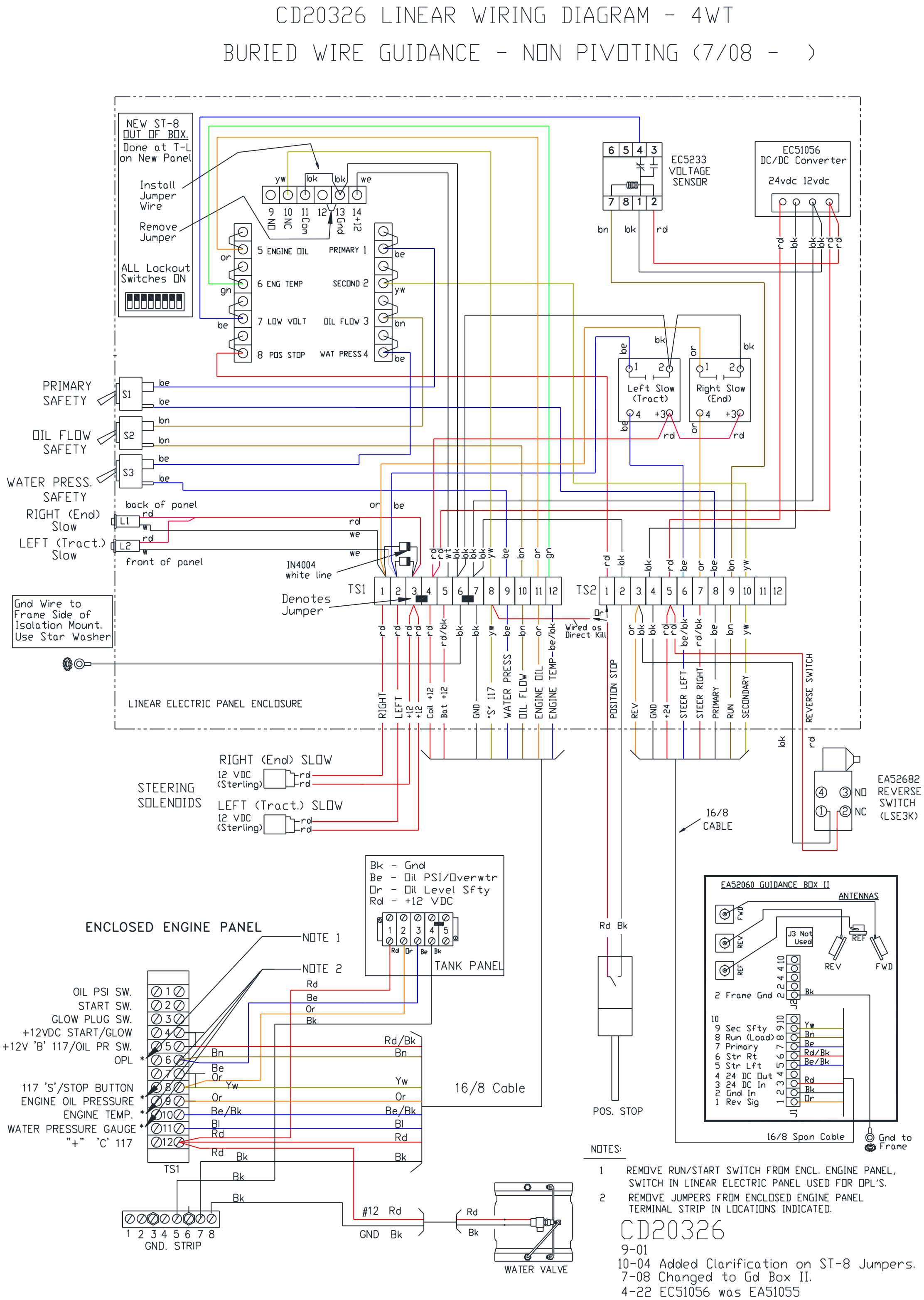
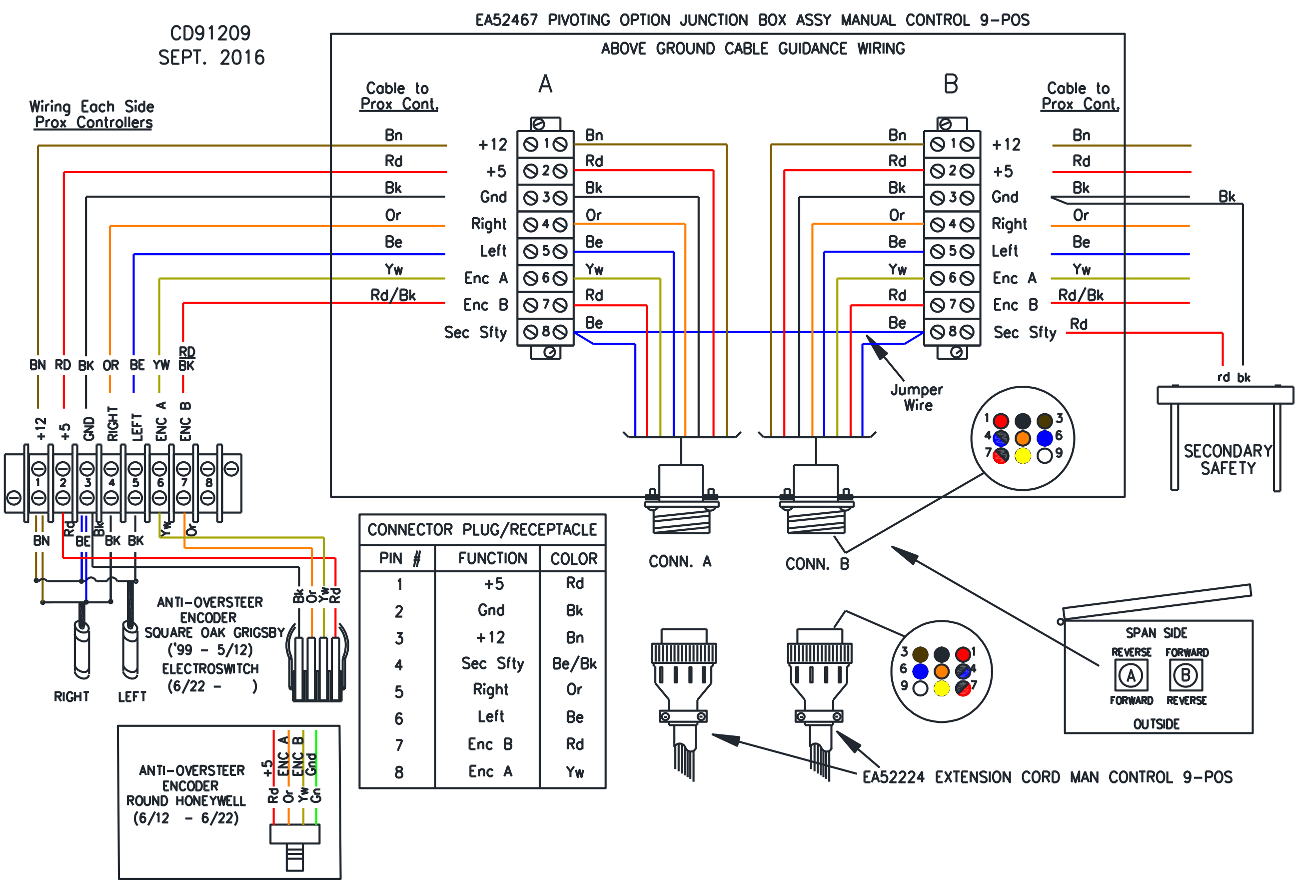


Figure 28. Wiring Diagram – Buried Wire Non-Pivoting (July 2008 -).

Pivoting Junction Box – Above Ground Cable (Figure 29) – Optional

1. Route 16/7 Guidance controller cables through controller tube and out the hole in the side of the tube and into Junction box. Place a grommet in the Control tube hole to prevent wire chafing.
2. Route Secondary Safety cable into Junction box.
3. Allow 6" to 12" extra cable for wiring connections. Cut off excess cable and tie any loose cables to the control tube using wire ties.
4. Wire Guidance Controllers and Secondary Safety as shown below.
5. Wire Junction Box Extension Cables into the Linear Electric Panel as shown in Figure 30. NOTE that the Cable labeled Forward is wired to terminals 14-20 and the Cable labeled Reverse is wired to terminals 7-13 on the Logic Board.
6. The Extension Cables attach to the Pivoting Junction Box according to which side of the tractor the controller tube is mounted.

NOTE: CHANGED SEPT. 2016 to 9-Position Sealed Connectors.



CONNECTOR PLUG/RECEPTACLE

PIN #	FUNCTION	COLOR
1	+5	Rd
2	Gnd	Bk
3	+12	Bn
4	Sec Sfty	Be/Bk
5	Right	Or
6	Left	Be
7	Enc B	Rd
8	Enc A	Yw

Figure 29. Optional Pivoting Junction Box Wiring

CD20331 Above Ground Cable Wiring Logic Board III - Pivoting
(Feb 2016 on = Linear Logic Board III)
(June 2022 on = Electroswitch Encoder

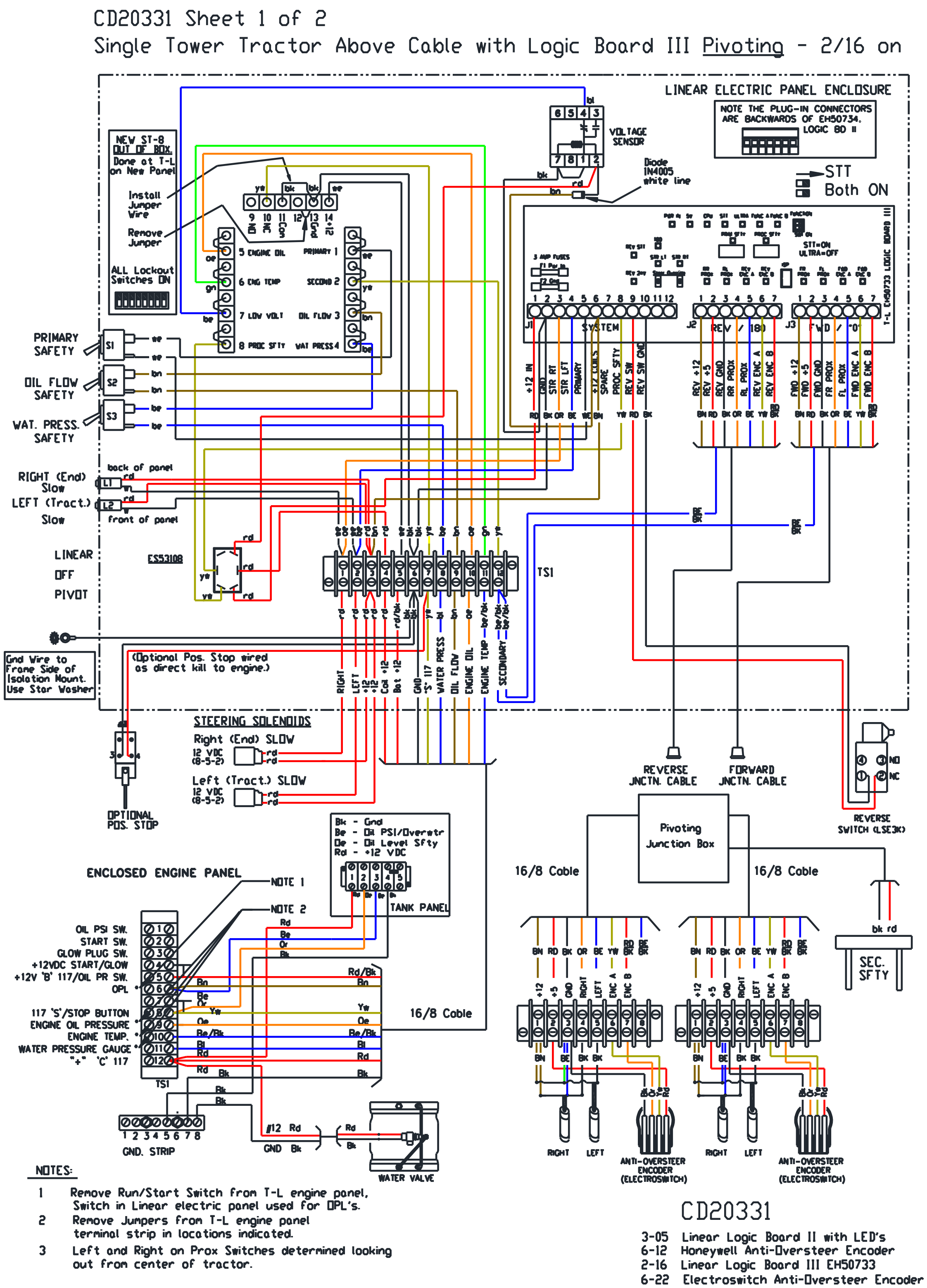


Figure 30. Wiring Diagram - Above Ground Cable with Pivoting Option

Buried Wire Guidance -Pivoting (July 2008 -   )

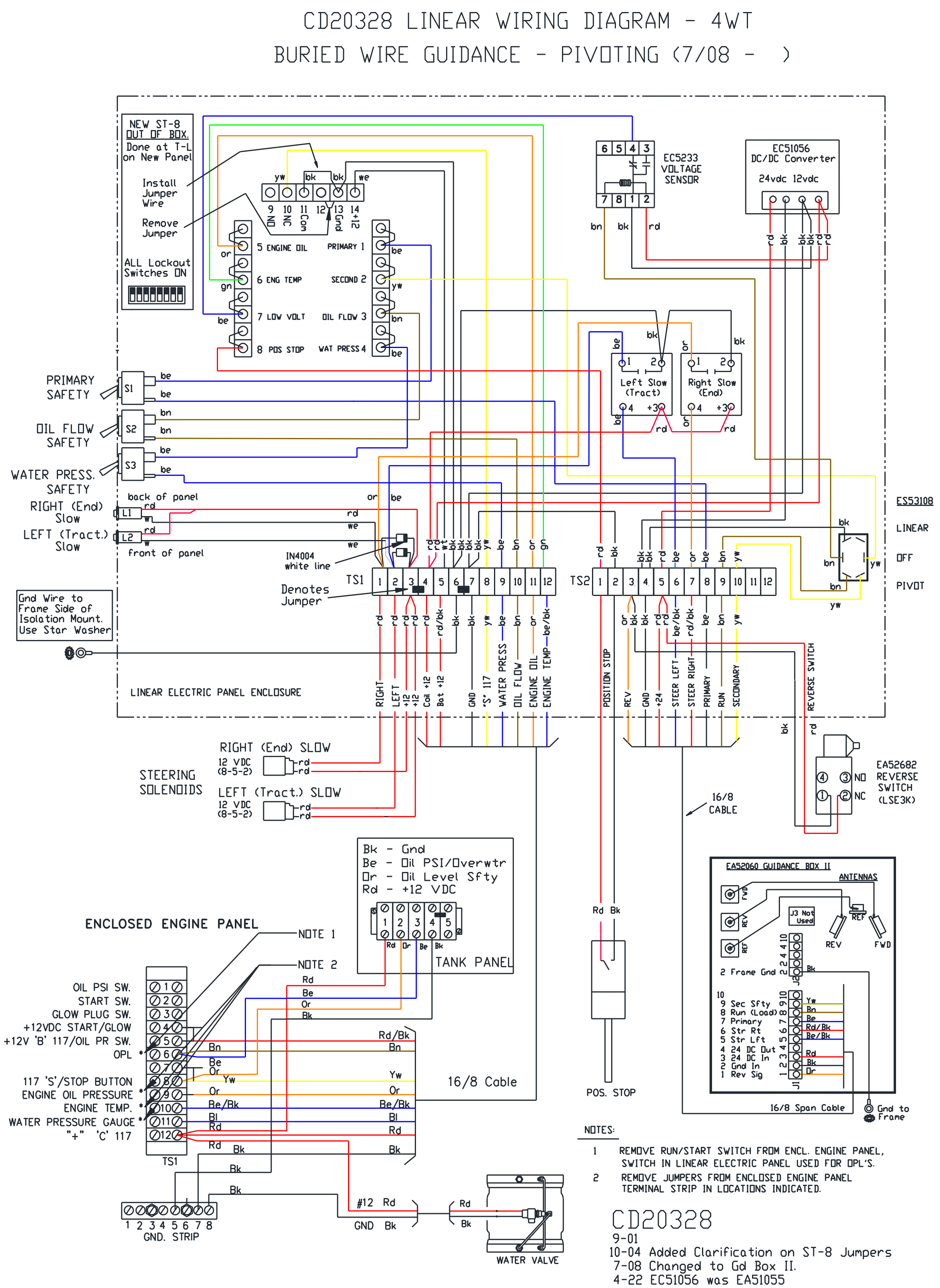


Figure 31. Wiring Diagram – Buried Wire with Pivoting Option (July 2008 -   ).

Wiring of Overwater Timer and Speed Sensor Motor (Figures 31a, 31b)

A Speed Sensor installed in one of the Hydraulic Motors is used to determine if the system has stopped. The Speed Sensor is wired into an Overwater Timer Box which is located in the Hydraulic Pumping Unit Panel. See the “General Information” section for Sensor Motor location.

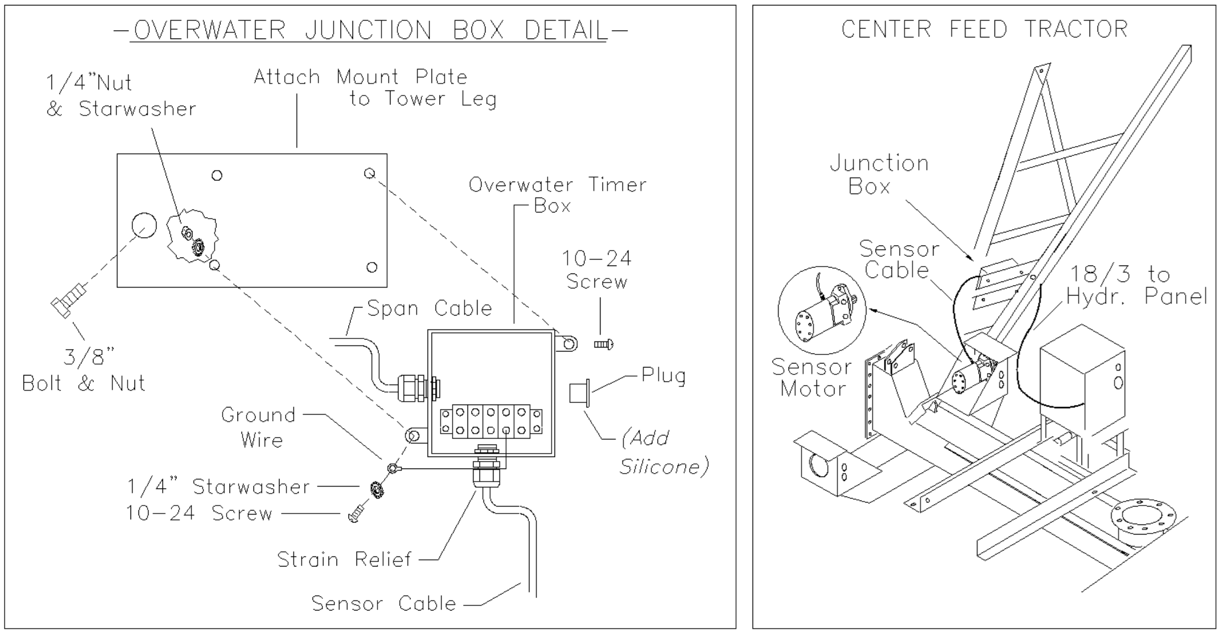


Figure 31a. Installation of Overwater Junction Box

1. Attach the Overwater Junction Box in the appropriate location.

2. Route the Speed Sensor Cable from the Hydraulic Motor to the Overwater Junction Box and route the 18/3 cable from the Junction Box to the Hydraulic Pumping Unit Tank Panel.

3. Strip Cable and wire into the Overwater Timer Box as shown in Figure 31b.
NOTE: Push on Orange tab with small screwdriver, then insert wire. Wire is held by spring tension.

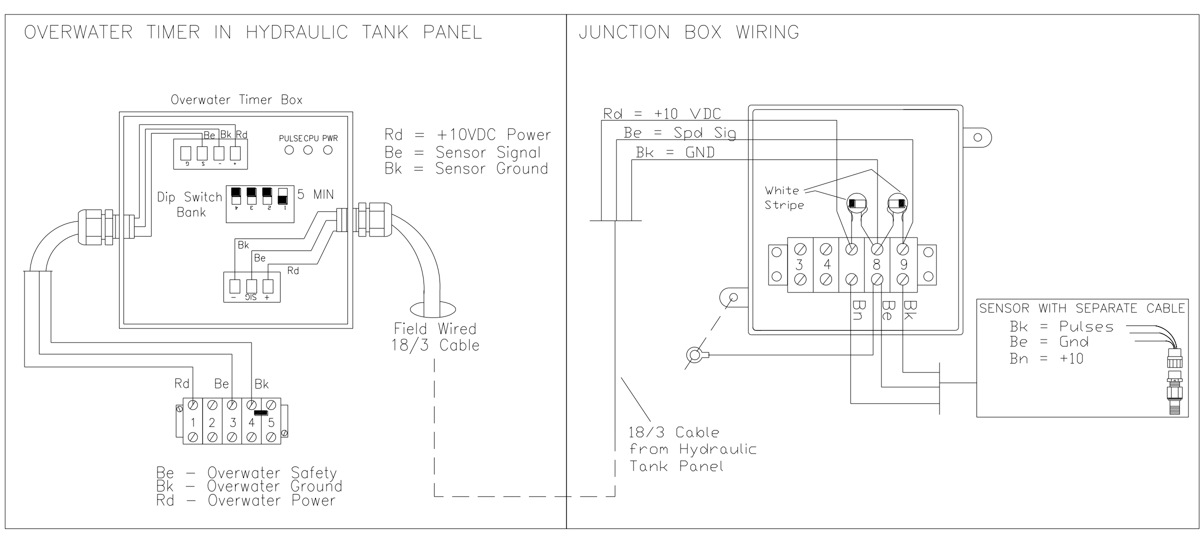


Figure 31b. Wiring of Overwater Timer

4. Note Overwater Timer Settings as shown in the table following. The Factory Setting should be 5 minutes. Test Overwater Timer for proper operation!

DIP Switch Setting				Time Delay (Min.)
1	2	3	4	
OFF	OFF	OFF	OFF	1
ON	**OFF**	**OFF**	**OFF**	**5**
OFF	ON	OFF	OFF	10
ON	ON	OFF	OFF	15
OFF	OFF	ON	OFF	20
ON	OFF	ON	OFF	25
OFF	ON	ON	OFF	30
ON	ON	ON	OFF	45
OFF	OFF	OFF	ON	60

Table 1. DIP Switch settings for various time delays.

Wiring of Hydraulic Tank Panel and Enclosed Engine Panel (Figure 31c)

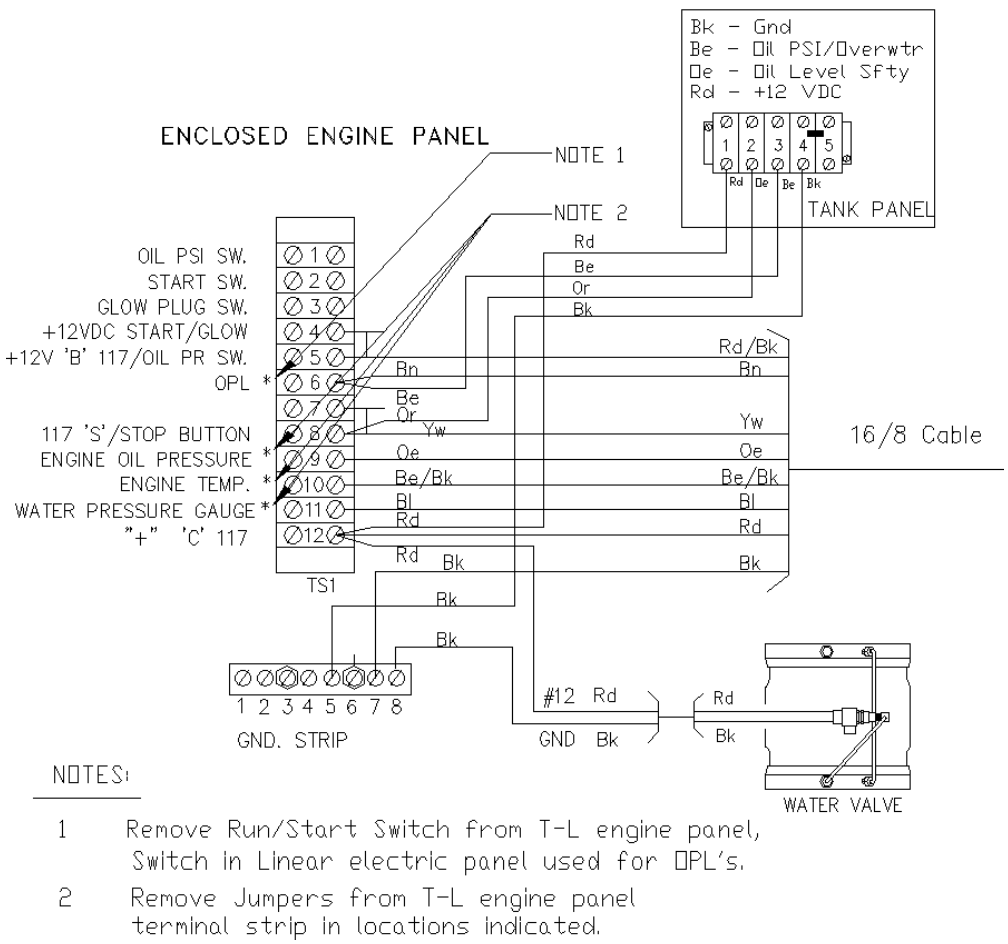


Figure 31c. Wiring of Hydraulic Tank Panel and Enclosed Engine Panel

Wire Location (Figures 32 and 33)

Items necessary for this procedure are:

- Transit, 100 foot Tape measure
- Trencher, 24"
- Vibratory Cable Plow
- 14 Gauge Stranded Guidance Wire twice the length of field plus 400 feet
- Twisted pair from power source to guidance loop
- Tools for splicing 14 gauge wire (electrical and friction tape, rosin core solder, soldering gun or propane torch, heat shrink)

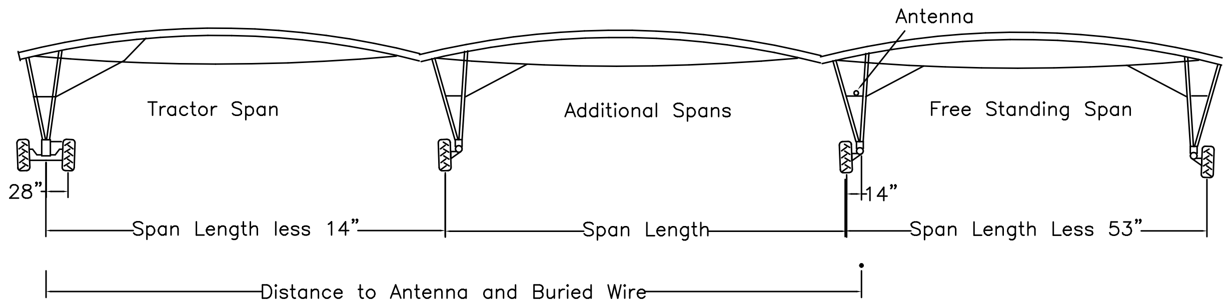


Figure 32. Buried Wire Location for End Feed with Center Guidance

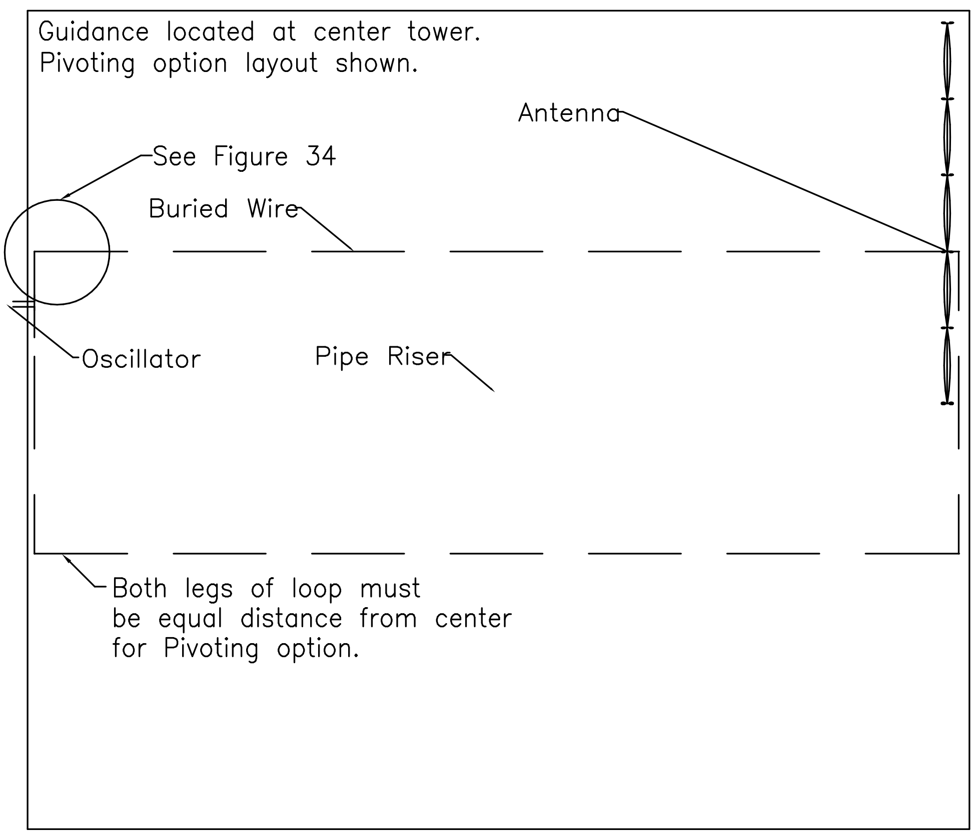


Figure 33. Buried Wire Layout for End Feed with Center Guidance

Burying The Guidance Wire

The Guidance wire loop should run to the edge of the field. A correction curve is required at the 90° bend at each corner of the Guidance wire run to compensate for the influence of the signal from the wire across the end of the field (See Detail below, shown is for standard antenna heights, 84” and 30” burial depth). NOTE: Correction curve not needed if Guidance Wire can be extended beyond field edge.

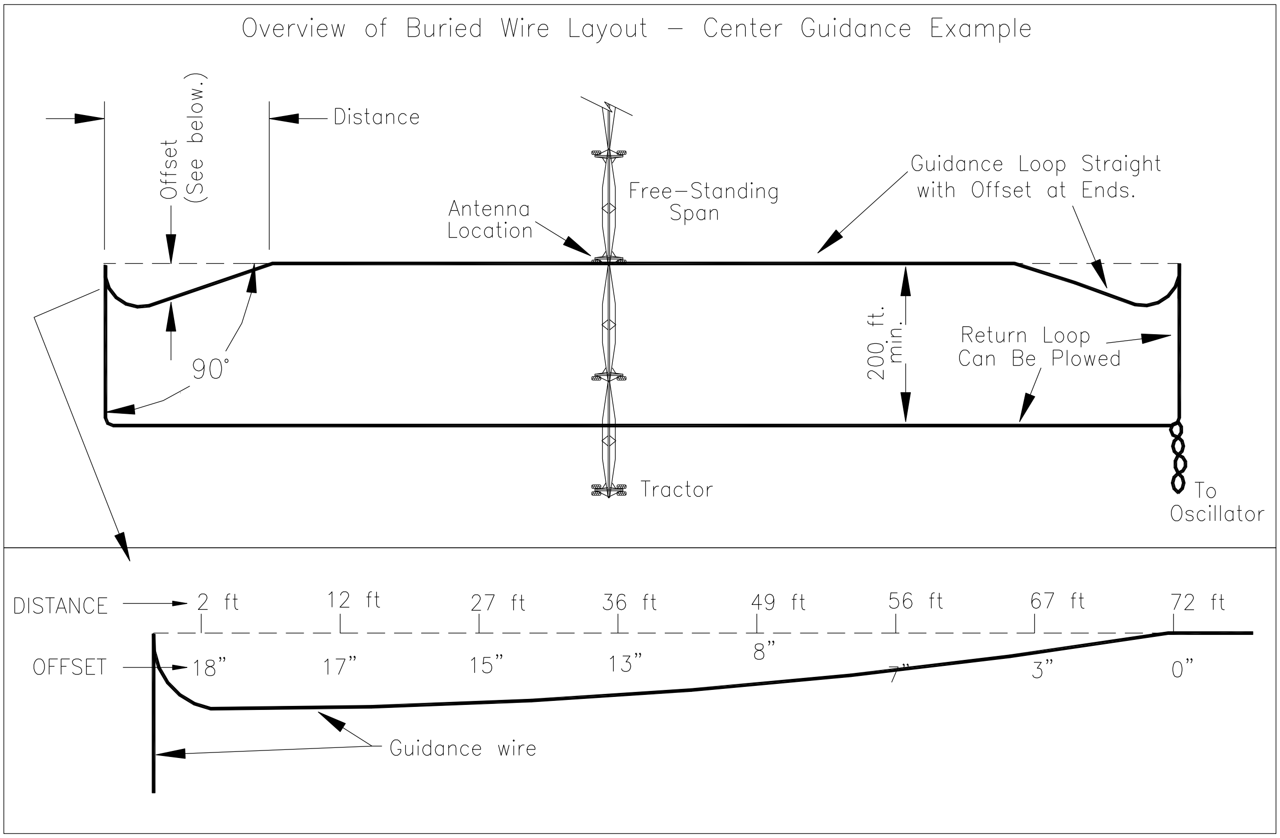


Figure 34. Correction Curve Dimensions at Field Edge (7/08).

The Guidance will safety out when the Steering antenna reaches the end of the loop. The Oscillator should be connected in along the end of the loop or on the non-guiding portion of the loop. If the Oscillator is located along the Guidance portion of the loop a twisted pair must be used for the connection.

Guidance wire location must be precisely measured the entire length of the field. It is critical that the Guidance wire is straight, even small deflections can cause guidance problems. The Guidance wire should be placed in a 24 inch wide by 30 inch deep trench, pulled tight by hand and final straightness checked with a transit. Cover the wire with a shallow layer of dirt by hand to hold the wire in place for final burial.

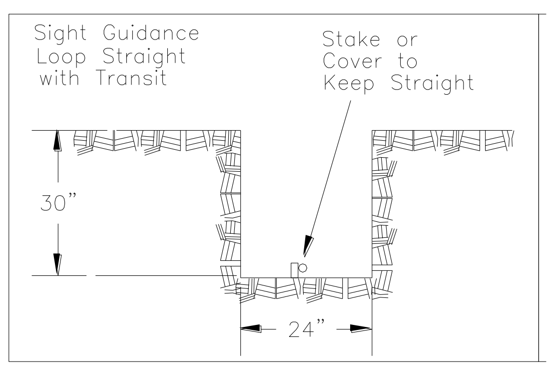


Figure 35. Trench Dimensions for Buried Wire

Wire Integrity

Perform "Initial Checks" on wire before installation. After installation perform "Checks at Splice pit". (See later pages).

IMPORTANT: As the buried wire is installed, ohm the wire with an electrical meter to ensure continuity. If continuity is lost or an unusually high reading is observed, mark the location for future reference. To test initially, measure copper to copper, copper to shield and shield to shield. When ready to bury the wire, tie the shield to the copper on the middle of the roll. Keep an ohm meter on the wire at the start point. NOTE: Some manufacturers of the Armored Cable do not guarantee continuity of the shield (armor). If this is the case, you will not be able to perform this check as the cable is buried. Once buried, check copper to copper.

Care should be taken while installing the buried wire to keep from over-stressing it. Great care should also be taken to eliminate excessive bending or kinking of the wire as it is installed. Over-stressing or excessive bending and kinking can cause the stainless steel shield in the wire to separate and "etch" the copper wire, thus causing a break or a short.

The Guidance Loop must be installed in an open trench so the wire can be sighted in perfectly straight. The return loop, if not used for guidance, can be plowed in.

When installing the guidance wire, ensure the wire flows freely through the burial equipment and that the wire is never bent sharper than an 18" radius. (see Figure below) Normal burial depth for the guidance wire is 30". In areas where there is rough ground and possibility of erosion, bury the wire 38". Over the years, the shallowing cable may be damaged by normal tillage practices.

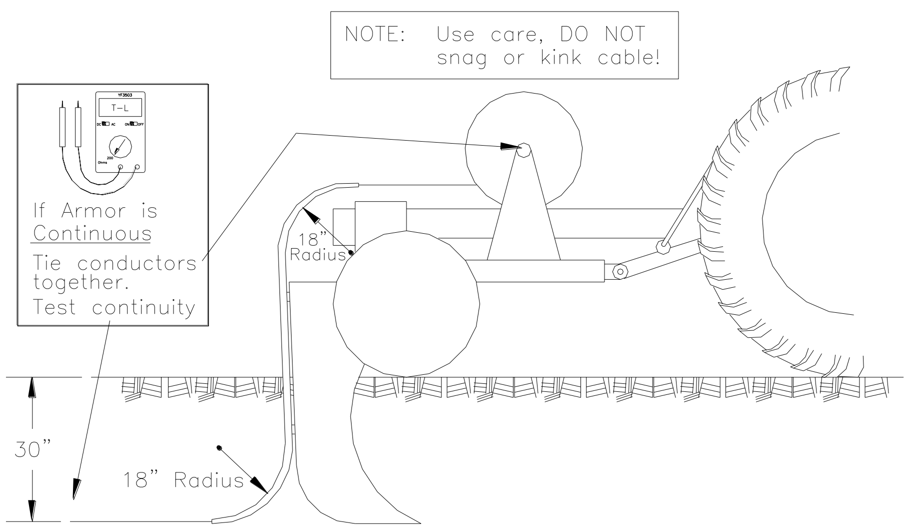
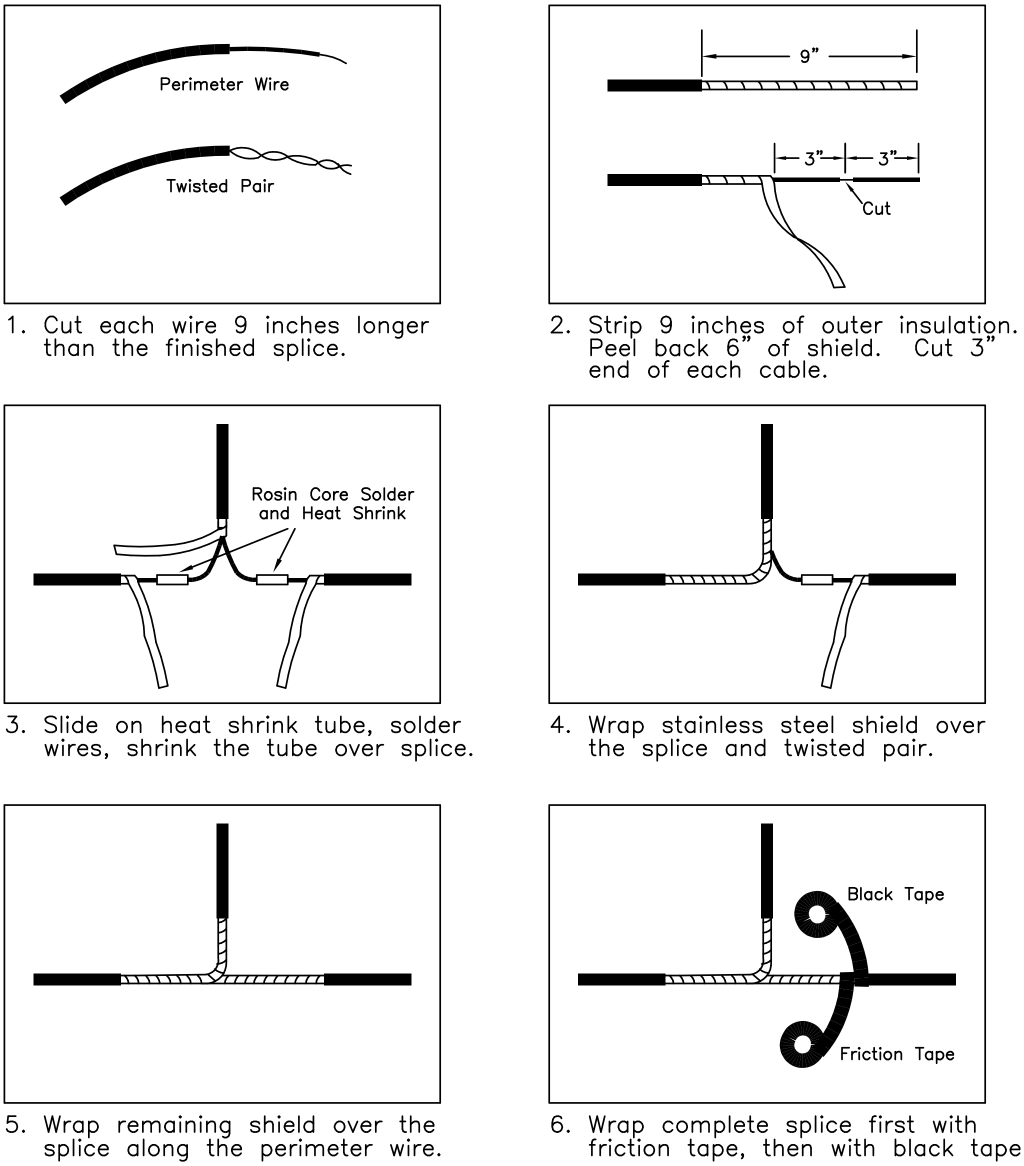


Figure 36. Guidance Wire Burial with Plow (Return Loop Only)

Splicing The Buried Wire (Figure 37)

After the perimeter wire and twisted pair are buried and installation has been checked, splice them together per Figure 37.

Tips: Remember the wire is buried 30".
At the point of the splice, a hole must be dug to allow room to work.
All splices must be soldered, using rosin core solder, and covered with heat shrink tubing.
Installation and wiring of the Oscillator is shown on the next page.



1. Cut each wire 9 inches longer than the finished splice.
2. Strip 9 inches of outer insulation. Peel back 6” of shield. Cut 3” end of each cable.
3. Slide on heat shrink tube, solder wires, shrink the tube over splice.
4. Wrap stainless steel shield over the splice and twisted pair.
5. Wrap remaining shield over the splice along the perimeter wire.
6. Wrap complete splice first with friction tape, then with black tape.

Figure 37. Splicing the Buried Wire

Oscillator Wiring Prior to Nov. 2008 (Figure 38)

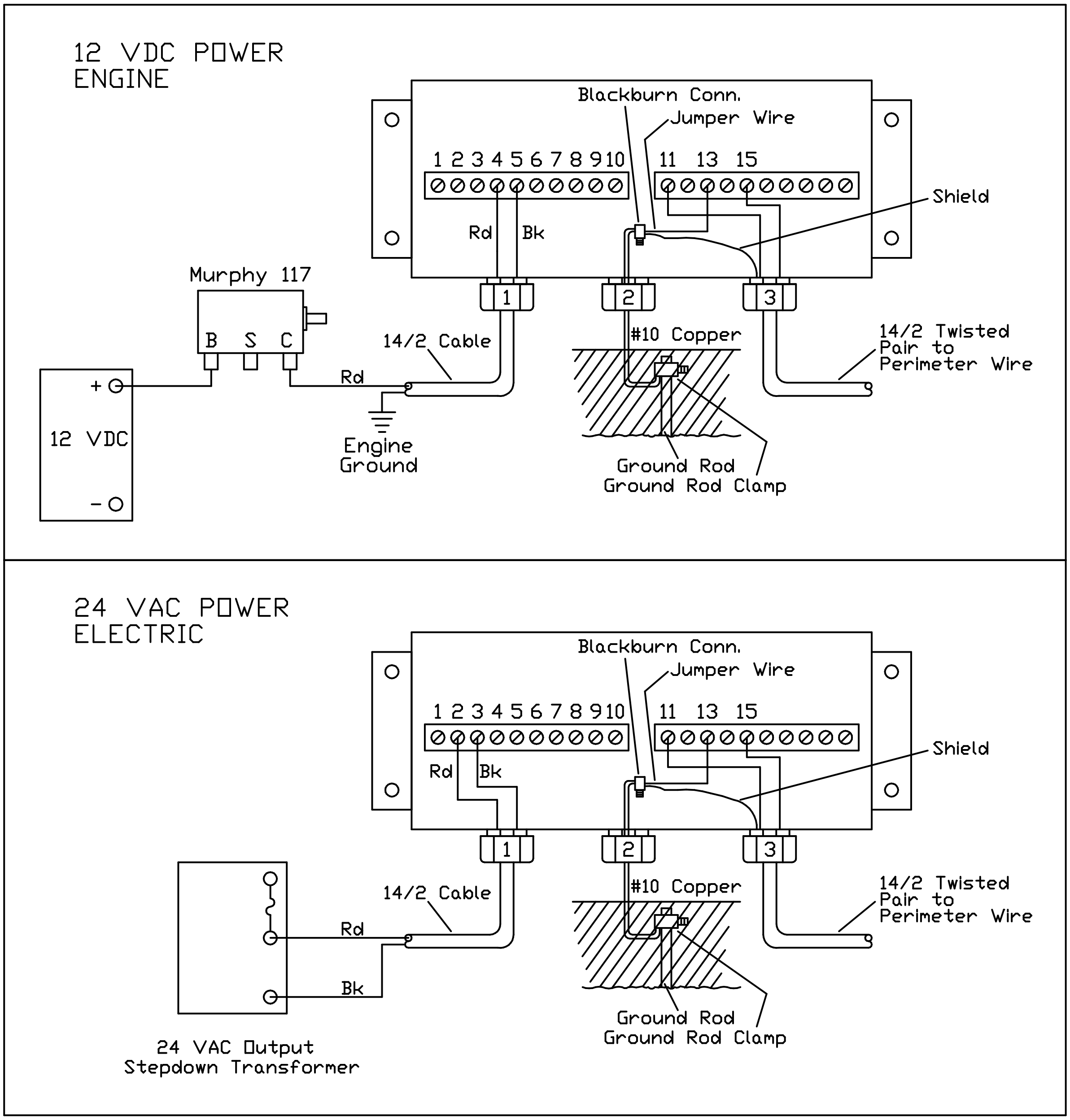


Figure 38. Oscillator Wiring – Engine or Electric Supply Prior to Nov. 2008.

Wiring of Oscillator Box Assy II (Nov. 1, 2008 -)

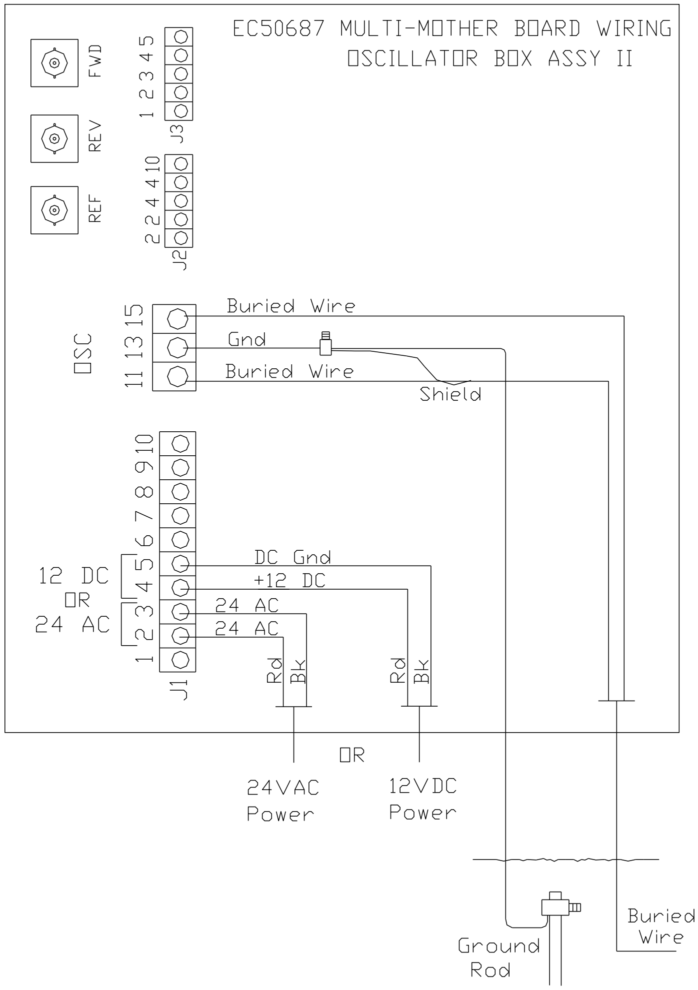
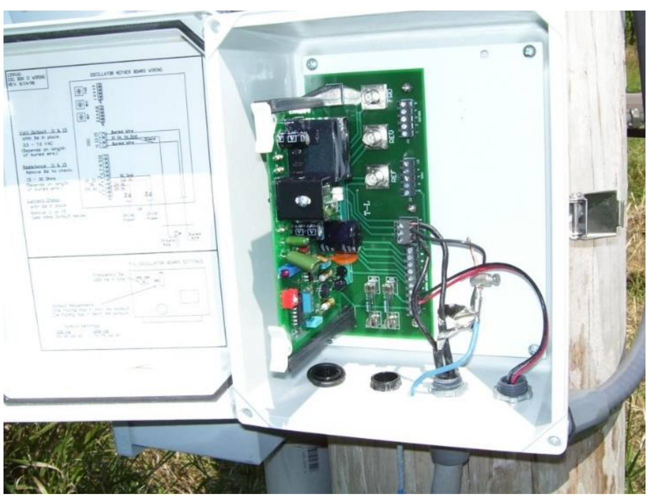


Figure 38a. Oscillator Box Assy II Wiring (Nov. 1, 2008 -)

Example, Oscillator Box II Wired for 24 VAC Operation



INSTALLATION DATA OF BURIED WIRE

A. Initial Checks

Before burial of wire, calculate the resistance of each spool. Then measure and record the actual reading by testing the resistance with an Ohm meter.

Perimeter Wire Spool resistance calculation

$$\frac{2 \times \text{Length of run (feet)} \times 2.525}{1000} = \text{OHMS}$$

CALCULATED ________OHMS

ACTUAL ________OHMS

Twisted Pair Wire Spool resistance calculation, if used.

$$\frac{[2 \times \text{Length of run (feet)} + 400\text{ft}\,] \times 2.525}{1000} = \text{OHMS}$$

CALCULATED ________OHMS

ACTUAL ________OHMS

B. Checks At Splice Pit

After burial of wire and before splicing the wire together check the wire resistance again. This check will insure that the wire was not stressed or broken during installation. Make sure that the ends of the copper are not shorted to the stainless shield.

1. **Perimeter Wire** copper to copper ACTUAL ________OHMS

2. **Twisted Pair** Wire, short copper to copper at the oscillator to get reading make sure the leads are not shorted against the stainless. ACTUAL ________OHMS

3. **Left side Copper conductor to Ground,** check for short to ground:
   Using ground rod shipped with system, push ground rod into bottom of pit 18-24"
   A properly insulated wire will read 20,000 to 200,000 ohms. ________OHMS

4. **Right side Copper conductor to Ground,** check for short to ground:
   Same as #3, but with right side copper conductor. ________OHMS

5. **Left side Stainless Shield to Ground,** check for short to ground:
   Same as #3, but with left side shield. ________OHMS

6. **Right side Stainless Shield to Ground: ,** check for short to ground:
   Same as #3, but with right side shield. ________OHMS

7. **Left side Copper conductor to Stainless Shield:**
   Make sure that the ends of the copper are not shorted to the stainless shield on the perimeter wire. Check ohms from copper to stainless.
   A properly insulated wire will read 20,000 to 200,000 ohms. ________OHMS

8. **Left side Copper conductor to Stainless Shield:**
   Same as #5, but with opposite end of wire. ________OHMS

9. **Shield to Shield of perimeter wire:** Some manufacturers of the Armored Cable do not guarantee continuity of the shield (armor). You may not be able to perform this check.
   Shield to shield resistance ranges from 0.10 ohms/ft to 0.15 ohms/ft
   using 0.13 ohm as average x 9200 ft loop = 1200 ohms estimate. ________OHMS

10. **Splice the Wire.** After the checks have been completed at the splice pit the wire can be spliced together as explained on page 25.

C. Checks At Oscillator

Remove the Oscillator circuit board for initial checks. Wire the twisted pair into the Oscillator mother board as shown. Make sure ground rod is set securely into the ground.

1. **Left side Copper conductor to Ground,** check for short to ground:
   A properly insulated wire will read 20,000 to 200,000 ohms. _______OHMS

2. **Right side Copper conductor to Ground,** check for short to ground:
   Same as #1, but with right side copper conductor. _______OHMS

3. **Copper to Copper:**
   Terminal 11 to terminal 15 if wire is already connected in.
   Resistance should equal the total of the Perimeter wire and twisted pair from steps 1 and 2 at the Splice pit. **_______OHMS***

4. **Attach Oscillator circuit board to mother board.**

5. **Check power to Oscillator:**
   If 24 VAC - 24 volt AC minimum between terminal #2 and #3.
   If 12 VDC - 12 volt DC minimum between terminal #4 and #5.

6. **Oscillator Milliamp Output:**
   Remove one of the buried wire leads (#15 or #11) from the terminal and connect a digital meter set on 200 mA range between the lead and the terminal where the wire was connected. This makes a series circuit and allows all current to pass through the meter. Repeat for opposite wire.

   Right Side Terminal #15 **_______mA***
   Left Side Terminal #11 _______mA

   - The larger of the two readings should be 90 mA $\pm$ 10 mA.
   - A difference of 5 to 10 mA between the two readings is acceptable.
   - A difference of 15 mA or more may indicate a problem with the wire.
   - A difference of 15 mA up indicates a bad skinned area and must be corrected.

7. **Oscillator Voltage Output:**
   Reconnect the buried wire leads. Check the voltage drop between terminals #11 and #15. Record the AC volt reading. **_______Vac***

   - 0 - 2 Vac indicates a bad Oscillator Board or short between the twisted pair.
   - 2 - 7 Vac is normal.
   - 10 Vac or higher indicates high resistance from a bad splice or break in the wire.

8. **Record Measurements for future reference.** It is handy to record the measurements (marked with an “*” above) inside the lid of the Oscillator enclosure.

D. Check Reference Signal

With the system <u>located over the guidance wire</u>, remove the reference antenna lead from the guidance control box (marked REF). Energize the buried wire. Connect the antenna lead to the voltmeter and dummy load (BNC - double banana plug/51K ohm resistor across plugs). Record AC mV reading.

- 13 - 16 mVAC is normal.
- 11 - 13 mVAC indicates that signal is weak and could shut system down.
- 9 - 10 mVAC indicates the signal strength is too weak for the safety circuit to hold in.
- 2 mVAC or less indicates no signal from the Oscillator.